

**Supplementary materials for the paper “Rank and Factor Loadings Estimation in Time Series Tensor Factor Model by Pre-averaging”- Additional simulation results, data analysis, and proof of Theorem 3.1, Theorem 4.1, Theorem 4.2, Theorem 5.1, Theorem 5.2 and Theorem 5.3**

We first present some extra simulation results and another piece of real data analysis in Section 1. The proofs of all theorems in the original paper are presented in Section 3.

## 1 Extra Simulation Results and Data Analysis

### 1.1 Effect of $M_0$ in pre-averaging and projection

We test empirically the effect of using different  $M_0$  in the pre-averaging procedure. We fix  $M = 5$ , and consider  $M_0 = 200, 400$  and  $800$ , respectively (refer to Section 3 for the meaning of these parameters). For each  $M_0$ , we first calculate the pre-averaging estimator  $\hat{\mathbf{Q}}_{k,pre}$  for each  $k \in [K]$ , and obtain  $\check{\mathbf{q}}_k^{(0)} = \hat{\mathbf{q}}_{k,pre} = \hat{\mathbf{Q}}_{k,pre,(1)}$ . Then we calculate  $\check{\mathbf{q}}_k^{(m)}$  according to the Algorithm for Iterative Projection Direction Refinement in Section 4.1 for  $m = 1, \dots, 29$ . Finally, we obtain the iterative projection estimator  $\check{\mathbf{U}}_k$  by utilising  $\check{\mathbf{q}}_k^{(29)}$  as the projection direction for each  $k \in [K]$ .

For estimation accuracy, we use the measure  $\|\hat{\mathbf{Q}}_k \hat{\mathbf{Q}}_k^T - \mathbf{U}_k \mathbf{U}_k^T\|$ , i.e. the spectral norm of the difference of the estimated and true underlying projection matrices, where  $\mathbf{U}_k$  is obtained using the singular value decomposition of  $\mathbf{A}_k$ . For ease of presentation, we only display results under  $t_3$  distributed errors for each setting, and consider the following three settings of different dimensions:

- i.  $K = 2, T = 100, d_1 = d_2 = 40$ ;
- ii.  $K = 2, T = 200, d_1 = d_2 = 80$ ;
- iii.  $K = 3, T = 200, d_1 = d_2 = d_3 = 15$ .

We omit the results for  $\mathbf{A}_2$  (and  $\mathbf{A}_3$ ) and only display the estimation accuracy for  $\mathbf{A}_1$  in Figures 1 to 4, since the results for  $\mathbf{A}_1$  and  $\mathbf{A}_2$  (and  $\mathbf{A}_3$ ) are very similar.

From Figure 1 to 3, both the pre-averaging and iterative projection estimators have better estimation accuracy as  $T$  and  $d_k$  increase. Regarding the effect of  $M_0$ , the first thing to observe is that in each sub-setting (a), the initial pre-averaging estimator performs slightly better with larger  $M_0$ , while in sub-setting (b),  $M_0$  does not significantly affect the accuracy of the pre-averaging estimator. This is natural, as the mean of each element of  $\mathbf{A}_k$  in sub-setting (a) is 0 while it is non-zero in sub-setting (b), where pre-averaging estimators take advantage. In fact, in sub-setting (b), all entries of  $\mathbf{A}_k$  have the same sign, so  $s_{k,max} \asymp d_k^2$  (as the strongest factor in  $\mathbf{A}_k$  is pervasive) can be easily achieved regardless of  $M_0$ . For sub-setting (a), around half of the entries of  $\mathbf{A}_k$  are positive and the other half are negative, so it is more difficult to achieve the theoretical maximum order  $s_{k,max} \asymp d_k^2$  (this happens when the majority of entries in the chosen sample  $S_k$  are of the same sign). However, in this case, we can still easily achieve  $s_{k,max} \asymp d_k$  with a small  $M_0$  (as discussed in Section 3.4 of the original paper), which is good enough to serve as an initial estimator for the Algorithm for Iterative Projection Direction Refinement.

The second thing to observe is that in most settings and dimensions, the differences in performance of the initial pre-averaging estimators (caused by changing  $M_0$ ) do not affect the estimation accuracy of the subsequent iterative projection estimators. This is because in our iterative projection, we only need to

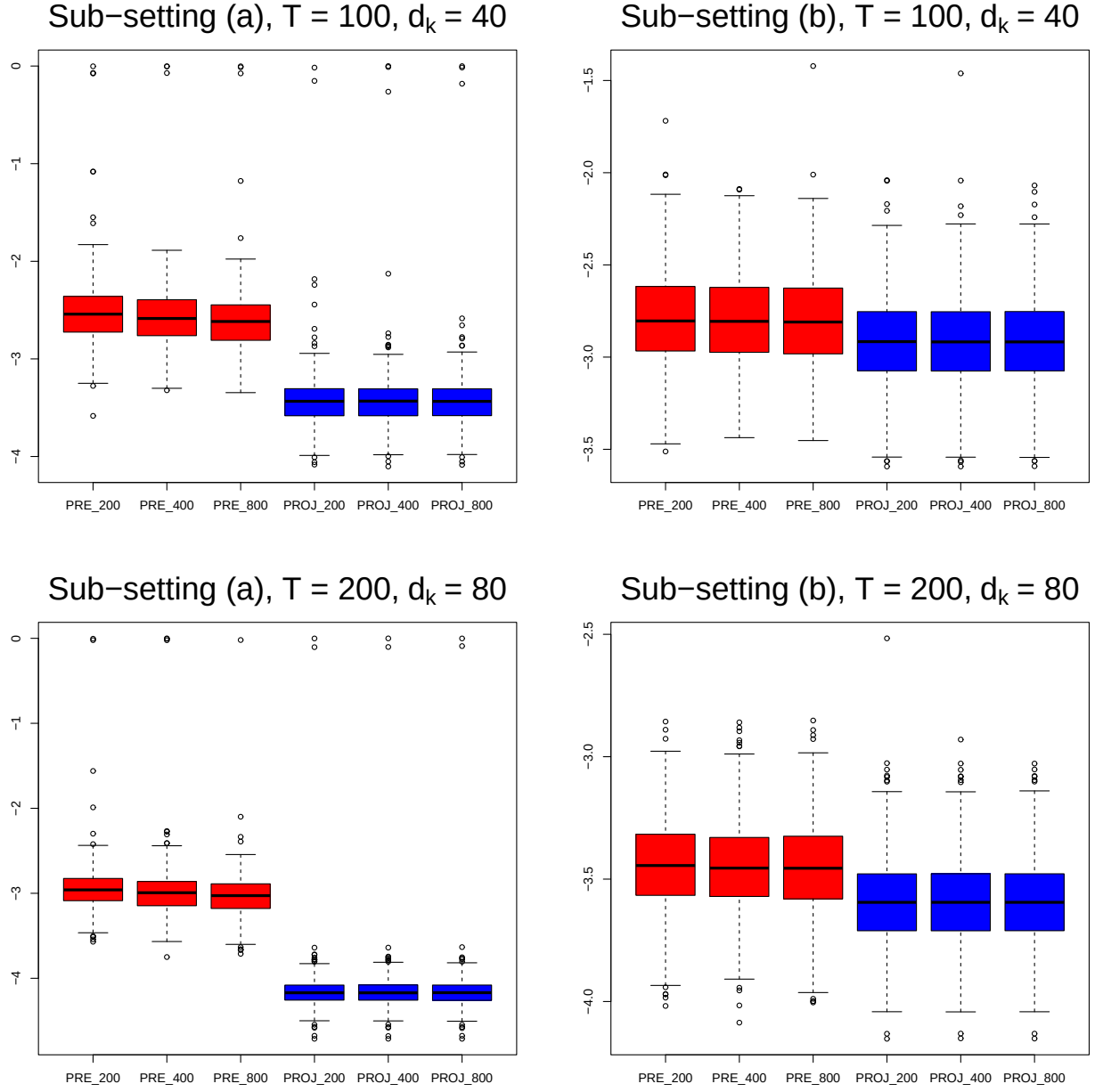


Figure 1: Plot of estimation error  $\|\hat{\mathbf{Q}}_1 \hat{\mathbf{Q}}_1^T - \mathbf{U}_1 \mathbf{U}_1^T\|$  (in log-scale) for Setting (I),  $K = 2$ . *Left:* Sub-setting (a). *Right:* Sub-setting (b).

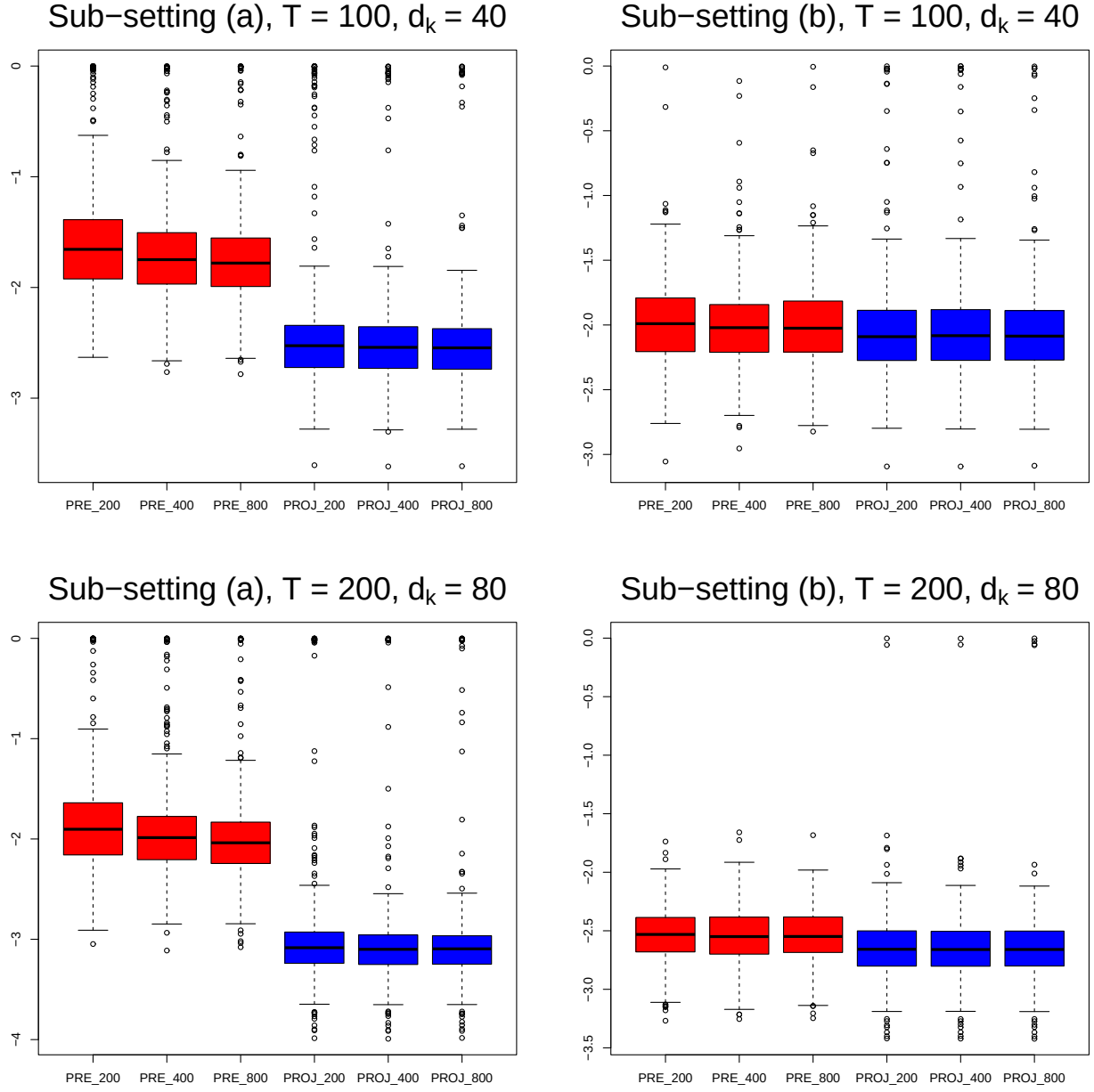


Figure 2: Plot of estimation error  $\|\hat{\mathbf{Q}}_1 \hat{\mathbf{Q}}_1^T - \mathbf{U}_1 \mathbf{U}_1^T\|$  (in log-scale) for Setting (II),  $K = 2$ . *Left:* Sub-setting (a). *Right:* Sub-setting (b).

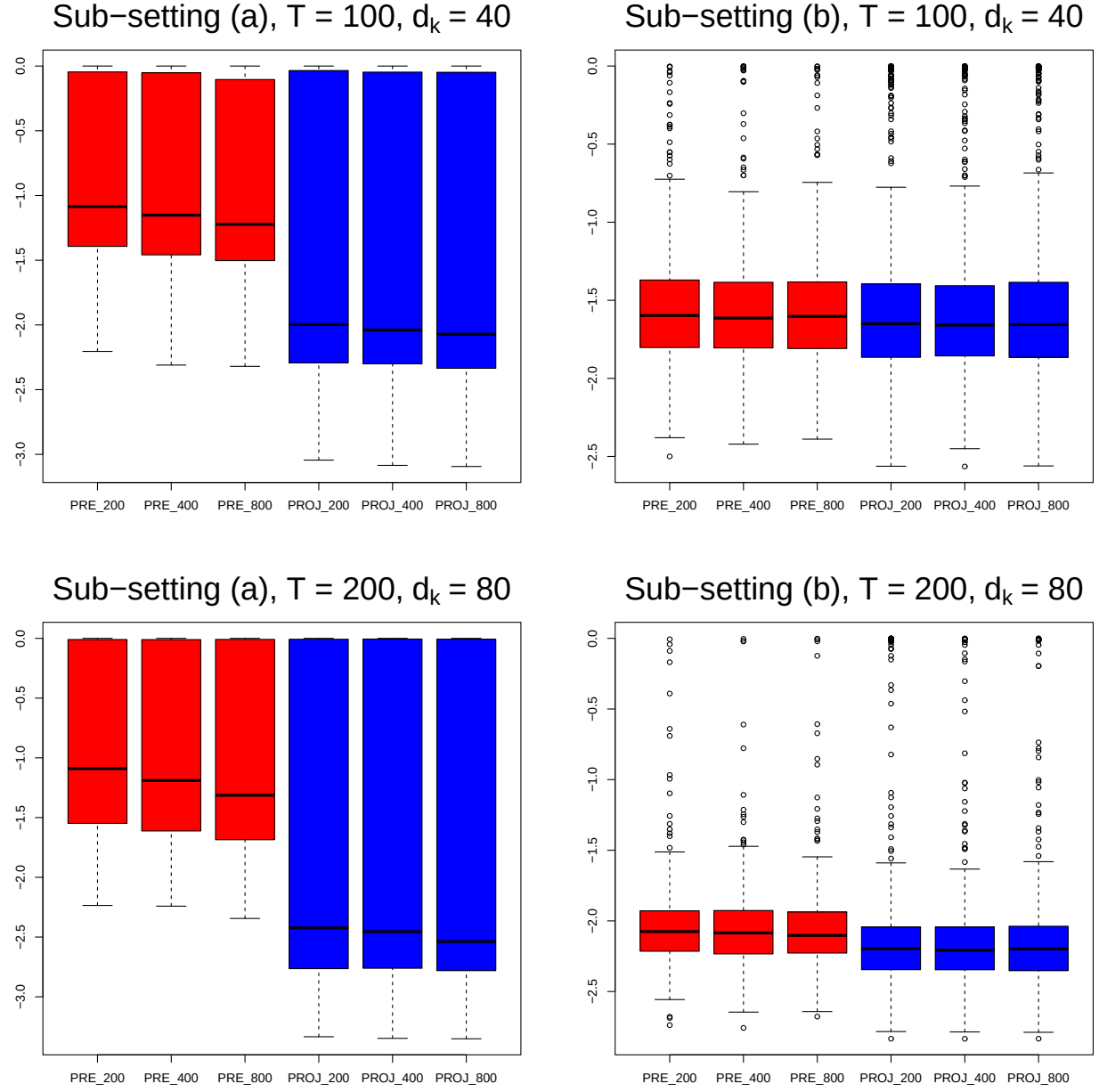


Figure 3: Plot of estimation error  $\|\hat{\mathbf{Q}}_1 \hat{\mathbf{Q}}_1^T - \mathbf{U}_1 \mathbf{U}_1^T\|$  (in log-scale) for Setting (III),  $K = 2$ . *Left:* Sub-setting (a). *Right:* Sub-setting (b).

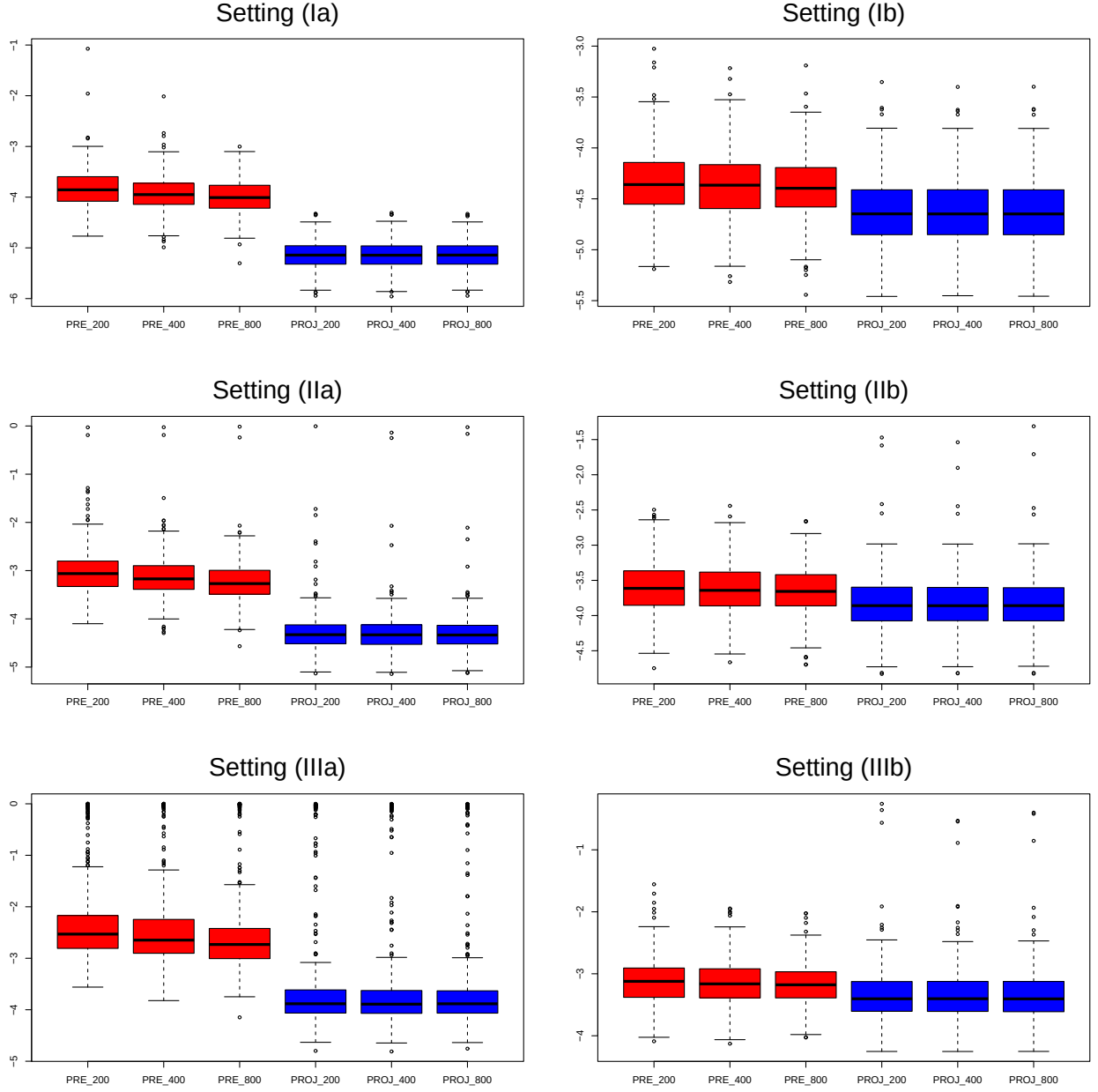


Figure 4: Plot of estimation error  $\|\hat{\mathbf{Q}}_1\hat{\mathbf{Q}}_1^T - \mathbf{U}_1\mathbf{U}_1^T\|$  (in log-scale) for  $K = 3, d_1 = d_2 = d_3 = 15$ .

utilize the strongest factor direction, which is estimated most accurately by the pre-averaging procedure (see Theorem 3.1 and Section 4.1 of the original paper for more details). Hence, while increasing  $M_0$  can lead to better performances of  $\hat{\mathbf{Q}}_{k,pre}$  as a whole, a small  $M_0$  is usually sufficient to provide an accurate initial direction  $\hat{\mathbf{q}}_k^{(0)} = \hat{\mathbf{q}}_{k,pre} = \hat{\mathbf{Q}}_{k,pre,(1)}$  for projection. This is especially true in Setting (Ia) and (IIa) where the strongest factor is pervasive. The only exception is Setting (IIIa) with  $K = 2$ , where even the strongest factor is too weak to be estimated accurately, so increasing  $M_0$  may lead to slightly better estimators after projection. However, such an issue is relieved when  $K = 3$ , since we have directions from more modes to be projected, which largely improves the estimation accuracy. In fact, a comparison between Figure 4 and Figure 1 to 3 reveals that both the pre-averaging and iterative projection estimator perform better as  $K$  increases.

Hence in general, increasing  $M_0$  does not significantly affect the performance of our final estimator obtained by iterative projection in most scenarios. To save some computational time without sacrificing estimation accuracy, we use  $M_0 = 200$  for the pre-averaging procedure in all of the subsequent simulations.

## 1.2 More simulation results

We present some simulation results that are not included in the original paper. See the details of the settings and the methods being compared in Section 6 of the original paper.

From Figure 5 to 7, we have similar conclusions as in the original paper that our iterative projection estimator performs the best, or at least on par with other competitors in all settings and dimensions. Pre-averaging estimator and the iterative projection estimator take advantage of non-zero means in a factor loading matrix and also can perform better when there are weak factors, while on par with other estimators designed for pervasive factors when all factors are indeed pervasive.

Table 1 records the average computational time of factor loading estimations under different methods and dimensions. For PROJ, we record the total computational time of the initial pre-averaging procedure and the iterative projection algorithm. It can be seen that the computational cost of our method is generally on par with iTIPUP and PE, while our method can be faster when  $d_k$  is small, but become slower when  $d_k$  grows large. For other methods, MRTS is extremely computational expensive, and it gives poor estimation accuracy as well.  $\alpha$ -PCA is the fastest because it does not require any iterations, but it also performs poorly as can be seen in Figure 2. The computational cost of RTFA also grows quickly when  $K$  and  $d_k$  increase, and it does not give good estimates when  $K = 3$  (see Figure 6), which prompt us to not consider using it for  $K = 4$ .

|         |                     | PROJ  | iTIPUP | PE    | $\alpha$ -PCA | RTFA   | MRTS  |
|---------|---------------------|-------|--------|-------|---------------|--------|-------|
| $K = 2$ | $T = 100, d_k = 40$ | 1.15  | 2.01   | 1.99  | 0.08          | 3.88   | 9.32  |
|         | $T = 200, d_k = 80$ | 4.63  | 4.63   | 4.55  | 0.32          | 15.65  | 49.73 |
| $K = 3$ | $T = 100, d_k = 15$ | 3.26  | 5.79   | 5.77  | /             | 22.72  | /     |
|         | $T = 200, d_k = 25$ | 10.66 | 10.05  | 10.04 | /             | 151.65 | /     |
| $K = 4$ | $T = 200, d_k = 15$ | 43.80 | 34.16  | 33.69 | /             | /      | /     |

Table 1: Mean of the run time (in seconds) for factor loading estimations under different methods and different dimensions.

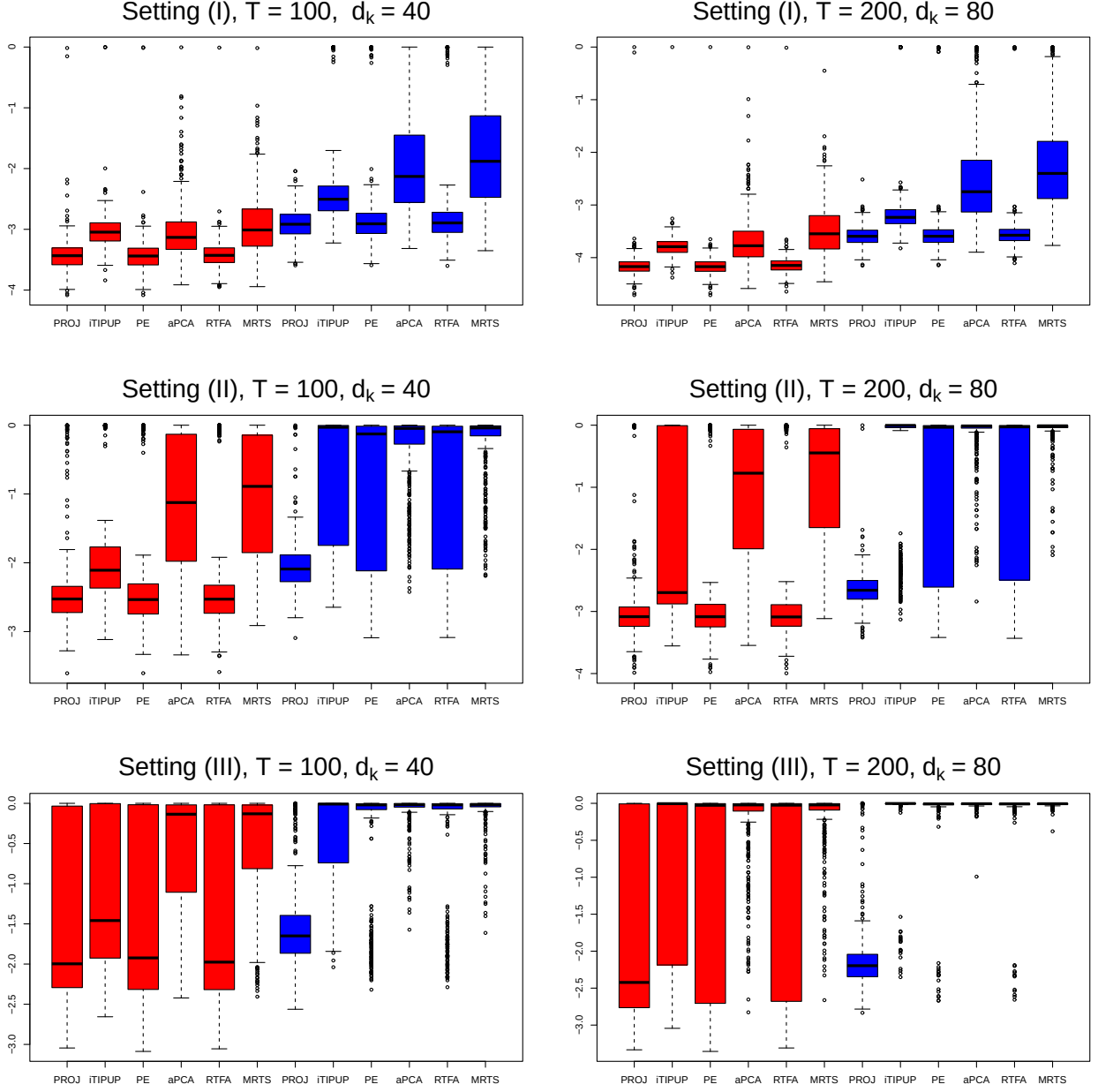


Figure 5: Plot of estimation error  $\|\hat{\mathbf{Q}}_1 \hat{\mathbf{Q}}_1^T - \mathbf{U}_1 \mathbf{U}_1^T\|$  (in log-scale) for  $K = 2$ ,  $t_3$ -distributed errors. In each panel, the left six boxplots (in red) represent sub-setting (a), while the right six boxplots (in blue) represent sub-setting (b).

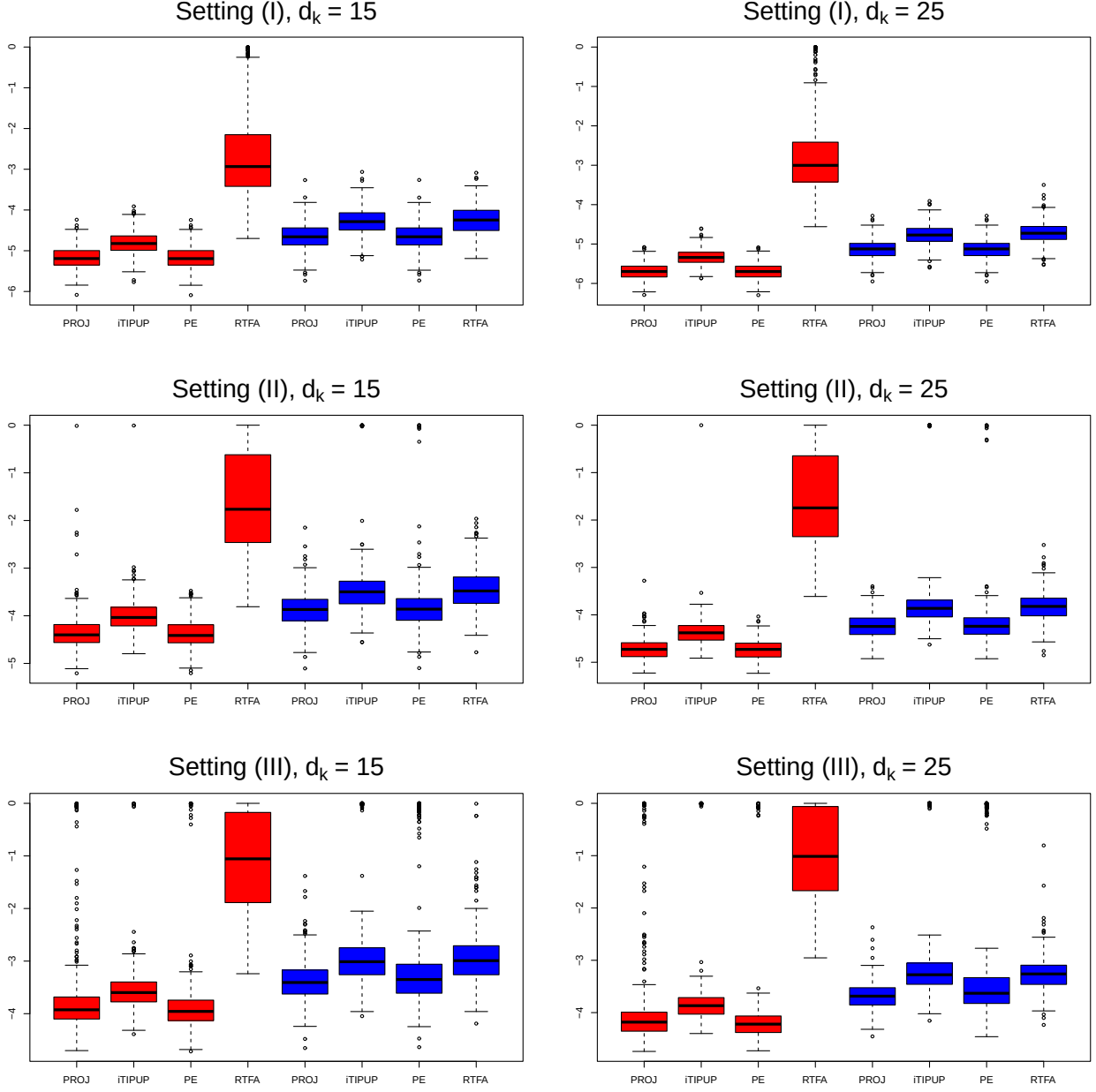


Figure 6: Plot of estimation error  $\|\hat{\mathbf{Q}}_1 \hat{\mathbf{Q}}_1^T - \mathbf{U}_1 \mathbf{U}_1^T\|$  (in log-scale) for  $K = 3$ , normally distributed errors. In each panel, the left four boxplots (in red) represent sub-setting (a), while the right four boxplots (in blue) represent sub-setting (b).



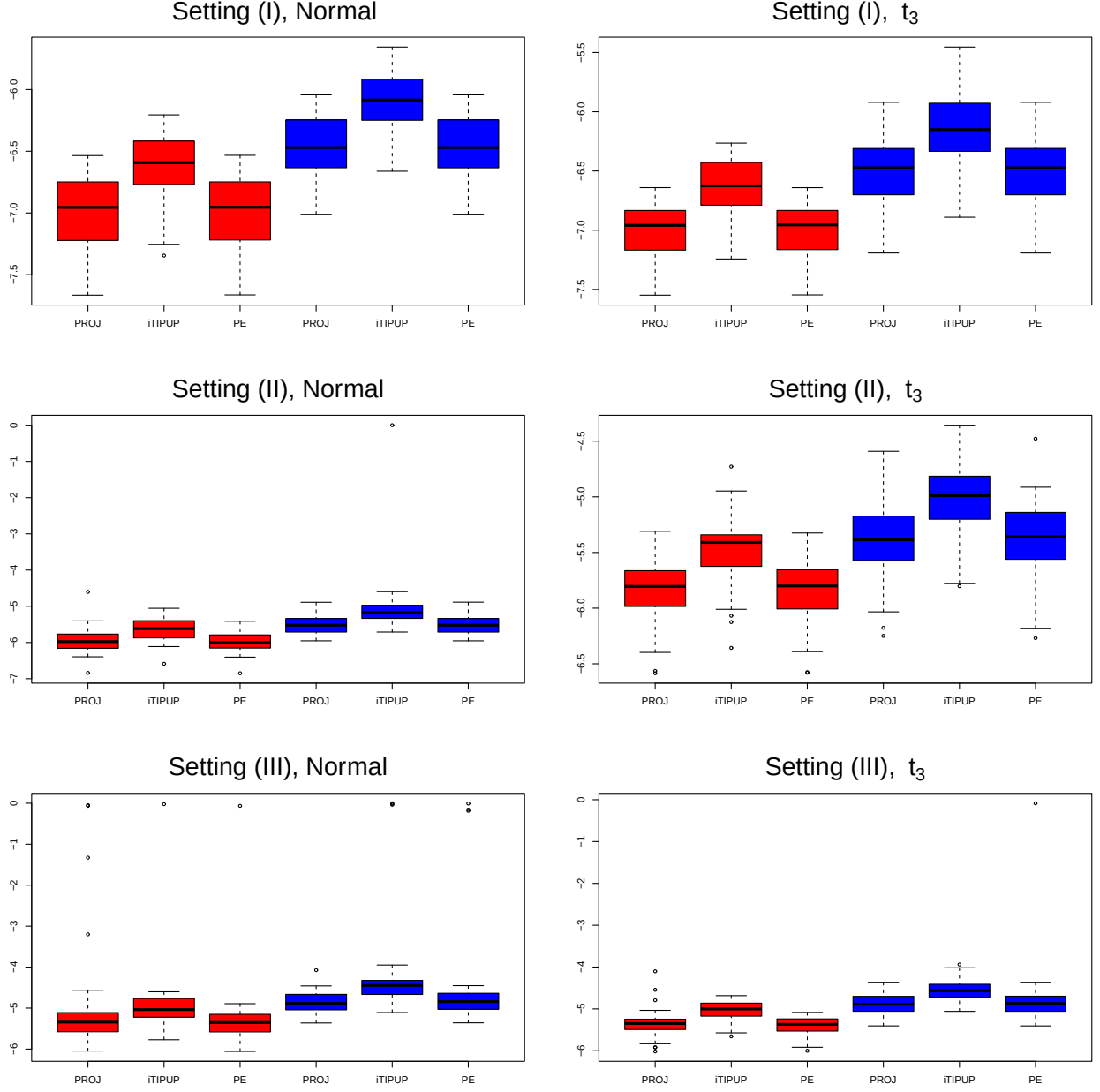


Figure 7: Plot of estimation error  $\|\hat{\mathbf{Q}}_1 \hat{\mathbf{Q}}_1^T - \mathbf{U}_1 \mathbf{U}_1^T\|$  (in log-scale) for  $K = 4, T = 200, d_k = 15$ . In each panel, the left three boxplots (in red) represent sub-setting (a), while the right three boxplots (in blue) represent sub-setting (b).

### 1.3 Fama-French portfolio returns

We analyse a set of Fama-French portfolio returns data formed on size and operating profitability. Stocks are categorized into 10 different sizes (market equity, using NYSE market equity deciles) and 10 different operating profitability (OP) levels (using NYSE OP deciles. OP is annual revenues minus cost of goods sold, interest expense, and selling, general, and administrative expenses divided by book equity for the last fiscal year end). These levels, and hence the stocks in each category, are allocated at the end of June each year. Moreover, the stocks in each of the  $10 \times 10$  categories form exactly two portfolios, one being value weighted, and the other of equal weight. Hence, there are two sets of  $10 \times 10$  portfolios with their time series of returns observed. We use monthly data from July 1973 to June 2021, so that  $T = 576$ , and each data tensor we have thus has size  $10 \times 10 \times 576$ . For more details, please visit

[https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/Data\\_Library/det\\_100\\_port\\_szme\\_op.html](https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/Data_Library/det_100_port_szme_op.html)

Since the market factor is certainly pervasive in financial returns, we use the following CAPM to remove its effects and facilitate detection of potentially weaker factors:

$$\mathbf{y}_t = \bar{\mathbf{y}} + \boldsymbol{\beta}(x_t - \bar{x}) + \mathbf{e}_t,$$

where  $\mathbf{y}_t \in \mathbb{R}^{100}$  contains the returns of the 100 portfolios at time  $t$ ,  $x_t$  is the return for the NYSE composite index at time  $t$ , and  $\boldsymbol{\beta} = (\beta_1, \dots, \beta_{100})^T$  is the vector of  $\beta$ 's for the 100 portfolios. Least squares estimation leads us to

$$\hat{\boldsymbol{\beta}} = \frac{\sum_{t=1}^T (x_t - \bar{x})(\mathbf{y}_t - \bar{\mathbf{y}})}{\sum_{t=1}^T (x_t - \bar{x})^2}.$$

Hence the data we analyse is  $\{\mathbf{y}_t - \bar{\mathbf{y}} - \hat{\boldsymbol{\beta}}(x_t - \bar{x})\}_{t=1, \dots, 576}$ , with each observed vector reshaped into a  $10 \times 10$  tensor.

|                | BCorTh      |             | iTIP-ER     |             | PE-ER       |             | $\alpha$ -PCA-ER |             | RTFA-ER     |             | MRTS-ER     |             |
|----------------|-------------|-------------|-------------|-------------|-------------|-------------|------------------|-------------|-------------|-------------|-------------|-------------|
|                | $\hat{r}_1$ | $\hat{r}_2$ | $\hat{r}_1$ | $\hat{r}_2$ | $\hat{r}_1$ | $\hat{r}_2$ | $\hat{r}_1$      | $\hat{r}_2$ | $\hat{r}_1$ | $\hat{r}_2$ | $\hat{r}_1$ | $\hat{r}_2$ |
| Value Weighted | 2           | 2           | 2           | 2           | 2           | 2           | 2                | 2           | 2           | 2           | 2           | 2           |
| Equal Weight   | 2           | 2           | 2           | 2           | 2           | 2           | 2                | 2           | 2           | 2           | 2           | 2           |

Table 2: Rank estimators for Fama-French Portfolios.

Table 2 shows that all of the rank estimators we consider, including BCorTh, give  $\hat{r}_1 = \hat{r}_2 = 2$  for both value weighted and equal weight portfolios. Using  $(\hat{r}_1, \hat{r}_2) = (2, 2)$ , we estimate the factor loading spaces by our iterative projection method and compared to iTIPUP and PE. Similar to Wang et al. (2019), we show the estimated loading matrices after a varimax rotation that maximizes the variance of the squared factor loadings, scaled by 30 for a cleaner view. To save space, we only show the results for value weighted portfolios in Table 3 and 4.

From Table 3, we can see that from PROJ and PE, OP1 itself (possibly OP2 as well) forms one group while OP6 to OP10 form another. iTIPUP also gives a clear grouping effect but is a bit different from the other two methods. Nevertheless, we can say that OP1 and OP2 possibly represents “low operating profitability”, while OP6 to OP10 are “high operating profitability”, and are governed by different factors. For Size, from the three methods, S1 to S5 form one group (“small size”) while another group contains at least S9 and S10 (“large size”), possibly S6 to S8 as well.

| Method | Factor | OP1 | OP2 | OP3 | OP4 | OP5 | OP6 | OP7 | OP8 | OP9 | OP10 |
|--------|--------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|
| PROJ   | 1      | 8   | -3  | -6  | -7  | -8  | -9  | -12 | -11 | -12 | -14  |
|        | 2      | 26  | 10  | 7   | 5   | 4   | 3   | 1   | 2   | 1   | 0    |
| iTIPUP | 1      | -25 | -14 | -9  | -1  | -2  | -1  | 0   | -3  | 1   | -2   |
|        | 2      | 12  | -13 | -5  | -8  | -7  | -8  | -10 | -11 | -9  | -9   |
| PE     | 1      | 28  | 8   | 4   | 2   | 1   | 0   | 0   | 0   | 0   | 2    |
|        | 2      | 5   | -5  | -7  | -9  | -10 | -10 | -11 | -11 | -12 | -12  |

Table 3: OP Factor Loading Matrices for Value Weighted Portfolios after rotation and scaling. Magnitudes larger than 8 are highlighted in red.

| Method | Factor | S1  | S2  | S3  | S4  | S5  | S6  | S7  | S8  | S9  | S10 |
|--------|--------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| PROJ   | 1      | -11 | -11 | -11 | -10 | -11 | -9  | -9  | -6  | -4  | 10  |
|        | 2      | -2  | 1   | 0   | 3   | 4   | 6   | 6   | 8   | 10  | 25  |
| iTIPUP | 1      | -11 | -13 | -13 | -13 | -11 | -9  | -3  | -2  | 0   | 9   |
|        | 2      | 6   | 2   | -2  | -2  | -5  | -9  | -12 | -12 | -14 | -16 |
| PE     | 1      | -14 | -15 | -12 | -10 | -9  | -6  | -4  | -1  | 2   | 9   |
|        | 2      | 5   | 1   | -2  | -5  | -7  | -10 | -11 | -13 | -14 | -15 |

Table 4: Size Factor Loading Matrices for Value Weighted Portfolios after rotation and scaling. Magnitudes larger than 8 are highlighted in red.

## 2 A Brief Introduction to TensorPreAve

Our methods are written into an R package **TensorPreAve** published on CRAN and GitHub. For a given data of tensor time series, the function **rank\_factors\_est** can estimate both the rank of core tensors and factor loading matrices simultaneously.

Alternatively, there are individual functions designed for each step of our estimation procedure. The pre-averaging method is implemented in function **pre\_est**, with default parameters chosen carefully to be applied in most scenarios and dimensions. We also provide an alternative way to tune the parameter  $j$  manually using the function **pre\_eigenplot** to observe the eigenvalues of a random sample (see Section 3.3 for more details). With initial estimated directions of the strongest factors obtained by the function **pre\_est**, users can feed them into the function **iter\_proj** to get the iterative projection estimator. Finally, the rank estimator (BCorTh) can be obtained by the function **bs\_cor\_rank** by inputting the refined directions from **iter\_proj**. The default number of bootstrapped samples is 50, but users can reduce it to 10 to save computational time when tensor dimension is large. Please refer to the vignettes and manual of **TensorPreAve** for more details.

## 3 Proof of Theorems

Before we present our proofs, we first express equivalent versions of Assumption (E2) and (F1) below, which makes the presentation of our proofs easier.

(E2) For  $t \in [T]$  and  $k \in [K]$ , defining  $\mathbf{f}_{e,t,\ell}^{(k)} = (f_{e,t,\ell,j}^{(k)})$  to be the  $\ell$ -th column of  $\text{mat}_k(\mathcal{F}_{e,t})$ , let

$$f_{e,t,\ell,j}^{(k)} = \sum_{q \geq 0} a_{e,q} z_{e,t-q,\ell,j}^{(k)}, \quad j \in [r_{e,k}], \ell \in [r_{e,-k}]; \quad \epsilon_{t,\ell,j}^{(k)} = \sum_{q \geq 0} a_{\epsilon,q} z_{\epsilon,t-q,\ell,j}^{(k)}, \quad j \in [d_k], \ell \in [d_{-k}],$$

where the coefficients  $a_{e,q}$  and  $a_{\epsilon,q}$  are such that  $\sum_{q \geq 0} a_{e,q}^2 = \sum_{q \geq 0} a_{\epsilon,q}^2 = 1$  and  $\sum_{q \geq 0} |a_{e,q}| \leq C$ ,  $\sum_{q \geq 0} |a_{\epsilon,q}| \leq C$  for some constant  $C$ . For each  $k \in [K]$ , the series of random variables  $\{z_{e,t,j}^{(k)}\}$  and  $\{z_{\epsilon,t,\ell,j}^{(k)}\}$  are independent of each other, all with i.i.d. elements having mean 0 and variance 1.

(F1) Let  $\mathbf{f}_{t,\ell}^{(k)} = (f_{t,\ell,j}^{(k)})$  be the  $\ell$ -th column vector in  $\text{mat}_k(\mathcal{F}_t)$ ,  $\ell \in [r_{-k}]$ . We assume that

$$f_{t,\ell,j}^{(k)} = \sum_{q \geq 0} a_{f,q} z_{f,t-q,\ell,j}^{(k)}, \quad j \in [r_k],$$

where we have  $\sum_{q \geq 0} a_{f,q}^2 = 1$  and  $\sum_{q \geq 0} |a_{f,q}| \leq C$  for some constant  $C$ . For each  $k \in [K]$ , the series of random variables  $\{z_{f,t,\ell,j}^{(k)}\}$  has i.i.d. elements having mean 0 and variance 1.

We next introduce some notations which will be used in the proof of Theorem 3.1. As introduced in Section 3, for a particular sample  $S_{-k,m} \subseteq [d_{-k}]$  with  $|S_{j,m}| \asymp d_j, j \neq k$ , denote  $\tilde{\mathbf{x}}_{t,k,m} = \sum_{i \in S_{-k,m}} \mathbf{x}_{t,-k,i}$ ,  $\tilde{\mathbf{f}}_{t,k,m} = \sum_{i \in S_{-k,m}} \mathbf{f}_{t,-k,i}$ ,  $\tilde{\mathbf{e}}_{t,k,m} = \sum_{i \in S_{-k,m}} \mathbf{e}_{t,-k,i}$  and  $\tilde{\mu}_{k,m} = \sum_{i \in S_{-k,m}} \mu_{-k,i}$ . Further, define  $\ddot{\mathbf{f}}_{t,k,m} = \tilde{\mathbf{f}}_{t,k,m} / s_{-k,m}^{\frac{1}{2}}$ , write  $\ddot{\mathbf{F}}_{k,m} = [\ddot{\mathbf{f}}_{1,k,m}, \dots, \ddot{\mathbf{f}}_{T,k,m}] \in \mathbb{R}^{r_k \times T}$  and  $\tilde{\mathbf{F}}_{k,m} = [\tilde{\mathbf{f}}_{1,k,m}, \dots, \tilde{\mathbf{f}}_{T,k,m}]$ . Similarly, denote  $\ddot{\mathbf{x}}_{t,k,m} = \tilde{\mathbf{x}}_{t,k,m} / s_{-k,m}^{\frac{1}{2}}$  and  $\ddot{\mathbf{X}}_{k,m} = (\ddot{\mathbf{x}}_{1,k,m}, \dots, \ddot{\mathbf{x}}_{T,k,m})^T \in \mathbb{R}^{T \times d_k}$ ,  $\tilde{\mathbf{X}}_{k,m} = (\tilde{\mathbf{x}}_{1,k,m}, \dots, \tilde{\mathbf{x}}_{T,k,m})^T$ . Also let  $\ddot{\mathbf{e}}_{t,k,m} = \tilde{\mathbf{e}}_{t,k,m} / s_{-k,m}^{\frac{1}{2}}$ ,  $\ddot{\mathbf{E}}_{k,m} = (\ddot{\mathbf{e}}_{1,k,m}, \dots, \ddot{\mathbf{e}}_{T,k,m})^T$  and  $\tilde{\mathbf{E}}_{k,m} = (\tilde{\mathbf{e}}_{1,k,m}, \dots, \tilde{\mathbf{e}}_{T,k,m})^T$ . Finally, let  $\ddot{\mu}_{k,m} = \tilde{\mu}_{k,m} / s_{-k,m}^{\frac{1}{2}}$ .

We present four important lemmas under our model assumptions. For convenience of presentation, we ignore the subscript  $m$  in Lemma 1, Lemma 2 and Lemma 4, since the results apply to all random samples  $S_{-k,m} \subseteq [d_{-k}]$  with  $|S_{j,m}| \asymp d_j, j \neq k$ .

**Lemma 1** Under Assumption (E1), (E2), (L1), (L2), (R1), we have

$$\lambda_1 \left( \frac{\ddot{\mathbf{E}}_k^T \ddot{\mathbf{E}}_k}{T} \right) = O_p \left( \frac{d_{-k}}{s_{-k}} \left( 1 + \frac{d_k}{T} \right) \right), \quad (\text{S.1})$$

$$\lambda_1 \left( \frac{\ddot{\mathbf{E}}_k^T (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{E}}_k}{T} \right) = O_p \left( \frac{d_{-k}}{s_{-k}} \left( 1 + \frac{d_k}{T} \right) \right). \quad (\text{S.2})$$

In addition, if Assumption (R2) is satisfied, then

$$P \left( \lambda_{\lfloor c \min(T, d_k) \rfloor} \left( \frac{\ddot{\mathbf{E}}_k^T \ddot{\mathbf{E}}_k}{T} \right) \geq C \left( \frac{d_{-k}}{s_{-k}} \left( 1 + \frac{d_k}{T} \right) \right) \right) = 1, \quad (\text{S.3})$$

$$P \left( \lambda_{\lfloor c \min(T, d_k) \rfloor} \left( \frac{\ddot{\mathbf{E}}_k^T (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{E}}_k}{T} \right) \geq C \left( \frac{d_{-k}}{s_{-k}} \left( 1 + \frac{d_k}{T} \right) \right) \right) = 1, \quad (\text{S.4})$$

for some  $c \in (0, 1]$  and  $C > 0$ .

In Lemma 1, (S.1) provides an upper bound for the largest eigenvalue of  $\ddot{\mathbf{E}}_k^T \ddot{\mathbf{E}}_k$  (and  $\ddot{\mathbf{E}}_k^T (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{E}}_k$ ), which facilitates the proof of Lemma 4. (S.3) suggests that at least  $\lfloor c \min(T, d_k) \rfloor$  largest eigenvalues of  $\ddot{\mathbf{E}}_k^T \ddot{\mathbf{E}}_k$  (and  $\ddot{\mathbf{E}}_k^T (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{E}}_k$ ) are of the same order, which guarantees the validity of using eigenvalue ratio to detect the existence of factors, which will be further discussed in Remark 1.

**Lemma 2** Under Assumption (E1), (E2), (F1), (L1), (L2), (R1), we have

$$\frac{\ddot{\mathbf{F}}_k \ddot{\mathbf{F}}_k^\top}{T} \rightarrow \Sigma_{F,k}$$

for some positive definite matrix  $\Sigma_{F,k}$ , with all eigenvalues bounded away from 0 and infinity. For  $j \in [r_k]$ ,

$$\lambda_j \left( \frac{\ddot{\mathbf{F}}_k \ddot{\mathbf{F}}_k^\top}{T} \right) \asymp 1, \quad (\text{S.5})$$

$$\lambda_j (\mathbf{A}_k^\top \mathbf{A}_k) \asymp d_k^{\alpha_{k,j}}, \quad (\text{S.6})$$

$$\lambda_j \left( \frac{\mathbf{A}_k \ddot{\mathbf{F}}_k \ddot{\mathbf{F}}_k^\top \mathbf{A}_k^\top}{T} \right) \asymp d_k^{\alpha_{k,j}}, \quad (\text{S.7})$$

$$\lambda_j \left( \frac{\ddot{\mathbf{F}}_k (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top) \ddot{\mathbf{F}}_k^\top}{T} \right) \asymp 1, \quad (\text{S.8})$$

$$\lambda_j \left( \frac{\mathbf{A}_k \ddot{\mathbf{F}}_k (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top) \ddot{\mathbf{F}}_k^\top \mathbf{A}_k^\top}{T} \right) \asymp d_k^{\alpha_{k,j}}. \quad (\text{S.9})$$

and for  $j \in [\min(z_k, r_k)]$ ,

$$\lambda_j \left( \frac{\ddot{\mathbf{X}}_k^\top (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top) \ddot{\mathbf{X}}_k}{T} \right) \asymp d_k^{\alpha_{k,j}}. \quad (\text{S.10})$$

In addition, if Assumption (R2) is satisfied, then for  $\min(z_k, r_k) + 1 \leq j \leq \lfloor c \min(T, d_k) \rfloor - r_k$ ,

$$\lambda_j \left( \frac{\ddot{\mathbf{X}}_k^\top (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top) \ddot{\mathbf{X}}_k}{T} \right) \asymp \frac{d_{-k}}{s_{-k}} \left( 1 + \frac{d_k}{T} \right). \quad (\text{S.11})$$

The weak serial dependence of factors and errors are quantified in the following lemma.

**Lemma 3** Define  $\mathbf{A}_{f,T}$  and  $\mathbf{A}_{e,T}$  similar to  $\mathbf{A}_{\epsilon,T}$  in Assumption (R2). Define  $\mathbf{A}_{f,T^2}$  to be the  $T^2 \times T^2$  fourth moment matrix of the MA process  $f_{t,l,j}^{(k)}$  for any  $k, l, j$ , and define  $\mathbf{A}_{e,T^2}$  and  $\mathbf{A}_{\epsilon,T^2}$  similarly. Then, under Assumption (E1), (E2) and (F1), we have the following results: For  $\mathbf{A}_T$  can either be  $\mathbf{A}_{f,T}$ ,  $\mathbf{A}_{e,T}$  or  $\mathbf{A}_{\epsilon,T}$ ,

$$\mathbf{1}_T^\top \mathbf{A}_T \mathbf{1}_T = O(T), \quad \|\mathbf{A}_T\|_F^2 = O(T). \quad (\text{S.12})$$

For  $\mathbf{A}_{T^2}$  can either be  $\mathbf{A}_{f,T^2}$ ,  $\mathbf{A}_{e,T^2}$  or  $\mathbf{A}_{\epsilon,T^2}$ ,

$$\mathbf{1}_{T^2}^\top \mathbf{A}_{T^2} \mathbf{1}_{T^2} = O(T^2), \quad \|\mathbf{A}_{T^2}\|_F^2 = O(T^2). \quad (\text{S.13})$$

Moreover,

$$\text{vec}(\mathbf{A}_{f,T})^\top \text{vec}(\mathbf{A}_{e,T}) = O(T), \quad \text{vec}(\mathbf{A}_{f,T})^\top \text{vec}(\mathbf{A}_{\epsilon,T}) = O(T), \quad (\text{S.14})$$

$$\mathbf{1}_{T^2}^\top \mathbf{A}_{f,T} \otimes \mathbf{A}_{e,T} \mathbf{1}_{T^2} = O(T^2), \quad \mathbf{1}_{T^2}^\top \mathbf{A}_{f,T} \otimes \mathbf{A}_{\epsilon,T} \mathbf{1}_{T^2} = O(T^2), \quad (\text{S.15})$$

$$\text{vec}(\mathbf{A}_{f,T} \otimes \mathbf{A}_{f,T})^\top \text{vec}(\mathbf{A}_{e,T^2}) = O(T^2), \quad \text{vec}(\mathbf{A}_{f,T} \otimes \mathbf{A}_{f,T})^\top \text{vec}(\mathbf{A}_{\epsilon,T^2}) = O(T^2), \quad (\text{S.16})$$

$$\text{vec}(\mathbf{A}_{f,T^2})^\top \text{vec}(\mathbf{A}_{e,T^2}) = O(T^2), \quad \text{vec}(\mathbf{A}_{f,T^2})^\top \text{vec}(\mathbf{A}_{\epsilon,T^2}) = O(T^2). \quad (\text{S.17})$$

Given a particular sample, we can obtain  $\hat{\mathbf{Q}}_{k,(z_k)}$  as the  $z_k$  largest eigenvectors of  $\hat{\Sigma}_{\tilde{\mathbf{x}}_k} := \frac{\tilde{\mathbf{x}}_k^\top (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top) \tilde{\mathbf{x}}_k}{T}$ , with normalisation  $\hat{\mathbf{Q}}_{k,(z_k)}^\top \hat{\mathbf{Q}}_{k,(z_k)} = \mathbf{I}_{z_k}$ . Let  $\tilde{\mathbf{V}}_k$  be the  $z_k \times z_k$  diagonal matrix of the first  $z_k$  largest eigenvalues of  $\hat{\Sigma}_{\tilde{\mathbf{x}}_k}$  in decreasing order, and  $\tilde{\mathbf{V}}_k := \tilde{\mathbf{V}}_k / s_{-k}$ . The theoretical properties of  $\hat{\mathbf{Q}}_{k,(z_k)}$  from a particular sample is presented in the following lemma.

**Lemma 4** Under Assumption (E1), (E2), (F1), (L1), (L2), (R1) and  $r_{e,k} = O(d_k)$ , for  $k \in [K]$ , let  $c_k := \frac{d_k^2}{s_{-k}^2} \left(1 + \frac{d_k^2}{T^2}\right) + \frac{d_{-k}}{s_{-k}} d_k^{\alpha_{k,1}} \min \left\{1 + \frac{d_k}{T}, \frac{r_k d_k}{T}\right\}$ , then

$$\|\hat{\mathbf{Q}}_{k,(z_k)} - \mathbf{Q}_k \ddot{\mathbf{H}}_k\|^2 = O_p \left( d_k^{-2\alpha_{k,z_k}} c_k \right), \quad (\text{S.18})$$

where  $\ddot{\mathbf{H}}_k = \frac{\mathbf{D}_k^{\frac{1}{2}} \ddot{\mathbf{F}}_k (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{F}}_k^T \mathbf{A}_k^T \hat{\mathbf{Q}}_{k,(z_k)} \ddot{\mathbf{V}}_k^{-1}}{T}$  has  $\text{rank}(\ddot{\mathbf{H}}_k) = z_k$ . Moreover, further assuming (L1'), there exists  $\hat{\mathbf{U}}_{k,(z_k)}$  with  $\hat{\mathbf{U}}_{k,(z_k)}^T \hat{\mathbf{U}}_{k,(z_k)} = \mathbf{I}_{z_k}$  such that  $\hat{\mathbf{Q}}_{k,(z_k)} = \hat{\mathbf{U}}_{k,(z_k)} \mathbf{P}_{k,(z_k)}$  with  $\mathbf{P}_{k,(z_k)}$  an orthogonal matrix, and

$$\|\hat{\mathbf{U}}_{k,(z_k)} - \mathbf{U}_{k,(z_k)}\|^2 = O_p \left( d_k^{-2\alpha_{k,z_k}} \left[ d_k^{2\alpha_{k,1}} \frac{r_k}{T} + c_k \right] \right), \quad (\text{S.19})$$

where  $\mathbf{U}_{k,(z_k)}$  is the matrix consisting of the first  $z_k$  columns of  $\mathbf{U}_k$ .

**Remark 1** Note that Lemma 2 implies the following result for eigenvalue ratio: For  $j \leq r_k - 1$ ,

$$\frac{\lambda_j \left( \frac{\ddot{\mathbf{X}}_k^T (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{X}}_k}{T} \right)}{\lambda_{j+1} \left( \frac{\ddot{\mathbf{X}}_k^T (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{X}}_k}{T} \right)} \asymp \frac{d_k^{\alpha_{k,j}}}{d_k^{\alpha_{k,j+1}}};$$

For  $r_k + 1 \leq j \leq \lfloor c \min(T, d_k) \rfloor - r_k - 1$ ,

$$\frac{\lambda_j \left( \frac{\ddot{\mathbf{X}}_k^T (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{X}}_k}{T} \right)}{\lambda_{j+1} \left( \frac{\ddot{\mathbf{X}}_k^T (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{X}}_k}{T} \right)} \asymp 1;$$

For  $j = r_k$ ,

$$\frac{\lambda_j \left( \frac{\ddot{\mathbf{X}}_k^T (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{X}}_k}{T} \right)}{\lambda_{j+1} \left( \frac{\ddot{\mathbf{X}}_k^T (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{X}}_k}{T} \right)} \asymp \frac{d_k^{\alpha_{k,r_k}}}{\frac{d_{-k}}{s_{-k}} \left(1 + \frac{d_k}{T}\right)} \rightarrow \infty.$$

Therefore,

$$\frac{\lambda_1 \left( \frac{\ddot{\mathbf{X}}_k^T (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{X}}_k}{T} \right)}{\lambda_{r_k+1} \left( \frac{\ddot{\mathbf{X}}_k^T (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{X}}_k}{T} \right)} \asymp \frac{d_k^{\alpha_{k,1}}}{\frac{d_{-k}}{s_{-k}} \left(1 + \frac{d_k}{T}\right)} \rightarrow \infty. \quad (\text{S.20})$$

(S.20) implies that at least one of ratio of subsequent eigenvalues  $\left\{ \frac{\lambda_j \left( \frac{\ddot{\mathbf{X}}_k^T (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{X}}_k}{T} \right)}{\lambda_{j+1} \left( \frac{\ddot{\mathbf{X}}_k^T (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{X}}_k}{T} \right)}, j \in [r_k] \right\}$  goes to infinity. This is essential to detect the existence of factors. Note that in the special case when all factors have the same strength, we should have  $\frac{\lambda_{r_k} \left( \frac{\ddot{\mathbf{X}}_k^T (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{X}}_k}{T} \right)}{\lambda_{r_k+1} \left( \frac{\ddot{\mathbf{X}}_k^T (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{X}}_k}{T} \right)} \rightarrow \infty$ , which recovers the result in Ahn and Horenstein (2013).

In contrast, when there is no factor exist, we should have that

$$\frac{\lambda_j \left( \frac{\ddot{\mathbf{X}}_k^T (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{X}}_k}{T} \right)}{\lambda_{j+1} \left( \frac{\ddot{\mathbf{X}}_k^T (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{X}}_k}{T} \right)} = \frac{\lambda_j \left( \frac{\ddot{\mathbf{E}}_k^T (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{E}}_k}{T} \right)}{\lambda_{j+1} \left( \frac{\ddot{\mathbf{E}}_k^T (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{E}}_k}{T} \right)} \asymp 1$$

for  $j = 1, \dots, \lfloor c \min(T, d_k) \rfloor$  by (S.3) in Lemma 1. Thus, every ratio (actually until  $\lfloor c \min(T, d_k) \rfloor$ ) of subsequent eigenvalues remains bounded. In this way, we can use eigenvalue ratio to detect the existence of factors accordingly: when factors exist, at least one of the eigenvalue ratio will goes to infinity; when there is no factor, all eigenvalue ratios (until  $\lfloor c \min(T, d_k) \rfloor$ ) remains bounded. Note that this is different from estimating the number of factors, since we can only know that there is factor exist, but we do not really know how many factors and what the factor strengths are. However, in the special case when all factors are of the same strength, we have  $\frac{\lambda_{r_k} \left( \frac{\tilde{\mathbf{x}}_k^T (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \tilde{\mathbf{x}}_k}{T} \right)}{\lambda_{r_k+1} \left( \frac{\tilde{\mathbf{x}}_k^T (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \tilde{\mathbf{x}}_k}{T} \right)} \rightarrow \infty$  and  $\frac{\lambda_j \left( \frac{\tilde{\mathbf{x}}_k^T (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \tilde{\mathbf{x}}_k}{T} \right)}{\lambda_{j+1} \left( \frac{\tilde{\mathbf{x}}_k^T (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \tilde{\mathbf{x}}_k}{T} \right)} \asymp 1$  for  $j \neq r_k$ , which is how Ahn and Horenstein (2013) estimates the number of factors. In the same spirit, the eye-ball test of Lam and Yao (2012) also uses eigenvalue ratios in a similar manner to deduce the number of factors in a statistical factor model.

*Proof of Lemma 1.* By Assumption (E1), (E2), (R1), we have

$$\begin{aligned} \tilde{\mathbf{e}}_{t,k} &= \mathbf{A}_{e,k} \tilde{\mathbf{f}}_{e,t,k} + \sum_{\ell \in S_k} (\boldsymbol{\Sigma}_{\epsilon,\ell}^{(k)})^{1/2} \boldsymbol{\epsilon}_{t,\ell}^{(k)} \\ &:= \tilde{\mathbf{e}}_{t,k,1} + \tilde{\mathbf{e}}_{t,k,2} \end{aligned}$$

where  $\tilde{\mathbf{e}}_{t,k,1} := \mathbf{A}_{e,k} \tilde{\mathbf{f}}_{e,t,k}$  and  $\tilde{\mathbf{e}}_{t,k,2} = \sum_{\ell \in S_k} (\boldsymbol{\Sigma}_{\epsilon,\ell}^{(k)})^{1/2} \boldsymbol{\epsilon}_{t,\ell}^{(k)}$ , and  $\tilde{\mathbf{f}}_{e,t,k} = \sum_{\ell \in S_k} \mathbf{f}_{t,\ell}^{(k)}$  is defined in a similar way as  $\tilde{\mathbf{f}}_{t,k}$ . Using such decomposition, the error matrix can be written as

$$\tilde{\mathbf{E}}_k = \tilde{\mathbf{E}}_{k,1} + \tilde{\mathbf{E}}_{k,2},$$

where  $\tilde{\mathbf{E}}_{k,1} = (\tilde{\mathbf{e}}_{1,k,1}, \dots, \tilde{\mathbf{e}}_{T,k,1})^T$  and  $\tilde{\mathbf{E}}_{k,2} = (\tilde{\mathbf{e}}_{1,k,2}, \dots, \tilde{\mathbf{e}}_{T,k,2})^T$ . Then we can deal with  $\tilde{\mathbf{E}}_{k,1}$  and  $\tilde{\mathbf{E}}_{k,2}$  separately using random matrix theory. We first look at  $\tilde{\mathbf{E}}_{k,2}$ . Similar to assumptions in Ahn and Horenstein (2013), we further write the matrix  $\tilde{\mathbf{E}}_{k,2}$  as

$$\tilde{\mathbf{E}}_{k,2} = L_{e,k,2}^{\frac{1}{2}} U_{e,k,2} R_{e,k,2}^{\frac{1}{2}},$$

where  $U_{e,k,2} \in \mathbb{R}^{T \times d_k}$  are iid random variables with uniformly bounded fourth moment by Assumption (R1), and  $L_{e,k,2} \in \mathbb{R}^{T \times T}$  and  $R_{e,k,2} \in \mathbb{R}^{d_k \times d_k}$  are time-serial and cross-sectional covariance matrices. Define  $\boldsymbol{\Sigma}_{\epsilon}^{(k)} := \sum_{\ell \in S_k} \boldsymbol{\Sigma}_{\epsilon,\ell}^{(k)}$ . By Assumption (E1),  $R_{e,k,2} = \boldsymbol{\Sigma}_{\epsilon}^{(k)}$ , and  $L_{e,k,2} = \mathbf{A}_{\epsilon,T}$ . Next, we need to bound the spectral norm of  $L_{e,k,2}$  and  $R_{e,k,2}$ . Assumption (E1) implies  $\|R_{e,k,2}\| = O(d_k)$ . For  $\mathbf{A}_{\epsilon,T}$ , since it is symmetric,  $\|\mathbf{A}_{\epsilon,T}\|_1 = \|\mathbf{A}_{\epsilon,T}\|_{\infty}$ , and

$$\begin{aligned} \|\mathbf{A}_{\epsilon,T}\|_1 &= \max_t \sum_{s=1}^T |(\mathbf{A}_{\epsilon,T})_{ts}| \\ &\leq 2 \sum_{v=0}^T \left| \sum_{q \geq 0} a_{\epsilon,q} a_{\epsilon,q+v} \right| \\ &\leq 2 \left( \sum_{q \geq 0} |a_{\epsilon,q}| \right)^2 \leq C \end{aligned} \tag{S.21}$$

by Assumption (E2). Thus,  $\|\mathbf{A}_{\epsilon,T}\| \leq \sqrt{\|\mathbf{A}_{\epsilon,T}\|_1 \|\mathbf{A}_{\epsilon,T}\|_{\infty}} = \|\mathbf{A}_{\epsilon,T}\|_1 \leq C$ . Bai and Yin (1993) and Latala (2005) show that  $\lambda_1 \left( U_{e,k,2} U_{e,k,2}^T / T \right) \rightarrow (1 + \sqrt{d_k/T})^2$ . Thus, similar to the result obtained by Ahn and Horenstein (2013) (see also Moon and Weidner (2015)), we have

$$\lambda_1 \left( \frac{\tilde{\mathbf{E}}_{k,2}^T \tilde{\mathbf{E}}_{k,2}}{T} \right) \leq \|L_{e,k,2}\| \|R_{e,k,2}\| \|U_{e,k,2} U_{e,k,2}^T / T\| = O_p \left( d_k \left( 1 + \frac{d_k}{T} \right) \right).$$

For  $\tilde{\mathbf{E}}_{k,1}$ , note that the common error has a similar structure as the factor model. Similar to the definition of  $s_k$  and  $s_{-k}$ , define  $s_{e,k} := \sum_{j \in S_{-k}} \left( \sum_{i=1}^{d_k} (\mathbf{A}_{e,k})_{ij} \right)^2$  and  $s_{e,-k} := \prod_{l=1; l \neq k}^K s_{e,l}$ . Define  $\tilde{\mathbf{F}}_{e,k}$  similar as  $\tilde{\mathbf{F}}_k$ , then following the similar analysis as in Lemma 2, we have for  $j \in [r_{e,k}]$ ,

$$\lambda_j \left( \frac{\tilde{\mathbf{F}}_{e,k} \tilde{\mathbf{F}}_{e,k}^T}{T} \right) \asymp s_{e,-k}.$$

Thus,

$$\lambda_1 \left( \frac{\tilde{\mathbf{E}}_{k,1}^T \tilde{\mathbf{E}}_{k,1}}{T} \right) = \lambda_1 \left( \frac{\mathbf{A}_{e,k} \tilde{\mathbf{F}}_{e,k} \tilde{\mathbf{F}}_{e,k}^T \mathbf{A}_{e,k}^T}{T} \right) \leq \lambda_1 (\mathbf{A}_{e,k}^T \mathbf{A}_{e,k}) \lambda_1 \left( \frac{\tilde{\mathbf{F}}_{e,k} \tilde{\mathbf{F}}_{e,k}^T}{T} \right) = O_p(s_{e,-k}),$$

since  $\lambda_1 (\mathbf{A}_{e,k}^T \mathbf{A}_{e,k}) = O(1)$  by Assumption (E1). Next, note that by definition,

$$s_{e,k} \leq r_{e,k} \|\mathbf{A}_{e,k}\|_1^2 = O(r_{e,k}),$$

so  $s_{e,-k} = O(r_{e,-k})$ . Since we assume  $r_{e,k} = O(d_k)$ , we have  $\lambda_1 \left( \frac{\tilde{\mathbf{E}}_{k,1}^T \tilde{\mathbf{E}}_{k,1}}{T} \right) = O_p(d_k)$ .

Therefore,

$$\begin{aligned} \lambda_1 \left( \frac{\tilde{\mathbf{E}}_k^T \tilde{\mathbf{E}}_k}{T} \right) &\leq \left\| \frac{\tilde{\mathbf{E}}_{k,1}^T \tilde{\mathbf{E}}_{k,1}}{T} \right\| + \left\| \frac{\tilde{\mathbf{E}}_{k,1}^T \tilde{\mathbf{E}}_{k,2}}{T} \right\| + \left\| \frac{\tilde{\mathbf{E}}_{k,2}^T \tilde{\mathbf{E}}_{k,1}}{T} \right\| + \left\| \frac{\tilde{\mathbf{E}}_{k,2}^T \tilde{\mathbf{E}}_{k,2}}{T} \right\| \\ &= O_p \left( d_k \left( 1 + \frac{d_k}{T} \right) \right) + 2 \left\| \frac{\tilde{\mathbf{E}}_{k,1}^T \tilde{\mathbf{E}}_{k,2}}{T} \right\| \\ &\leq O_p \left( d_k \left( 1 + \frac{d_k}{T} \right) \right) + \sqrt{\left\| \frac{\tilde{\mathbf{E}}_{k,1}^T \tilde{\mathbf{E}}_{k,1}}{T} \right\| \left\| \frac{\tilde{\mathbf{E}}_{k,2}^T \tilde{\mathbf{E}}_{k,2}}{T} \right\|} \\ &= O_p \left( d_k \left( 1 + \frac{d_k}{T} \right) \right), \end{aligned}$$

which implies (S.1), and

$$\begin{aligned} \lambda_1 \left( \frac{\tilde{\mathbf{E}}_k^T (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \tilde{\mathbf{E}}_k}{T} \right) &\leq \left\| \frac{\tilde{\mathbf{E}}_k^T}{\sqrt{T}} \right\|^2 \left\| \mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T \right\| \\ &\leq 2 \lambda_1 \left( \frac{\tilde{\mathbf{E}}_k^T \tilde{\mathbf{E}}_k}{T} \right) \\ &= O_p \left( d_k \left( 1 + \frac{d_k}{T} \right) \right), \end{aligned} \tag{S.22}$$

which implies (S.2)

To show (S.3), we focus on  $\tilde{\mathbf{E}}_{k,2}$ . Note that Assumption (R2) is parallel to Assumption (D) in Ahn and Horenstein (2013). Following Lemma A.7. in Ahn and Horenstein (2013), we have

$$P \left( \lambda_{\lfloor c \min(T, d_k) \rfloor} \left( \frac{\tilde{\mathbf{E}}_{k,2}^T \tilde{\mathbf{E}}_{k,2}}{\max(T, d_k)} \right) \geq C d_k \right) = 1$$

for some  $c \in (0, 1]$  and  $C > 0$ . Thus,

$$P \left( \lambda_{\lfloor c \min(T, d_k) \rfloor} \left( \frac{\tilde{\mathbf{E}}_{k,2}^T \tilde{\mathbf{E}}_{k,2}}{T} \right) \geq C \left( d_k \left( 1 + \frac{d_k}{T} \right) \right) \right) = 1$$



for some  $c \in (0, 1]$  and  $C > 0$ . Finally, by Weyl's inequalities,

$$\begin{aligned} \sqrt{\lambda_{\lfloor c \min(T, d_k) \rfloor} \left( \frac{\tilde{\mathbf{E}}_k^T \tilde{\mathbf{E}}_k}{T} \right)} &\geq \sqrt{\lambda_{\lfloor c \min(T, d_k) \rfloor} \left( \frac{\tilde{\mathbf{E}}_{k,2}^T \tilde{\mathbf{E}}_{k,2}}{T} \right)} - \sqrt{\lambda_1 \left( \frac{\tilde{\mathbf{E}}_{k,1}^T \tilde{\mathbf{E}}_{k,1}}{T} \right)} \\ &\asymp \sqrt{d_k \left( 1 + \frac{d_k}{T} \right)}, \end{aligned}$$

which implies (S.3), since  $\lambda_1 \left( \frac{\tilde{\mathbf{E}}_{k,1}^T \tilde{\mathbf{E}}_{k,1}}{T} \right) = o_p \left( d_k \left( 1 + \frac{d_k}{T} \right) \right)$  under Assumption (R2). Similarly, following the same argument as Lemma A.7. in Ahn and Horenstein (2013), we can show that

$$P \left( \lambda_{\lfloor c \min(T, d_k) \rfloor} \left( \frac{\ddot{\mathbf{E}}_{k,2}^T (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{E}}_{k,2}}{T} \right) \geq C \left( d_k \left( 1 + \frac{d_k}{T} \right) \right) \right) = 1$$

for some  $c \in (0, 1]$  and  $C > 0$ , so

$$\begin{aligned} \sqrt{\lambda_{\lfloor c \min(T, d_k) \rfloor} \left( \frac{\tilde{\mathbf{E}}_k^T (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \tilde{\mathbf{E}}_k}{T} \right)} &\geq \sqrt{\lambda_{\lfloor c \min(T, d_k) \rfloor} \left( \frac{\tilde{\mathbf{E}}_{k,2}^T (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \tilde{\mathbf{E}}_{k,2}}{T} \right)} \\ &\quad - \sqrt{\lambda_1 \left( \frac{\tilde{\mathbf{E}}_{k,1}^T (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \tilde{\mathbf{E}}_{k,1}}{T} \right)} \\ &\asymp \sqrt{d_k \left( 1 + \frac{d_k}{T} \right)}, \end{aligned}$$

which implies (S.4). This completes the proof of Lemma 1.  $\square$

*Proof of Lemma 2.* We consider the case  $K = 3$  without loss of generality. By Assumption (F1), we know each element of  $\mathcal{F}_t$  are independent with mean 0 and variance 1. Suppose we want to estimate  $\mathbf{A}_1$ . For  $i, j \in [r_1]$ , we have

$$\begin{aligned} E \left( \tilde{f}_{t,1,i} \tilde{f}_{t,1,j} \right) &= \sum_{p=1}^{r_2} \sum_{q=1}^{r_3} \sum_{u=1}^{r_2} \sum_{v=1}^{r_3} \left( \sum_{\ell \in S_2} (\mathbf{A}_2)_{lp} \sum_{\ell \in S_3} (\mathbf{A}_3)_{lq} \sum_{\ell \in S_2} (\mathbf{A}_2)_{lu} \sum_{\ell \in S_3} (\mathbf{A}_3)_{lv} \right) E(f_{t,ipq} f_{t,juv}) \\ &= \sum_{p=1}^{r_2} \sum_{q=1}^{r_3} \left( \sum_{\ell \in S_2} (\mathbf{A}_2)_{lp} \right)^2 \left( \sum_{\ell \in S_3} (\mathbf{A}_3)_{lq} \right)^2 \\ &\asymp s_2 s_3 = s_{-1} \end{aligned}$$

when  $i = j$ , and 0 otherwise. Thus,  $E \left( \tilde{f}_{t,1,i} \tilde{f}_{t,1,j} \right) \asymp 1$  for  $i = j$ , and 0 otherwise, which implies that  $\Sigma_{F,k}$  is a diagonal matrix with all diagonal elements bounded away from 0 and infinity.

Next, we show (S.5) – (S.7). To apply random matrix theory, similar to Ahn and Horenstein (2013), we can further write the matrix  $\ddot{\mathbf{F}}_k$  as

$$\ddot{\mathbf{F}}_k = L_{f,k}^{\frac{1}{2}} U_{f,k} R_{f,k}^{\frac{1}{2}},$$

where  $U_{f,k} \in \mathbb{R}^{r_k \times T}$  are iid random variables with uniformly bounded fourth moment by Assumption (R1), and  $L_{f,k} \in \mathbb{R}^{r_k \times r_k}$  and  $R_{f,k} \in \mathbb{R}^{T \times T}$  are cross-sectional and time-serial covariance matrices. By Assumption (F1),  $L_{f,k} = I_{r_k}$ , and  $R_{f,k} = \mathbf{A}_{f,T}$ . Following similar analysis as (S.21), we have  $\|R_{f,k}\|_1 \leq C$ . Thus,  $\|R_{f,k}\| \leq \sqrt{\|R_{f,k}\|_1 \|R_{f,k}\|_\infty} = \|R_{f,k}\|_1 \leq C$ . Since we assume  $r_k = o(T)$ , Bai and Yin (1993) and Latala (2005) show that  $\lambda_1 \left( U_{f,k} U_{f,k}^T / T \right) - (1 + \sqrt{r_k/T})^2 \rightarrow 0$  and  $\lambda_{r_k} \left( U_{f,k} U_{f,k}^T / T \right) - (1 - \sqrt{r_k/T})^2 \rightarrow 0$ .

Similar to the result obtained by Ahn and Horenstein (2013) (see also Moon and Weidner (2015)), since the largest eigenvalues of  $R_{f,k}$  and  $L_{f,k}$  are bounded, we have

$$\lambda_1 \left( \ddot{\mathbf{F}}_k \ddot{\mathbf{F}}_k^T / T \right) \leq \|L_{f,k}\| \|R_{f,k}\| \|U_{f,k} U_{f,k}^T / T\| = O_p(1).$$

Similarly, since  $\lambda_1(R_{f,k}) = \|R_{f,k}\| \geq 1$ , we have

$$\lambda_{r_k} \left( \ddot{\mathbf{F}}_k \ddot{\mathbf{F}}_k^T / T \right) = \lambda_{r_k} \left( U_{f,k} R_{f,k} U_{f,k}^T / T \right) = \lambda_{r_k} \left( U_{f,k}^T U_{f,k} R_{f,k} / T \right) \geq \lambda_{r_k} \left( U_{f,k}^T U_{f,k} / T \right) \lambda_1(R_{f,k}) \geq 1$$

as  $r_k, T \rightarrow \infty$ . Thus, we have  $\lambda_j \left( \frac{\ddot{\mathbf{F}}_k \ddot{\mathbf{F}}_k^T}{T} \right) \asymp 1$  for  $j \in [r_k]$ .

Next, for  $j \in [r_k]$ , we know  $\lambda_j \left( \frac{\mathbf{A}_k \ddot{\mathbf{F}}_k \ddot{\mathbf{F}}_k^T \mathbf{A}_k^T}{T} \right) = \lambda_j \left( \frac{\ddot{\mathbf{F}}_k \ddot{\mathbf{F}}_k^T \mathbf{A}_k^T \mathbf{A}_k}{T} \right)$  and

$$\lambda_j(\mathbf{A}_k^T \mathbf{A}_k) \lambda_{r_k} \left( \frac{\ddot{\mathbf{F}}_k \ddot{\mathbf{F}}_k^T}{T} \right) \leq \lambda_j \left( \frac{\ddot{\mathbf{F}}_k \ddot{\mathbf{F}}_k^T \mathbf{A}_k^T \mathbf{A}_k}{T} \right) \leq \lambda_j(\mathbf{A}_k^T \mathbf{A}_k) \lambda_1 \left( \frac{\ddot{\mathbf{F}}_k \ddot{\mathbf{F}}_k^T}{T} \right).$$

Since  $\lambda_j \left( \frac{\ddot{\mathbf{F}}_k \ddot{\mathbf{F}}_k^T}{T} \right) \asymp 1$  for  $j \in [r_k]$ , it follows that  $\lambda_j \left( \frac{\ddot{\mathbf{F}}_k \ddot{\mathbf{F}}_k^T \mathbf{A}_k^T \mathbf{A}_k}{T} \right) \asymp \lambda_j(\mathbf{A}_k^T \mathbf{A}_k)$  for  $j \in [r_k]$ . Similarly,  $\lambda_j(\mathbf{A}_k^T \mathbf{A}_k) = \lambda_j \left( \mathbf{D}_k^{\frac{1}{2}} \mathbf{D}_k^{-\frac{1}{2}} \mathbf{A}_k^T \mathbf{A}_k \mathbf{D}_k^{-\frac{1}{2}} \mathbf{D}_k^{\frac{1}{2}} \right) = \lambda_j \left( \mathbf{D}_k \mathbf{D}_k^{-\frac{1}{2}} \mathbf{A}_k^T \mathbf{A}_k \mathbf{D}_k^{-\frac{1}{2}} \right)$  and

$$\lambda_j(\mathbf{D}_k) \lambda_{r_k} \left( \mathbf{D}_k^{-\frac{1}{2}} \mathbf{A}_k^T \mathbf{A}_k \mathbf{D}_k^{-\frac{1}{2}} \right) \leq \lambda_j \left( \mathbf{D}_k \mathbf{D}_k^{-\frac{1}{2}} \mathbf{A}_k^T \mathbf{A}_k \mathbf{D}_k^{-\frac{1}{2}} \right) \leq \lambda_j(\mathbf{D}_k) \lambda_1 \left( \mathbf{D}_k^{-\frac{1}{2}} \mathbf{A}_k^T \mathbf{A}_k \mathbf{D}_k^{-\frac{1}{2}} \right).$$

By Assumption (L1),  $\lambda_j(\mathbf{D}_k) \asymp d_k^{\alpha_{k,j}}$  and  $\lambda_j \left( \mathbf{D}_k^{-\frac{1}{2}} \mathbf{A}_k^T \mathbf{A}_k \mathbf{D}_k^{-\frac{1}{2}} \right) \asymp 1$  for  $j \in [r_k]$  (this also holds when elements of  $\mathbf{A}_k$  are random and independent). Thus, for  $j \in [r_k]$ , we have  $\lambda_j \left( \frac{\mathbf{A}_k \ddot{\mathbf{F}}_k \ddot{\mathbf{F}}_k^T \mathbf{A}_k^T}{T} \right) \asymp \lambda_j(\mathbf{A}_k^T \mathbf{A}_k) \asymp d_k^{\alpha_{k,j}}$ .

Next, we will show (S.8) – (S.9) accordingly. To start with, note that

$$\begin{aligned} \lambda_1 \left( \frac{\ddot{\mathbf{F}}_k (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{F}}_k^T}{T} \right) &\leq \left\| \frac{\ddot{\mathbf{F}}_k \ddot{\mathbf{F}}_k^T}{T} \right\| + \left\| \frac{\ddot{\mathbf{F}}_k \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T \ddot{\mathbf{F}}_k^T}{T} \right\| \\ &\leq \left\| \frac{\ddot{\mathbf{F}}_k \ddot{\mathbf{F}}_k^T}{T} \right\| + \left\| \frac{\ddot{\mathbf{F}}_k}{\sqrt{T}} \right\|^2 \left\| \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T \right\| = 2 \left\| \frac{\ddot{\mathbf{F}}_k \ddot{\mathbf{F}}_k^T}{T} \right\| \asymp 1. \end{aligned}$$

And

$$\begin{aligned} \lambda_{r_k} \left( \frac{\ddot{\mathbf{F}}_k (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{F}}_k^T}{T} \right) &= \lambda_{r_k} \left( \frac{\ddot{\mathbf{F}}_k \ddot{\mathbf{F}}_k^T}{T} - \frac{\ddot{\mathbf{F}}_k \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T \ddot{\mathbf{F}}_k^T}{T} \right) \\ &\geq \lambda_{r_k} \left( \frac{\ddot{\mathbf{F}}_k \ddot{\mathbf{F}}_k^T}{T} \right) - \lambda_1 \left( \frac{\ddot{\mathbf{F}}_k \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T \ddot{\mathbf{F}}_k^T}{T^2} \right). \end{aligned} \quad (\text{S.23})$$

We next obtain an upper bound for  $\lambda_1 \left( \frac{\ddot{\mathbf{F}}_k \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T \ddot{\mathbf{F}}_k^T}{T^2} \right)$ . Note that  $\ddot{\mathbf{F}}_k \mathbf{1}_T \in \mathbb{R}^{r_k \times 1}$  is a random vector, with its  $i$ -th element to be  $\sum_{t=1}^T \ddot{f}_{t,i}$ . For its moment bounds,  $E \left( \sum_{t=1}^T \ddot{f}_{t,i} \right)^2 = \mathbf{1}_T^T \mathbf{A}_{f,T} \mathbf{1}_T = O(T)$  and  $E \left( \sum_{t=1}^T \ddot{f}_{t,i} \right)^4 = \mathbf{1}_{T^2}^T \mathbf{A}_{f,T^2} \mathbf{1}_{T^2} = O(T^2)$  by Lemma 3. Therefore, following similar argument in previous analysis, we can decompose  $\ddot{\mathbf{F}}_k \mathbf{1}_T = \sqrt{T} \mathbf{I}_{r_k} U$ , where  $U \in \mathbb{R}^{r_k \times 1}$  has iid random entries with bounded second and fourth moments. Thus,

$$\lambda_1 \left( \frac{\ddot{\mathbf{F}}_k \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T \ddot{\mathbf{F}}_k^T}{T^2} \right) = \frac{1}{T^2} \lambda_1 \left( \ddot{\mathbf{F}}_k \mathbf{1}_T \mathbf{1}_T^T \ddot{\mathbf{F}}_k^T \right) \leq \frac{1}{T^2} O_p(r_k T) = O_p \left( \frac{r_k}{T} \right) = o_p(1),$$

which implies  $\lambda_{r_k} \left( \frac{\ddot{\mathbf{F}}_k (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{F}}_k^T}{T} \right) \succeq 1$  by (S.23). Therefore,  $\lambda_j \left( \frac{\ddot{\mathbf{F}}_k (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{F}}_k^T}{T} \right)$  for  $j \in [r_k]$ .

Next, for  $j \in [r_k]$ , it is easy to see

$$\begin{aligned} \lambda_j (\mathbf{A}_k^\top \mathbf{A}_k) \lambda_{r_k} \left( \frac{\ddot{\mathbf{F}}_k (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top) \ddot{\mathbf{F}}_k^\top}{T} \right) &\leq \lambda_j \left( \frac{\mathbf{A}_k \ddot{\mathbf{F}}_k (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top) \ddot{\mathbf{F}}_k^\top \mathbf{A}_k^\top}{T} \right) \\ &\leq \lambda_j (\mathbf{A}_k^\top \mathbf{A}_k) \lambda_1 \left( \frac{\ddot{\mathbf{F}}_k (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top) \ddot{\mathbf{F}}_k^\top}{T} \right), \end{aligned}$$

which implies  $\lambda_j \left( \frac{\mathbf{A}_k \ddot{\mathbf{F}}_k (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top) \ddot{\mathbf{F}}_k^\top \mathbf{A}_k^\top}{T} \right) \asymp d_k^{\alpha_{k,j}}$  for  $j \in [r_k]$ .

Observe that  $(\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top)^2 = \mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top$ , and  $\ddot{\mathbf{X}}_k^\top (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top) = (\mathbf{A}_k \ddot{\mathbf{F}}_k + \ddot{\mathbf{E}}_k^\top) (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top)$ . Hence to show (S.10), following the similar argument as Theorem 1 in Freyaldenhoven (2022), for  $j \in [r_k]$ ,

$$\begin{aligned} \sqrt{\lambda_j \left( \frac{\ddot{\mathbf{X}}_k^\top (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top) \ddot{\mathbf{X}}_k}{T} \right)} &\geq \sqrt{\lambda_j \left( \frac{\mathbf{A}_k \ddot{\mathbf{F}}_k (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top) \ddot{\mathbf{F}}_k^\top \mathbf{A}_k^\top}{T} \right)} - \sqrt{\lambda_1 \left( \frac{\ddot{\mathbf{E}}_k^\top (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top) \ddot{\mathbf{E}}_k}{T} \right)} \\ &\geq \sqrt{d_k^{\alpha_{k,j}}} - \sqrt{O_p \left( \frac{d_k}{s_k} \left( 1 + \frac{d_k}{T} \right) \right)} \asymp \sqrt{d_k^{\alpha_{k,j}}}. \end{aligned} \quad (\text{S.24})$$

where the second line follows from Lemma 1 and the last line follows from Assumption (L2). Similarly, for  $j \in [r_k]$ ,

$$\begin{aligned} \lambda_j \left( \frac{\ddot{\mathbf{X}}_k^\top (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top) \ddot{\mathbf{X}}_k}{T} \right) &\leq \lambda_j \left( \frac{\mathbf{A}_k \ddot{\mathbf{F}}_k (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top) \ddot{\mathbf{F}}_k^\top \mathbf{A}_k^\top}{T} \right) + \lambda_1 \left( \frac{\ddot{\mathbf{E}}_k^\top (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top) \ddot{\mathbf{E}}_k}{T} \right) \\ &\quad + 2 \sqrt{\lambda_j \left( \frac{\mathbf{A}_k \ddot{\mathbf{F}}_k (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top) \ddot{\mathbf{F}}_k^\top \mathbf{A}_k^\top}{T} \right)} \sqrt{\lambda_1 \left( \frac{\ddot{\mathbf{E}}_k^\top (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top) \ddot{\mathbf{E}}_k}{T} \right)} \\ &\asymp d_k^{\alpha_{k,j}} + O_p \left( \frac{d_k}{s_k} \left( 1 + \frac{d_k}{T} \right) \right) \asymp d_k^{\alpha_{k,j}}. \end{aligned} \quad (\text{S.25})$$

Therefore,  $\lambda_j \left( \frac{\ddot{\mathbf{X}}_k^\top (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top) \ddot{\mathbf{X}}_k}{T} \right) \asymp d_k^{\alpha_{k,j}}$  for  $j \in [r_k]$ .

Finally, to show (S.11), we apply similar technique as Lemma A.8 in Ahn and Horenstein (2013). For simplicity of expression, in the following proof, we write  $\mathbf{M}_T = \mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top$  and  $\mathbf{M} = (\ddot{\mathbf{F}}_k \mathbf{M}_T)^\top (\ddot{\mathbf{F}}_k \mathbf{M}_T \ddot{\mathbf{F}}_k^\top)^{-1} \ddot{\mathbf{F}}_k \mathbf{M}_T$ . Then  $\text{rank}(\mathbf{M}) \leq r_k$ , and

$$\ddot{\mathbf{X}}_k^\top \mathbf{M}_T \ddot{\mathbf{X}}_k = (\mathbf{A}_k \ddot{\mathbf{F}}_k + \ddot{\mathbf{E}}_k^\top \mathbf{M}) \mathbf{M}_T (\mathbf{A}_k \ddot{\mathbf{F}}_k + \ddot{\mathbf{E}}_k^\top \mathbf{M})^\top + \ddot{\mathbf{E}}_k^\top (\mathbf{M}_T - \mathbf{M} \mathbf{M}_T \mathbf{M}^\top) \ddot{\mathbf{E}}_k.$$

Hence, for  $r_k + 1 \leq j \leq \lfloor c \min(T, d_k) \rfloor - r_k$ ,

$$\begin{aligned} \lambda_j \left( \ddot{\mathbf{E}}_k^\top (\mathbf{M}_T - \mathbf{M} \mathbf{M}_T \mathbf{M}^\top) \ddot{\mathbf{E}}_k \right) &\leq \lambda_j \left( \ddot{\mathbf{X}}_k^\top \mathbf{M}_T \ddot{\mathbf{X}}_k \right) \\ &\leq \lambda_{j-r_k} \left( \ddot{\mathbf{E}}_k^\top (\mathbf{M}_T - \mathbf{M} \mathbf{M}_T \mathbf{M}^\top) \ddot{\mathbf{E}}_k \right) \\ &\quad + \lambda_{r_k+1} \left[ (\mathbf{A}_k \ddot{\mathbf{F}}_k + \ddot{\mathbf{E}}_k^\top \mathbf{M}) \mathbf{M}_T (\mathbf{A}_k \ddot{\mathbf{F}}_k + \ddot{\mathbf{E}}_k^\top \mathbf{M})^\top \right] \\ &\leq \lambda_{j-r_k} \left( \ddot{\mathbf{E}}_k^\top (\mathbf{M}_T - \mathbf{M} \mathbf{M}_T \mathbf{M}^\top) \ddot{\mathbf{E}}_k \right), \end{aligned}$$

since  $(\mathbf{A}_k \ddot{\mathbf{F}}_k + \ddot{\mathbf{E}}_k^\top \mathbf{M}) \mathbf{M}_T (\mathbf{A}_k \ddot{\mathbf{F}}_k + \ddot{\mathbf{E}}_k^\top \mathbf{M})^\top$  is positive semi-definite and has rank at most  $r_k$ . Similarly, we can show that

$$\begin{aligned} \lambda_{j-r_k} \left( \ddot{\mathbf{E}}_k^\top (\mathbf{M}_T - \mathbf{M} \mathbf{M}_T \mathbf{M}^\top) \ddot{\mathbf{E}}_k \right) &\leq \lambda_{j-r_k} \left( \ddot{\mathbf{E}}_k^\top (\mathbf{M}_T - \mathbf{M} \mathbf{M}_T \mathbf{M}^\top) \ddot{\mathbf{E}}_k + \ddot{\mathbf{E}}_k^\top \mathbf{M} \mathbf{M}_T \mathbf{M}^\top \ddot{\mathbf{E}}_k \right) \\ &= \lambda_{j-r_k} \left( \ddot{\mathbf{E}}_k^\top \mathbf{M}_T \ddot{\mathbf{E}}_k \right), \end{aligned}$$

$$\begin{aligned}\lambda_{j+r_k} \left( \ddot{\mathbf{E}}_k^T \mathbf{M}_T \ddot{\mathbf{E}}_k \right) &\leq \lambda_j \left( \ddot{\mathbf{E}}_k^T (\mathbf{M}_T - \mathbf{M} \mathbf{M}_T^T) \ddot{\mathbf{E}}_k \right) + \lambda_{r_k+1} \left( \ddot{\mathbf{E}}_k^T \mathbf{M} \mathbf{M}_T^T \ddot{\mathbf{E}}_k \right) \\ &= \lambda_j \left( \ddot{\mathbf{E}}_k^T (\mathbf{M}_T - \mathbf{M} \mathbf{M}_T^T) \ddot{\mathbf{E}}_k \right).\end{aligned}$$

Therefore, for  $r_k + 1 \leq j \leq \lfloor c \min(T, d_k) \rfloor - r_k$ ,

$$\lambda_{j+r_k} \left( \ddot{\mathbf{E}}_k^T \mathbf{M}_T \ddot{\mathbf{E}}_k \right) \leq \lambda_j \left( \ddot{\mathbf{X}}_k^T \mathbf{M}_T \ddot{\mathbf{X}}_k \right) \leq \lambda_{j-r_k} \left( \ddot{\mathbf{E}}_k^T \mathbf{M}_T \ddot{\mathbf{E}}_k \right),$$

which implies (S.11) with the result of Lemma 1. This completes the proof of Lemma 2.  $\square$

*Proof of Lemma 3.* For  $\mathbf{A}_T$  can either be  $\mathbf{A}_{f,T}$ ,  $\mathbf{A}_{e,T}$  or  $\mathbf{A}_{\epsilon,T}$ , we have already shown that  $\|\mathbf{A}_T\|_1 = O(1)$  in the Proof of Lemma 1 and Lemma 2 (see (S.21) for example). Hence

$$\mathbf{1}_T^T \mathbf{A}_T \mathbf{1}_T \leq \sum_{t,s=1}^T |(\mathbf{A}_T)_{ts}| \leq T \|\mathbf{A}_T\|_1 = O(T).$$

In addition, it is not difficult to see every entry of  $\mathbf{A}_T$  has absolute value bounded above by 1, hence,

$$\|\mathbf{A}_T\|_F^2 = \sum_{t,s=1}^T |(\mathbf{A}_T)_{ts}|^2 \leq \sum_{t,s=1}^T |(\mathbf{A}_T)_{ts}| \leq T \|\mathbf{A}_T\|_1 = O(T).$$

Next, for  $\mathbf{A}_{T^2}$  can either be  $\mathbf{A}_{f,T^2}$ ,  $\mathbf{A}_{e,T^2}$  or  $\mathbf{A}_{\epsilon,T^2}$ , we have

$$\|\mathbf{A}_{T^2}\|_1 = \max_{t_1, s_1} \sum_{t_2, s_2=1}^T |(\mathbf{A}_{T^2})_{t_1 s_1, t_2 s_2}| \leq 2 \max_{q_1, q_2} \sum_{q_3, q_4} |a_{q_1} a_{q_2} a_{q_3} a_{q_4}| \leq 2 \left( \sum_{q \geq 0} |a_q| \right)^4 \leq C.$$

Hence,

$$\mathbf{1}_T^T \mathbf{A}_{T^2} \mathbf{1}_T \leq \sum_{t,s=1}^T |(\mathbf{A}_{T^2})_{ts}| \leq T^2 \|\mathbf{A}_{T^2}\|_1 = O(T^2).$$

Similarly, we can also observe that every entry of  $\mathbf{A}_{T^2}$  has absolute value bounded above by 1. To see this, take  $f_{t,l,j}^{(k)}$  for example, note that for the MA process  $f_t := f_{t,l,j}^{(k)}$  for any  $(l, j, k)$ ,  $|E(f_{t_1} f_{t_2} f_{t_3} f_{t_4})| \leq 0.5 [E(f_{t_1}^2 f_{t_2}^2) + E(f_{t_3}^2 f_{t_4}^2)] \leq 0.25 [E(f_{t_1}^4) + E(f_{t_2}^4) + E(f_{t_3}^4) + E(f_{t_4}^4)] \leq 1$ , because  $E(f_{t_i}^4) = \sum_{q \geq 0} a_q^4 \leq \left( \sum_{q \geq 0} a_q^2 \right)^2 = 1$  for any  $t_i$ . Therefore,

$$\|\mathbf{A}_{T^2}\|_F^2 = \sum_{t,s=1}^T |(\mathbf{A}_{T^2})_{ts}|^2 \leq \sum_{t,s=1}^T |(\mathbf{A}_{T^2})_{ts}| \leq T^2 \|\mathbf{A}_{T^2}\|_1 = O(T^2).$$

(S.14) – (S.17) can be easily implied by (S.12) and (S.13). To see this, first note that for any matrices  $A$  and  $B$ ,  $\|A \otimes B\|_F = \|A\|_F \|B\|_F$ , and  $|\text{vec}(A)^T \text{vec}(B)| \leq \|A\|_F \|B\|_F$  by Cauchy-Schwarz, which implies (S.14), (S.16) and (S.17). Finally,  $\mathbf{1}_{T^2}^T \mathbf{A}_{f,T} \otimes \mathbf{A}_{e,T} \mathbf{1}_{T^2} \leq T^2 \|\mathbf{A}_{f,T} \otimes \mathbf{A}_{e,T}\|_1 = T^2 \|\mathbf{A}_{f,T}\|_1 \|\mathbf{A}_{e,T}\|_1 = O(T^2)$ , which gives (S.15). This completes the proof of Lemma 3.  $\square$

*Proof of Lemma 4.* Note that we have

$$\begin{aligned}\frac{\ddot{\mathbf{X}}_k^T (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{X}}_k}{T} &= \frac{\left( \mathbf{A}_k \ddot{\mathbf{F}}_k + \ddot{\mathbf{E}}_k^T \right) (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \left( \mathbf{A}_k \ddot{\mathbf{F}}_k + \ddot{\mathbf{E}}_k^T \right)^T}{T} \\ &= \mathbf{A}_k \frac{\ddot{\mathbf{F}}_k (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{F}}_k^T}{T} \mathbf{A}_k^T + R_k,\end{aligned}\tag{S.26}$$

where

$$R_k = \frac{\mathbf{A}_k \ddot{\mathbf{F}}_k (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{E}}_k}{T} + \frac{\ddot{\mathbf{E}}_k^T (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{F}}_k^T \mathbf{A}_k^T}{T} + \frac{\ddot{\mathbf{E}}_k^T (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{E}}_k}{T}.\tag{S.27}$$

We estimate  $\widehat{\mathbf{Q}}_{k,(z_k)}$  as the first  $z_k$  eigenvectors of (S.26). Let's define  $\mathbf{U}_{k,(z_k)}$  to be the matrix consisting of the first  $z_k$  columns of  $\mathbf{U}_k$ , and  $\mathbf{B}$  be its orthogonal complement. Then  $\mathbf{U}_{k,(z_k)}$  is an invariant subspace for  $\mathbf{A}_k \mathbf{A}_k^\top$  and

$$\begin{bmatrix} \mathbf{U}_{k,(z_k)}^\top \\ \mathbf{B}^\top \end{bmatrix} \mathbf{A}_k \mathbf{A}_k^\top \begin{bmatrix} \mathbf{U}_{k,(z_k)} & \mathbf{B} \end{bmatrix} = \begin{bmatrix} \mathbf{G}_{k,(z_k)} & 0 \\ 0 & \Lambda_{k,(z_k)} \end{bmatrix},$$

where  $\mathbf{G}_{k,(z_k)}$  is a  $z_k \times z_k$  diagonal matrix consisting the largest  $z_k$  eigenvalues of  $\mathbf{A}_k^\top \mathbf{A}_k$ , and  $\Lambda_{k,(z_k)}$  is a  $(d_k - z_k) \times (d_k - z_k)$  diagonal matrix where the first  $r_k - z_k$  entries are the  $z_k + 1$  to  $r_k$  eigenvalues of  $\mathbf{A}_k^\top \mathbf{A}_k$ , and the remaining entries are all 0's. In this way, we can apply Lemma 3 of Lam et al. (2011) and know that there exists  $\widehat{\mathbf{U}}_{k,(z_k)}$  (with  $\widehat{\mathbf{U}}_{k,(z_k)}^\top \widehat{\mathbf{U}}_{k,(z_k)} = \mathbf{I}_{z_k}$ ) such that  $\widehat{\mathbf{Q}}_{k,(z_k)} = \widehat{\mathbf{U}}_{k,(z_k)} \mathbf{P}_{k,(z_k)}$  where  $\mathbf{P}_{k,(z_k)}$  is orthogonal matrix, so that

$$\begin{aligned} \|\widehat{\mathbf{U}}_{k,(z_k)} - \mathbf{U}_{k,(z_k)}\| &\leq \frac{8 \left\| \mathbf{A}_k \left[ \frac{\ddot{\mathbf{F}}_k (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top) \ddot{\mathbf{F}}_k^\top}{T} - \mathbf{I}_{r_k} \right] \mathbf{A}_k^\top + R_k \right\|}{\text{sep}(\mathbf{G}_{k,z_k}, \Lambda_{k,z_k})} \\ &\preceq \frac{\left\| \mathbf{A}_k \left[ \frac{\ddot{\mathbf{F}}_k (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top) \ddot{\mathbf{F}}_k^\top}{T} - \mathbf{I}_{r_k} \right] \mathbf{A}_k^\top \right\| + \|R_k\|}{d_k^{\alpha_{k,z_k}}}. \end{aligned} \quad (\text{S.28})$$

Next, we bound the norms on the numerator of (S.28). For the first term,

$$\begin{aligned} \left\| \mathbf{A}_k \left[ \frac{\ddot{\mathbf{F}}_k (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top) \ddot{\mathbf{F}}_k^\top}{T} - \mathbf{I}_{r_k} \right] \mathbf{A}_k^\top \right\| &\leq \|\mathbf{A}_k\|^2 \left\| \frac{\ddot{\mathbf{F}}_k (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top) \ddot{\mathbf{F}}_k^\top}{T} - \mathbf{I}_{r_k} \right\| \\ &\asymp d_k^{\alpha_{k,1}} \left\| \frac{\ddot{\mathbf{F}}_k (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top) \ddot{\mathbf{F}}_k^\top}{T} - \mathbf{I}_{r_k} \right\|. \end{aligned}$$

By Assumption (F1),  $\{f_{t,l,j}^{(k)}\}$  is a linear process with absolutely summable autocovariance sequence. In Lemma 2, we have shown that elements of  $\ddot{\mathbf{F}}_k$  retain the covariance structure of  $\{f_{t,l,j}^{(k)}\}$ , so each row of  $\ddot{\mathbf{F}}_k$  is a linear process with absolutely summable autocovariance sequence, which satisfies Assumption (R2) in Lam (2021). Thus, applying the result from Lemma 3 (or more specifically, equation (8.26)) of Lam (2021) implies

$$\left\| \frac{\ddot{\mathbf{F}}_k (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top) \ddot{\mathbf{F}}_k^\top}{T} - \mathbf{I}_{r_k} \right\| = O_p \left( \sqrt{\frac{r_k}{T}} \right), \quad (\text{S.29})$$

which further gives

$$\left\| \mathbf{A}_k \left[ \frac{\ddot{\mathbf{F}}_k (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top) \ddot{\mathbf{F}}_k^\top}{T} - \mathbf{I}_{r_k} \right] \mathbf{A}_k^\top \right\| = O_p \left( d_k^{\alpha_{k,1}} \sqrt{\frac{r_k}{T}} \right).$$

Next, we bound the norm of  $R_k$ . First, note that  $\|\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top\| \leq \|\mathbf{I}_T\| + \|\frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top\| = 2$ . Bounding the squared norm of each term on the right hand side of (S.27), we have

$$\begin{aligned} \left\| \frac{\mathbf{A}_k \ddot{\mathbf{F}}_k (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top) \ddot{\mathbf{F}}_k^\top}{T} \right\|^2 &\leq \left\| \mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top \right\|^2 \left\| \frac{\mathbf{A}_k \ddot{\mathbf{F}}_k}{T^{\frac{1}{2}}} \right\|^2 \left\| \frac{\ddot{\mathbf{F}}_k^\top}{T^{\frac{1}{2}}} \right\|^2 \\ &\leq 4 \lambda_1 \left( \frac{\mathbf{A}_k \ddot{\mathbf{F}}_k \ddot{\mathbf{F}}_k^\top \mathbf{A}_k^\top}{T} \right) \lambda_1 \left( \frac{\ddot{\mathbf{F}}_k^\top \ddot{\mathbf{F}}_k}{T} \right) \\ &= O_p(d_k^{\alpha_{k,1}}) O_p \left( \frac{d_k}{s_k} \left( 1 + \frac{d_k}{T} \right) \right), \end{aligned} \quad (\text{S.30})$$

where the last line follows from Lemma 1 and Lemma 2. Similarly,

$$\left\| \frac{\ddot{\mathbf{E}}_k^T (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{F}}_k^T \mathbf{A}_k^T}{T} \right\|^2 = O_p(d_k^{\alpha_{k,1}}) O_p\left(\frac{d_k}{s_k} \left(1 + \frac{d_k}{T}\right)\right).$$

As for the last term,

$$\left\| \frac{\ddot{\mathbf{E}}_k^T (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{E}}_k}{T} \right\|^2 \leq 4\lambda_1 \left( \frac{\ddot{\mathbf{E}}_k^T \ddot{\mathbf{E}}_k}{T} \right)^2 = O_p\left(\frac{d_k^2}{s_k^2} \left(1 + \frac{d_k^2}{T^2}\right)\right) \quad (\text{S.31})$$

by Lemma 1 and Lemma 2.

For the first two terms on the right hand side of (S.27), there may exist potentially better bounds. To see this, note that we can equivalently write

$$\begin{aligned} \left\| \frac{\mathbf{A}_k \ddot{\mathbf{F}}_k (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{E}}_k}{T} \right\|^2 &\leq \|\mathbf{A}_k\|^2 \left\| \frac{\ddot{\mathbf{F}}_k \ddot{\mathbf{E}}_k}{T} \right\|^2 + \|\mathbf{A}_k\|^2 \left\| \frac{\ddot{\mathbf{F}}_k \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T \ddot{\mathbf{E}}_k}{T} \right\|^2 \\ &= O_p(d_k^{\alpha_{k,1}}) \left( \left\| \frac{\ddot{\mathbf{F}}_k \ddot{\mathbf{E}}_k}{T} \right\|^2 + \left\| \frac{\ddot{\mathbf{F}}_k \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T \ddot{\mathbf{E}}_k}{T} \right\|^2 \right). \end{aligned}$$

Since  $\frac{\ddot{\mathbf{F}}_k \ddot{\mathbf{E}}_k}{T}$  and  $\frac{\ddot{\mathbf{F}}_k \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T \ddot{\mathbf{E}}_k}{T}$  are  $r_k \times d_k$  matrices with each element having a certain rate of convergence, we actually can have a better rate by simply counting the numbers of elements of them. More specifically, the  $(i, j)$  element of  $\ddot{\mathbf{F}}_k \ddot{\mathbf{E}}_k$  is

$$(\ddot{\mathbf{F}}_k \ddot{\mathbf{E}}_k)_{ij} = \sum_{t=1}^T \ddot{f}_{t,k,i} \ddot{e}_{t,k,j},$$

where  $\ddot{f}_{t,k,i}$  is the  $i$ -th entry of  $\ddot{\mathbf{f}}_{t,k}$ , and  $\ddot{e}_{t,k,j}$  are defined similarly. We now focus on bounding the Frobenius norm of  $\ddot{\mathbf{F}}_k \ddot{\mathbf{E}}_k$ . Note that  $\|\ddot{\mathbf{F}}_k \ddot{\mathbf{E}}_k\|_F^2 = \sum_{(i,j) \in (r_k, d_k)} (\ddot{\mathbf{F}}_k \ddot{\mathbf{E}}_k)_{ij}^2$  and

$$\begin{aligned} \|\ddot{\mathbf{F}}_k \ddot{\mathbf{E}}_k\|_F^2 &= E\|\ddot{\mathbf{F}}_k \ddot{\mathbf{E}}_k\|_F^2 + O_p\left(\sqrt{\text{Var}(\|\ddot{\mathbf{F}}_k \ddot{\mathbf{E}}_k\|_F^2)}\right) \\ &= E\|\ddot{\mathbf{F}}_k \ddot{\mathbf{E}}_k\|_F^2 + O_p\left(\sqrt{E\left(\|\ddot{\mathbf{F}}_k \ddot{\mathbf{E}}_k\|_F^4\right) - \left(E\|\ddot{\mathbf{F}}_k \ddot{\mathbf{E}}_k\|_F^2\right)^2}\right). \end{aligned} \quad (\text{S.32})$$

Thus, we need to obtain bounds for  $E\|\ddot{\mathbf{F}}_k \ddot{\mathbf{E}}_k\|_F^2$  and  $E\|\ddot{\mathbf{F}}_k \ddot{\mathbf{E}}_k\|_F^4$ . We start with  $E\|\ddot{\mathbf{F}}_k \ddot{\mathbf{E}}_k\|_F^2$ . For each entry of  $\ddot{\mathbf{F}}_k \ddot{\mathbf{E}}_k$ , we have

$$E(\ddot{\mathbf{F}}_k \ddot{\mathbf{E}}_k)_{ij}^2 = E\left(\sum_{t=1}^T \ddot{f}_{t,k,i} \ddot{e}_{t,k,j}\right)^2 = \sum_{t=1}^T \sum_{s=1}^T E(\ddot{f}_{t,k,i} \ddot{f}_{s,k,i}) E(\ddot{e}_{t,k,j} \ddot{e}_{s,k,j}).$$

By Assumption (F1),

$$E(\ddot{f}_{t,k,i} \ddot{f}_{s,k,i}) = \sum_{q \geq 0} a_{f,q} a_{f,q-|t-s|} = (\mathbf{A}_{f,T})_{ts}. \quad (\text{S.33})$$

Similarly, by Assumption (E1) and (E2),

$$E(\ddot{e}_{t,k,j} \ddot{e}_{s,k,j}) = \frac{1}{s_k} \left[ s_{e,-k} (\mathbf{A}_{e,k} \mathbf{A}_{e,k}^T)_{jj} (\mathbf{A}_{e,T})_{ts} + \left( \boldsymbol{\Sigma}_\epsilon^{(k)} \right)_{jj} (\mathbf{A}_{\epsilon,T})_{ts} \right]. \quad (\text{S.34})$$

Hence,

$$\begin{aligned} E(\ddot{\mathbf{F}}_k \ddot{\mathbf{E}}_k)_{ij}^2 &= \frac{1}{s_k} \sum_{t=1}^T \sum_{s=1}^T (\mathbf{A}_{f,T})_{ts} \left[ s_{e,-k} (\mathbf{A}_{e,k} \mathbf{A}_{e,k}^T)_{jj} (\mathbf{A}_{e,T})_{ts} + \left( \boldsymbol{\Sigma}_\epsilon^{(k)} \right)_{jj} (\mathbf{A}_{\epsilon,T})_{ts} \right] \\ &= \frac{1}{s_k} \left[ s_{e,-k} (\mathbf{A}_{e,k} \mathbf{A}_{e,k}^T)_{jj} \text{vec}(\mathbf{A}_{f,T})^T \text{vec}(\mathbf{A}_{e,T}) + \left( \boldsymbol{\Sigma}_\epsilon^{(k)} \right)_{jj} \text{vec}(\mathbf{A}_{f,T})^T \text{vec}(\mathbf{A}_{\epsilon,T}) \right], \end{aligned}$$

and

$$\begin{aligned}
E \left\| \ddot{\mathbf{F}}_k \ddot{\mathbf{E}}_k \right\|_F^2 &= \sum_{i=1}^{r_k} \sum_{j=1}^{d_k} E \left( \ddot{\mathbf{F}}_k \ddot{\mathbf{E}}_k \right)_{ij}^2 \\
&= \frac{r_k}{s_{-k}} \left[ s_{e,-k} \text{tr}(\mathbf{A}_{e,k} \mathbf{A}_{e,k}^T) \text{vec}(\mathbf{A}_{f,T})^T \text{vec}(\mathbf{A}_{e,T}) + \text{tr}(\boldsymbol{\Sigma}_\epsilon^{(k)}) \text{vec}(\mathbf{A}_{f,T})^T \text{vec}(\mathbf{A}_{e,T}) \right] \\
&= O_p \left( r_k d_k T \frac{d_{-k}}{s_{-k}} \right), \tag{S.35}
\end{aligned}$$

where the last line follows from Assumption (E1) and Lemma 3. Next, for  $E \left\| \ddot{\mathbf{F}}_k \ddot{\mathbf{E}}_k \right\|_F^4$ , note that

$$E \left\| \ddot{\mathbf{F}}_k \ddot{\mathbf{E}}_k \right\|_F^4 = \sum_{i_1=1}^{r_k} \sum_{j_1=1}^{d_k} \sum_{i_2=1}^{r_k} \sum_{j_2=1}^{d_k} E \left[ (\ddot{\mathbf{F}}_k \ddot{\mathbf{E}}_k)_{i_1 j_1}^2 (\ddot{\mathbf{F}}_k \ddot{\mathbf{E}}_k)_{i_2 j_2}^2 \right],$$

and

$$\begin{aligned}
E \left[ (\ddot{\mathbf{F}}_k \ddot{\mathbf{E}}_k)_{i_1 j_1}^2 (\ddot{\mathbf{F}}_k \ddot{\mathbf{E}}_k)_{i_2 j_2}^2 \right] &= E \left[ \left( \sum_{t=1}^T \ddot{f}_{t,k,i_1} \ddot{e}_{t,k,j_1} \right)^2 \left( \sum_{t=1}^T \ddot{f}_{t,k,i_2} \ddot{e}_{t,k,j_2} \right)^2 \right] \\
&= \sum_{t_1=1}^T \sum_{s_1=1}^T \sum_{t_2=1}^T \sum_{s_2=1}^T E \left( \ddot{f}_{t_1,k,i_1} \ddot{f}_{s_1,k,i_1} \ddot{f}_{t_2,k,i_2} \ddot{f}_{s_2,k,i_2} \right) \\
&\quad \cdot E \left( \ddot{e}_{t_1,k,j_1} \ddot{e}_{s_1,k,j_1} \ddot{e}_{t_2,k,j_2} \ddot{e}_{s_2,k,j_2} \right).
\end{aligned}$$

We deal with the terms separately. For the term related to  $\ddot{f}$ , we separate the cases for  $i_1 = i_2$  and  $i_1 \neq i_2$ . When  $i_1 = i_2$ , by Assumption (F1), we have

$$E \left( \ddot{f}_{t_1,k,i_1} \ddot{f}_{s_1,k,i_1} \ddot{f}_{t_2,k,i_2} \ddot{f}_{s_2,k,i_2} \right) = E \left( \ddot{f}_{t_1,k,i_1} \ddot{f}_{s_1,k,i_1} \ddot{f}_{t_2,k,i_1} \ddot{f}_{s_2,k,i_1} \right) = (\mathbf{A}_{f,T^2})_{t_1 s_1, t_2 s_2},$$

where  $(\mathbf{A}_{f,T^2})_{t_1 s_1, t_2 s_2}$  is the  $(t_1 s_1, t_2 s_2)$  entry of  $\mathbf{A}_{f,T^2}$ . When  $i_1 \neq i_2$ ,

$$E \left( \ddot{f}_{t_1,k,i_1} \ddot{f}_{s_1,k,i_1} \ddot{f}_{t_2,k,i_2} \ddot{f}_{s_2,k,i_2} \right) = E \left( \ddot{f}_{t_1,k,i_1} \ddot{f}_{s_1,k,i_1} \right) E \left( \ddot{f}_{t_2,k,i_2} \ddot{f}_{s_2,k,i_2} \right) = (\mathbf{A}_{f,T})_{t_1 s_1} (\mathbf{A}_{f,T})_{t_2 s_2}.$$

Next, for the term related to  $\ddot{e}$ , by Assumption (E1) and (E2),

$$\begin{aligned}
&E \left( \ddot{e}_{t_1,k,j_1} \ddot{e}_{s_1,k,j_1} \ddot{e}_{t_2,k,j_2} \ddot{e}_{s_2,k,j_2} \right) \\
&\leq \frac{1}{s_{-k}^2} \left[ s_{e,-k}^2 (\mathbf{A}_{e,k} \mathbf{A}_{e,k}^T)_{j_1 j_1} (\mathbf{A}_{e,k} \mathbf{A}_{e,k}^T)_{j_2 j_2} (\mathbf{A}_{e,T^2})_{t_1 s_1, t_2 s_2} + \sum_{l=1}^{d_k} (\boldsymbol{\Sigma}_{\epsilon,l}^{(k)})_{j_1 j_1} (\boldsymbol{\Sigma}_{\epsilon,l}^{(k)})_{j_2 j_2} (\mathbf{A}_{\epsilon,T^2})_{t_1 s_1, t_2 s_2} \right] \\
&\leq \frac{1}{s_{-k}^2} \left[ s_{e,-k}^2 (\mathbf{A}_{e,k} \mathbf{A}_{e,k}^T)_{j_1 j_1} (\mathbf{A}_{e,k} \mathbf{A}_{e,k}^T)_{j_2 j_2} (\mathbf{A}_{e,T^2})_{t_1 s_1, t_2 s_2} + (\boldsymbol{\Sigma}_\epsilon^{(k)})_{j_1 j_1} (\boldsymbol{\Sigma}_\epsilon^{(k)})_{j_2 j_2} (\mathbf{A}_{\epsilon,T^2})_{t_1 s_1, t_2 s_2} \right]. \tag{S.36}
\end{aligned}$$

Hence, when  $i_1 = i_2$ ,

$$\begin{aligned}
&E \left[ (\ddot{\mathbf{F}}_k \ddot{\mathbf{E}}_k)_{i_1 j_1}^2 (\ddot{\mathbf{F}}_k \ddot{\mathbf{E}}_k)_{i_2 j_2}^2 \right] \\
&\leq \frac{1}{s_{-k}^2} \left[ s_{e,-k}^2 (\mathbf{A}_{e,k} \mathbf{A}_{e,k}^T)_{j_1 j_1} (\mathbf{A}_{e,k} \mathbf{A}_{e,k}^T)_{j_2 j_2} \text{vec}(\mathbf{A}_{f,T^2})^T \text{vec}(\mathbf{A}_{e,T^2}) \right. \\
&\quad \left. + (\boldsymbol{\Sigma}_\epsilon^{(k)})_{j_1 j_1} (\boldsymbol{\Sigma}_\epsilon^{(k)})_{j_2 j_2} \text{vec}(\mathbf{A}_{f,T^2})^T \text{vec}(\mathbf{A}_{\epsilon,T^2}) \right],
\end{aligned}$$

and when  $i_1 \neq i_2$ ,

$$\begin{aligned} & E \left[ (\ddot{\mathbf{F}}_k \ddot{\mathbf{E}}_k)_{i_1 j_1}^2 (\ddot{\mathbf{F}}_k \ddot{\mathbf{E}}_k)_{i_2 j_2}^2 \right] \\ & \leq \frac{1}{s_{-k}^2} \left[ s_{e,-k}^2 (\mathbf{A}_{e,k} \mathbf{A}_{e,k}^\top)_{j_1 j_1} (\mathbf{A}_{e,k} \mathbf{A}_{e,k}^\top)_{j_2 j_2} \text{vec}(\mathbf{A}_{f,T} \otimes \mathbf{A}_{f,T})^\top \text{vec}(\mathbf{A}_{e,T^2}) \right. \\ & \quad \left. + (\boldsymbol{\Sigma}_\epsilon^{(k)})_{j_1 j_1} (\boldsymbol{\Sigma}_\epsilon^{(k)})_{j_2 j_2} \text{vec}(\mathbf{A}_{f,T} \otimes \mathbf{A}_{f,T})^\top \text{vec}(\mathbf{A}_{e,T^2}) \right]. \end{aligned}$$

Thus, by Lemma 3, we have for any  $(i_1, j_1, i_2, j_2)$ ,

$$\begin{aligned} & E \left[ (\ddot{\mathbf{F}}_k \ddot{\mathbf{E}}_k)_{i_1 j_1}^2 (\ddot{\mathbf{F}}_k \ddot{\mathbf{E}}_k)_{i_2 j_2}^2 \right] \\ & \leq O \left( T^2 \frac{1}{s_{-k}^2} \right) \left[ s_{e,-k}^2 (\mathbf{A}_{e,k} \mathbf{A}_{e,k}^\top)_{j_1 j_1} (\mathbf{A}_{e,k} \mathbf{A}_{e,k}^\top)_{j_2 j_2} + (\boldsymbol{\Sigma}_\epsilon^{(k)})_{j_1 j_1} (\boldsymbol{\Sigma}_\epsilon^{(k)})_{j_2 j_2} \right]. \end{aligned}$$

Therefore,

$$\begin{aligned} E \left\| \ddot{\mathbf{F}}_k \ddot{\mathbf{E}}_k \right\|_F^4 &= \sum_{i_1=1}^{r_k} \sum_{j_1=1}^{d_k} \sum_{i_2=1}^{r_k} \sum_{j_2=1}^{d_k} E \left[ (\ddot{\mathbf{F}}_k \ddot{\mathbf{E}}_k)_{i_1 j_1}^2 (\ddot{\mathbf{F}}_k \ddot{\mathbf{E}}_k)_{i_2 j_2}^2 \right] \\ &\leq r_k^2 O \left( T^2 \frac{s_{e,-k}^2}{s_{-k}^2} \right) \left\{ [\text{tr}(\mathbf{A}_{e,k} \mathbf{A}_{e,k}^\top)]^2 + [\text{tr}(\boldsymbol{\Sigma}_\epsilon^{(k)})]^2 \right\} = O \left( r_k^2 T^2 d_k^2 \frac{d_{-k}^2}{s_{-k}^2} \right), \end{aligned} \quad (\text{S.37})$$

by Assumption (E1). (S.37) together with (S.35) and (S.32) imply  $\left\| \ddot{\mathbf{F}}_k \ddot{\mathbf{E}}_k \right\|_F^2 = O_p \left( r_k d_k T \frac{d_{-k}}{s_{-k}} \right)$ .

Next, we will show  $\left\| \ddot{\mathbf{F}}_k \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top \ddot{\mathbf{E}}_k \right\|_F^2 = O_p \left( r_k d_k \frac{d_{-k}}{s_{-k}} \right)$  in a similar manner. First, note that

$$\left( \ddot{\mathbf{F}}_k \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top \ddot{\mathbf{E}}_k \right)_{ij} = \frac{1}{T} \left( \sum_{t=1}^T \ddot{f}_{t,k,i} \right) \left( \sum_{t=1}^T \ddot{e}_{t,k,j} \right),$$

and

$$\begin{aligned} E \left( \ddot{\mathbf{F}}_k \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top \ddot{\mathbf{E}}_k \right)_{ij}^2 &= \frac{1}{T^2} \sum_{t_1=1}^T \sum_{s_1=1}^T \sum_{t_2=1}^T \sum_{s_2=1}^T E \left( \ddot{f}_{t_1,k,i} \ddot{f}_{s_1,k,i} \right) E \left( \ddot{e}_{t_2,k,j} \ddot{e}_{s_2,k,j} \right) \\ &= \frac{1}{T^2} \sum_{t_1, t_2, s_1, s_2=1}^T (\mathbf{A}_{f,T})_{t_1 s_1} \frac{1}{s_{-k}} \left[ s_{e,-k} (\mathbf{A}_{e,k} \mathbf{A}_{e,k}^\top)_{jj} (\mathbf{A}_{e,T})_{t_2 s_2} + (\boldsymbol{\Sigma}_\epsilon^{(k)})_{jj} (\mathbf{A}_{e,T})_{t_2 s_2} \right] \\ &= \frac{1}{T^2 s_{-k}} \left[ s_{e,-k} (\mathbf{A}_{e,k} \mathbf{A}_{e,k}^\top)_{jj} \mathbf{1}_T^\top \mathbf{A}_{f,T} \otimes \mathbf{A}_{e,T} \mathbf{1}_T + (\boldsymbol{\Sigma}_\epsilon^{(k)})_{jj} \mathbf{1}_T^\top \mathbf{A}_{f,T} \otimes \mathbf{A}_{e,T} \mathbf{1}_T \right] \\ &= O \left( \frac{1}{s_{-k}} \right) \left[ s_{e,-k} (\mathbf{A}_{e,k} \mathbf{A}_{e,k}^\top)_{jj} + (\boldsymbol{\Sigma}_\epsilon^{(k)})_{jj} \right] \end{aligned}$$

where the second line follows from (S.33) and (S.34), and the last line follows from Lemma 3. Hence,

$$\begin{aligned} E \left\| \ddot{\mathbf{F}}_k \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top \ddot{\mathbf{E}}_k \right\|_F^2 &= \sum_{i=1}^{r_k} \sum_{j=1}^{d_k} O \left( \frac{1}{s_{-k}} \right) \left[ s_{e,-k} (\mathbf{A}_{e,k} \mathbf{A}_{e,k}^\top)_{jj} + (\boldsymbol{\Sigma}_\epsilon^{(k)})_{jj} \right] \\ &= O \left( r_k \frac{1}{s_{-k}} \right) \left[ s_{e,-k} \text{tr}(\mathbf{A}_{e,k} \mathbf{A}_{e,k}^\top) + \text{tr}(\boldsymbol{\Sigma}_\epsilon^{(k)}) \right] = O \left( r_k d_k \frac{d_{-k}}{s_{-k}} \right). \end{aligned} \quad (\text{S.38})$$

Next, we know

$$E \left\| \ddot{\mathbf{F}}_k \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top \ddot{\mathbf{E}}_k \right\|_F^4 = \frac{1}{T^4} \sum_{i_1=1}^{r_k} \sum_{j_1=1}^{d_k} \sum_{i_2=1}^{r_k} \sum_{j_2=1}^{d_k} E \left[ (\ddot{\mathbf{F}}_k \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top \ddot{\mathbf{E}}_k)_{i_1 j_1}^2 (\ddot{\mathbf{F}}_k \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top \ddot{\mathbf{E}}_k)_{i_2 j_2}^2 \right],$$



where

$$E \left[ (\ddot{\mathbf{F}}_k \mathbf{1}_T \mathbf{1}_T^\top \ddot{\mathbf{E}}_k)_{i_1 j_1}^2 (\ddot{\mathbf{F}}_k \mathbf{1}_T \mathbf{1}_T^\top \ddot{\mathbf{E}}_k)_{i_2 j_2}^2 \right] = E \left[ \left( \sum_{t=1}^T \ddot{f}_{t,k,i_1} \right)^2 \left( \sum_{t=1}^T \ddot{f}_{t,k,i_2} \right)^2 \right] E \left[ \left( \sum_{t=1}^T \ddot{e}_{t,k,j_1} \right)^2 \left( \sum_{t=1}^T \ddot{e}_{t,k,j_2} \right)^2 \right].$$

For the terms related to  $\ddot{f}$ , when  $i_1 = i_2$ ,

$$\begin{aligned} E \left[ \left( \sum_{t=1}^T \ddot{f}_{t,k,i_1} \right)^2 \left( \sum_{t=1}^T \ddot{f}_{t,k,i_2} \right)^2 \right] &= E \left[ \left( \sum_{t=1}^T \ddot{f}_{t,k,i_1} \right)^4 \right] = \sum_{t_1, t_2, s_1, s_2=1}^T E \left[ \ddot{f}_{t_1,k,i_1} \ddot{f}_{s_1,k,i_1} \ddot{f}_{t_2,k,i_1} \ddot{f}_{s_2,k,i_1} \right] \\ &= \sum_{t_1, t_2, s_1, s_2=1}^T (\mathbf{A}_{f,T^2})_{t_1 s_1, t_2 s_2} = \mathbf{1}_{T^2}^\top \mathbf{A}_{f,T^2} \mathbf{1}_{T^2} = O(T^2) \end{aligned}$$

by Lemma 3. When  $i_1 \neq i_2$ ,

$$\begin{aligned} E \left[ \left( \sum_{t=1}^T \ddot{f}_{t,k,i_1} \right)^2 \left( \sum_{t=1}^T \ddot{f}_{t,k,i_2} \right)^2 \right] &= E \left[ \left( \sum_{t=1}^T \ddot{f}_{t,k,i_1} \right)^2 \right] E \left[ \left( \sum_{t=1}^T \ddot{f}_{t,k,i_2} \right)^2 \right] \\ &= \left[ \sum_{t=1}^T \sum_{s=1}^T E \left( \ddot{f}_{t,k,i_1} \ddot{f}_{s,k,i_1} \right) \right]^2 = (\mathbf{1}_T^\top \mathbf{A}_{f,T} \mathbf{1}_T)^2 = O(T^2) \end{aligned}$$

by Lemma 3. Therefore,  $E \left[ \left( \sum_{t=1}^T \ddot{f}_{t,k,i_1} \right)^2 \left( \sum_{t=1}^T \ddot{f}_{t,k,i_2} \right)^2 \right] = O(T^2)$  for any  $(i_1, i_2)$ . Finally, for terms with  $\ddot{e}$ ,

$$\begin{aligned} E \left[ \left( \sum_{t=1}^T \ddot{e}_{t,k,j_1} \right)^2 \left( \sum_{t=1}^T \ddot{e}_{t,k,j_2} \right)^2 \right] &= \sum_{t_1=1}^T \sum_{s_1=1}^T \sum_{t_2=1}^T \sum_{s_2=1}^T E \left[ \ddot{e}_{t_1,k,j_1} \ddot{e}_{s_1,k,j_1} \ddot{e}_{t_2,k,j_2} \ddot{e}_{s_2,k,j_2} \right] \\ &\leq \sum_{t_1, t_2, s_1, s_2=1}^T \frac{1}{s_{-k}^2} \left[ s_{e,-k}^2 (\mathbf{A}_{e,k} \mathbf{A}_{e,k}^\top)_{j_1 j_1} (\mathbf{A}_{e,k} \mathbf{A}_{e,k}^\top)_{j_2 j_2} (\mathbf{A}_{e,T^2})_{t_1 s_1, t_2 s_2} + (\boldsymbol{\Sigma}_\epsilon^{(k)})_{j_1 j_1} (\boldsymbol{\Sigma}_\epsilon^{(k)})_{j_2 j_2} (\mathbf{A}_{\epsilon,T^2})_{t_1 s_1, t_2 s_2} \right] \\ &= \frac{1}{s_{-k}^2} \left[ s_{e,-k}^2 (\mathbf{A}_{e,k} \mathbf{A}_{e,k}^\top)_{j_1 j_1} (\mathbf{A}_{e,k} \mathbf{A}_{e,k}^\top)_{j_2 j_2} \mathbf{1}_{T^2}^\top \mathbf{A}_{e,T^2} \mathbf{1}_{T^2} + (\boldsymbol{\Sigma}_\epsilon^{(k)})_{j_1 j_1} (\boldsymbol{\Sigma}_\epsilon^{(k)})_{j_2 j_2} \mathbf{1}_{T^2}^\top \mathbf{A}_{\epsilon,T^2} \mathbf{1}_{T^2} \right] \\ &= O \left( \frac{T^2}{s_{-k}^2} \right) \left[ s_{e,-k}^2 (\mathbf{A}_{e,k} \mathbf{A}_{e,k}^\top)_{j_1 j_1} (\mathbf{A}_{e,k} \mathbf{A}_{e,k}^\top)_{j_2 j_2} + (\boldsymbol{\Sigma}_\epsilon^{(k)})_{j_1 j_1} (\boldsymbol{\Sigma}_\epsilon^{(k)})_{j_2 j_2} \right], \end{aligned}$$

where the third line follows from (S.36), and the last line from Lemma 3. Therefore,

$$E \left[ (\ddot{\mathbf{F}}_k \mathbf{1}_T \mathbf{1}_T^\top \ddot{\mathbf{E}}_k)_{i_1 j_1}^2 (\ddot{\mathbf{F}}_k \mathbf{1}_T \mathbf{1}_T^\top \ddot{\mathbf{E}}_k)_{i_2 j_2}^2 \right] = O \left( \frac{T^4}{s_{-k}^2} \right) \left[ s_{e,-k}^2 (\mathbf{A}_{e,k} \mathbf{A}_{e,k}^\top)_{j_1 j_1} (\mathbf{A}_{e,k} \mathbf{A}_{e,k}^\top)_{j_2 j_2} + (\boldsymbol{\Sigma}_\epsilon^{(k)})_{j_1 j_1} (\boldsymbol{\Sigma}_\epsilon^{(k)})_{j_2 j_2} \right].$$

Hence,

$$\begin{aligned} E \left\| \ddot{\mathbf{F}}_k \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top \ddot{\mathbf{E}}_k \right\|_F^4 &= O \left( \frac{1}{s_{-k}^2} \right) \sum_{i_1=1}^{r_k} \sum_{j_1=1}^{d_k} \sum_{i_2=1}^{r_k} \sum_{j_2=1}^{d_k} \left[ s_{e,-k}^2 (\mathbf{A}_{e,k} \mathbf{A}_{e,k}^\top)_{j_1 j_1} (\mathbf{A}_{e,k} \mathbf{A}_{e,k}^\top)_{j_2 j_2} + (\boldsymbol{\Sigma}_\epsilon^{(k)})_{j_1 j_1} (\boldsymbol{\Sigma}_\epsilon^{(k)})_{j_2 j_2} \right] \\ &= r_k^2 O \left( \frac{1}{s_{-k}^2} \right) \left\{ s_{e,-k}^2 [\text{tr}(\mathbf{A}_{e,k} \mathbf{A}_{e,k}^\top)]^2 + [\text{tr}(\boldsymbol{\Sigma}_\epsilon^{(k)})]^2 \right\} = O \left( r_k^2 d_k^2 \frac{d_{-k}^2}{s_{-k}^2} \right). \end{aligned} \quad (\text{S.39})$$

(S.39) together with (S.38) imply  $\left\| \ddot{\mathbf{F}}_k \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top \ddot{\mathbf{E}}_k \right\|_F^2 = O_p \left( r_k d_k \frac{d_{-k}}{s_{-k}} \right)$ . Therefore, we can conclude that

$$\left\| \frac{\mathbf{A}_k \ddot{\mathbf{F}}_k (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top) \ddot{\mathbf{E}}_k}{T} \right\|^2 = O_p(d_k^{\alpha_{k,1}}) \left( \left\| \frac{\ddot{\mathbf{F}}_k \ddot{\mathbf{E}}_k}{T} \right\|^2 + \left\| \frac{\ddot{\mathbf{F}}_k \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top \ddot{\mathbf{E}}_k}{T} \right\|^2 \right) = O_p \left( \frac{d_{-k}}{s_{-k}} \frac{r_k d_k}{T} d_k^{\alpha_{k,1}} \right). \quad (\text{S.40})$$

Compared with the original rate (S.30), we can obtain a potentially better rate (S.40) by directly bounding and counting the elements in a large matrix. In other words, the rate for  $\left\| \frac{\mathbf{A}_k \ddot{\mathbf{F}}_k (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top) \ddot{\mathbf{E}}_k}{T} \right\|^2$  will be the minimum between the two, which is

$$\left\| \frac{\mathbf{A}_k \ddot{\mathbf{F}}_k (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top) \ddot{\mathbf{E}}_k}{T} \right\|^2 = O_p \left( \frac{d_k}{s_k} d_k^{\alpha_{k,1}} \min \left\{ 1 + \frac{d_k}{T}, \frac{r_k d_k}{T} \right\} \right).$$

Combining the rates (S.30), (S.40) and (S.31) yields that

$$\|R_k\|^2 = \frac{d_k^2}{s_k^2} \left( 1 + \frac{d_k^2}{T^2} \right) + \frac{d_k}{s_k} d_k^{\alpha_{k,1}} \min \left\{ 1 + \frac{d_k}{T}, \frac{r_k d_k}{T} \right\} := c_k.$$

Hence,

$$\|\widehat{\mathbf{U}}_{k,(z_k)} - \mathbf{U}_{k,(z_k)}\|^2 = O_p \left( d_k^{-2\alpha_{k,z_k}} \left[ d_k^{2\alpha_{k,1}} \frac{r_k}{T} + c_k \right] \right).$$

Finally, to show (S.18), let  $\tilde{\mathbf{V}}_k$  be the  $z_k \times z_k$  diagonal matrix of the first  $z_k$  largest eigenvalues of  $\widehat{\Sigma}_{\tilde{\mathbf{x}}_k}$  in decreasing order, and  $\check{\mathbf{V}}_k := \tilde{\mathbf{V}}_k / s_k$ . Then it follows from (S.26) that

$$\begin{aligned} \widehat{\mathbf{Q}}_{k,(z_k)} - \mathbf{Q}_k \ddot{\mathbf{H}}_k &= R_k \widehat{\mathbf{Q}}_{k,(z_k)} \check{\mathbf{V}}_k^{-1}, \text{ implying} \\ \|\widehat{\mathbf{Q}}_{k,(z_k)} - \mathbf{Q}_k \ddot{\mathbf{H}}_k\|^2 &\leq \|R_k\|^2 \|\check{\mathbf{V}}_k^{-1}\|^2 O_p \left( d_k^{-2\alpha_{k,z_k}} c_k \right). \end{aligned}$$

This completes the proof of Lemma 4.

*Proof of Theorem 3.1.* Define  $s_{-k,pre} := \frac{1}{M} \sum_{m=1}^M s_{-k,m}$ . First, by the definition of  $s_{-k,m}$ , we have  $\lambda_j \left( \frac{\tilde{\mathbf{F}}_{k,m} \tilde{\mathbf{F}}_{k,m}^\top}{T} \right) \asymp s_{-k,m}$  for  $j \in [r_k]$  and for each  $m$ . Then,

$$\begin{aligned} \lambda_1 \left( \frac{1}{M} \sum_{m=1}^M \frac{\tilde{\mathbf{F}}_{k,m} \tilde{\mathbf{F}}_{k,m}^\top}{T} \right) &\leq \frac{1}{M} \sum_{m=1}^M \lambda_1 \left( \frac{\tilde{\mathbf{F}}_{k,m} \tilde{\mathbf{F}}_{k,m}^\top}{T} \right) \asymp s_{-k,pre}, \\ \lambda_{r_k} \left( \frac{1}{M} \sum_{m=1}^M \frac{\tilde{\mathbf{F}}_{k,m} \tilde{\mathbf{F}}_{k,m}^\top}{T} \right) &\geq \frac{1}{M} \sum_{m=1}^M \lambda_{r_k} \left( \frac{\tilde{\mathbf{F}}_{k,m} \tilde{\mathbf{F}}_{k,m}^\top}{T} \right) \asymp s_{-k,pre}. \end{aligned}$$

Therefore, for  $j \in [r_k]$ ,

$$\lambda_j \left( \frac{1}{M} \sum_{m=1}^M \frac{\tilde{\mathbf{F}}_{k,m} \tilde{\mathbf{F}}_{k,m}^\top}{T} \right) \asymp s_{-k,pre}.$$

Next, following the same analysis as in the proof of Lemma 2, it is easy to see that

$$\lambda_j \left( \frac{1}{M} \sum_{m=1}^M \frac{\mathbf{A}_k \tilde{\mathbf{F}}_{k,m} \tilde{\mathbf{F}}_{k,m}^\top \mathbf{A}_k^\top}{T} \right) \asymp s_{-k,pre} \asymp \lambda_j \left( \frac{1}{M} \sum_{m=1}^M \frac{\mathbf{A}_k \tilde{\mathbf{F}}_{k,m} (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top) \tilde{\mathbf{F}}_{k,m}^\top \mathbf{A}_k^\top}{T} \right),$$

for  $j \in [r_k]$ . On the other hand, by Lemma 1,

$$\lambda_1 \left( \frac{1}{M} \sum_{m=1}^M \frac{\tilde{\mathbf{E}}_{k,m}^\top \tilde{\mathbf{E}}_{k,m}}{T} \right) \leq \frac{1}{M} \sum_{m=1}^M \lambda_1 \left( \frac{\tilde{\mathbf{E}}_{k,m}^\top \tilde{\mathbf{E}}_{k,m}}{T} \right) = O_p \left( d_{-k,pre} \left( 1 + \frac{d_k}{T} \right) \right),$$

where  $d_{-k,pre} = \frac{1}{M} \sum_{m=1}^M d_{-k,m}$ . Similarly,

$$\lambda_1 \left( \frac{1}{M} \sum_{m=1}^M \frac{\tilde{\mathbf{E}}_{k,m}^\top (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top) \tilde{\mathbf{E}}_{k,m}}{T} \right) \leq O_p \left( d_{-k,pre} \left( 1 + \frac{d_k}{T} \right) \right).$$

If Assumptions (E1) – (R1) are satisfied for all  $M$  chosen samples, then  $\frac{d_{k,m}}{s_{k,m}}(1 + \frac{d_k}{T}) = o(d_k^{\alpha_k, z_k})$  for all  $m$ , which implies  $\frac{d_{k,pre}}{s_{k,pre}}(1 + \frac{d_k}{T}) = o(d_k^{\alpha_k, z_k})$ . Therefore, if we define the new normalized version  $\ddot{\mathbf{X}}_{k,m} = \tilde{\mathbf{X}}_{k,m}/s_{k,pre}^{\frac{1}{2}}$ , and  $\ddot{\mathbf{F}}_{k,m}$  and  $\ddot{\mathbf{E}}_{k,m}$  similarly, then following (S.24) and (S.25), (we can write  $\frac{1}{M} \sum_{m=1}^M \frac{\ddot{\mathbf{X}}_{k,m}^T (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{X}}_{k,m}}{T} = \mathbf{X}^T \mathbf{X}$  by Cholesky decomposition, and similarly define  $\mathbf{F}$  and  $\mathbf{E}$ , so that  $\mathbf{X} = \mathbf{F} + \mathbf{E}$ .) we have

$$\lambda_j \left( \frac{1}{M} \sum_{m=1}^M \frac{\ddot{\mathbf{X}}_{k,m}^T (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{X}}_{k,m}}{T} \right) \asymp 1$$

for  $j \in [r_k]$ . Then, we can make use of the decomposition

$$\begin{aligned} \frac{1}{M} \sum_{m=1}^M \frac{\ddot{\mathbf{X}}_{k,m}^T (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{X}}_{k,m}}{T} &= \frac{1}{M} \sum_{m=1}^M \frac{(\mathbf{A}_k \ddot{\mathbf{F}}_{k,m} + \ddot{\mathbf{E}}_{k,m}^T) (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) (\mathbf{A}_k \ddot{\mathbf{F}}_{k,m} + \ddot{\mathbf{E}}_{k,m}^T)^T}{T} \\ &= \mathbf{A}_k \left[ \frac{1}{M} \sum_{m=1}^M \frac{\ddot{\mathbf{F}}_{k,m} (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{F}}_{k,m}^T}{T} \right] \mathbf{A}_k^T + R_{k,pre}, \end{aligned} \quad (\text{S.41})$$

where

$$\begin{aligned} R_{k,pre} &= \frac{1}{M} \sum_{m=1}^M \left[ \frac{\mathbf{A}_k \ddot{\mathbf{F}}_{k,m} (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{E}}_{k,m}^T}{T} + \frac{\ddot{\mathbf{E}}_{k,m}^T (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{F}}_{k,m}^T \mathbf{A}_k^T}{T} \right. \\ &\quad \left. + \frac{\ddot{\mathbf{E}}_{k,m}^T (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{E}}_{k,m}}{T} \right]. \end{aligned} \quad (\text{S.42})$$

Then, following similar argument as in the proof of Lemma 4, we can apply Lemma 3 of Lam et al. (2011) and know that there exists  $\hat{\mathbf{U}}_{k,pre,(z_k)}$  (with  $\hat{\mathbf{U}}_{k,pre,(z_k)}^T \hat{\mathbf{U}}_{k,pre,(z_k)} = \mathbf{I}_{z_k}$ ) such that  $\hat{\mathbf{Q}}_{k,pre,(z_k)} = \hat{\mathbf{U}}_{k,pre,(z_k)} \mathbf{P}_{k,pre,(z_k)}$  where  $\mathbf{P}_{k,pre,(z_k)}$  is an orthogonal matrix, so that

$$\|\hat{\mathbf{U}}_{k,pre,(z_k)} - \mathbf{U}_{k,(z_k)}\| \preceq \frac{d_k^{\alpha_k, 1} \left\| \frac{1}{M} \sum_{m=1}^M \frac{\ddot{\mathbf{F}}_{k,m} (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{F}}_{k,m}^T}{T} - \mathbf{I}_{r_k} \right\| + \|R_{k,pre}\|}{d_k^{\alpha_k, z_k}}.$$

To proceed, note that by (S.29), we have  $\left\| \frac{\ddot{\mathbf{F}}_{k,m} (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{F}}_{k,m}^T}{T s_{k,m}} - \mathbf{I}_{r_k} \right\| = O_p(\sqrt{r_k/T})$  for each  $m$ , so

$$\left\| \frac{\ddot{\mathbf{F}}_{k,m} (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{F}}_{k,m}^T}{T} - \frac{s_{k,m}}{s_{k,pre}} \mathbf{I}_{r_k} \right\| = O_p\left(\frac{s_{k,m}}{s_{k,pre}} \sqrt{\frac{r_k}{T}}\right), \text{ and thus,}$$

$$\begin{aligned} \left\| \frac{1}{M} \sum_{m=1}^M \frac{\ddot{\mathbf{F}}_{k,m} (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{F}}_{k,m}^T}{T} - \mathbf{I}_{r_k} \right\| &= \frac{1}{M} \left\| \sum_{m=1}^M \frac{\ddot{\mathbf{F}}_{k,m} (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{F}}_{k,m}^T}{T} - M \mathbf{I}_{r_k} \right\| \\ &= \frac{1}{M} \left\| \sum_{m=1}^M \left[ \frac{\ddot{\mathbf{F}}_{k,m} (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{F}}_{k,m}^T}{T} - \frac{s_{k,m}}{s_{k,pre}} \mathbf{I}_{r_k} \right] \right\| \\ &\leq \frac{1}{M} \sum_{m=1}^M \left\| \frac{\ddot{\mathbf{F}}_{k,m} (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^T) \ddot{\mathbf{F}}_{k,m}^T}{T} - \frac{s_{k,m}}{s_{k,pre}} \mathbf{I}_{r_k} \right\| \\ &= \frac{1}{M} \sum_{m=1}^M O_p\left(\frac{s_{k,m}}{s_{k,pre}} \sqrt{\frac{r_k}{T}}\right) = O_p\left(\sqrt{\frac{r_k}{T}}\right). \end{aligned}$$

To bound  $\|R_{k,pre}\|^2$ , we can bound each term on the right hand side of (S.42) by using the similar

technique as in the proof of Lemma 4. We have that

$$\begin{aligned} \left\| \frac{1}{M} \sum_{m=1}^M \frac{\mathbf{A}_k \ddot{\mathbf{F}}_{k,m} (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top) \ddot{\mathbf{E}}_{k,m}}{T} \right\|^2 &\leq \frac{1}{M} \sum_{m=1}^M \left\| \frac{\mathbf{A}_k \ddot{\mathbf{F}}_{k,m} (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top) \ddot{\mathbf{E}}_{k,m}}{T} \right\|^2 \\ &= O_p \left( d_k^{\alpha_{k,1}} \min \left\{ 1 + \frac{d_k}{T}, \frac{r_k d_k}{T} \right\} \frac{\frac{1}{M} \sum_{m=1}^M d_{-k,m} s_{-k,m}}{s_{-k,pre}^2} \right), \end{aligned}$$

and similarly,

$$\left\| \frac{1}{M} \sum_{m=1}^M \frac{\ddot{\mathbf{E}}_{k,m}^\top (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top) \ddot{\mathbf{F}}_{k,m}^\top \mathbf{A}_k^\top}{T} \right\|^2 = O_p \left( d_k^{\alpha_{k,1}} \min \left\{ 1 + \frac{d_k}{T}, \frac{r_k d_k}{T} \right\} \frac{\frac{1}{M} \sum_{m=1}^M d_{-k,m} s_{-k,m}}{s_{-k,pre}^2} \right),$$

and

$$\begin{aligned} \left\| \frac{1}{M} \sum_{m=1}^M \frac{\ddot{\mathbf{E}}_{k,m}^\top (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top) \ddot{\mathbf{E}}_{k,m}}{T} \right\|^2 &\leq \frac{1}{M} \sum_{m=1}^M \left\| \frac{\ddot{\mathbf{E}}_{k,m}^\top (\mathbf{I}_T - \frac{1}{T} \mathbf{1}_T \mathbf{1}_T^\top) \ddot{\mathbf{E}}_{k,m}}{T} \right\|^2 \\ &= O_p \left( \left( 1 + \frac{d_k^2}{T^2} \right) \frac{\frac{1}{M} \sum_{m=1}^M d_{-k,m}^2}{s_{-k,pre}^2} \right). \end{aligned}$$

Combining the above results give

$$c_{k,pre} := \|R_{k,pre}\|^2 = O_p \left( \min \left\{ 1 + \frac{d_k}{T}, \frac{r_k d_k}{T} \right\} \frac{\frac{1}{M} \sum_{m=1}^M d_{-k,m} s_{-k,m}}{s_{-k,pre}^2} + d_k^{\alpha_{k,1}} \left( 1 + \frac{d_k^2}{T^2} \right) \frac{\frac{1}{M} \sum_{m=1}^M d_{-k,m}^2}{s_{-k,pre}^2} \right),$$

and thus,

$$\left\| \hat{\mathbf{U}}_{k,pre,(z_k)} - \mathbf{U}_{k,(z_k)} \right\|^2 = O_p \left( d_k^{-2\alpha_{k,z_k}} \left[ d_k^{2\alpha_{k,1}} \frac{r_k}{T} + c_{k,pre} \right] \right). \quad (\text{S.43})$$

Further, let  $\tilde{\mathbf{V}}_{k,pre}$  be the  $z_k \times z_k$  diagonal matrix of the first  $z_k$  largest eigenvalues of  $\hat{\Sigma}_{\tilde{\mathbf{x}}_{k,agg}}$  in decreasing order, and  $\ddot{\mathbf{V}}_{k,pre} := \tilde{\mathbf{V}}_{k,pre} / s_{-k,pre}$ . Then it follows from (S.41) that

$$\begin{aligned} \hat{\mathbf{Q}}_{k,pre,(z_k)} - \mathbf{Q}_k \ddot{\mathbf{H}}_{k,pre} &= R_{k,pre} \hat{\mathbf{Q}}_{k,pre,(z_k)} \ddot{\mathbf{V}}_{k,pre}^{-1}, \text{ implying} \\ \left\| \hat{\mathbf{Q}}_{k,pre,(z_k)} - \mathbf{Q}_k \ddot{\mathbf{H}}_{k,pre} \right\|^2 &\leq \|R_{k,pre}\|^2 \|\ddot{\mathbf{V}}_{k,pre}^{-1}\|^2 = O_p \left( d_k^{-2\alpha_{k,z_k}} c_{k,pre} \right). \end{aligned} \quad (\text{S.44})$$

Finally, (3.12) and (3.13) follow as we substitute  $d_{-k,m} \asymp d_k$  and  $s_{-k,m} \asymp s_{-k,max}$  into (S.44) and (S.43), which is guaranteed by our choice of  $S_{k,m}$  and  $M_0$  as discussed in Section 3.3 and 3.4. This completes the proof of Theorem 3.1.  $\square$

Before the proof of Theorem 4.1, we decompose

$$\begin{aligned}
\tilde{\Sigma}_{y,m+1}^{(k)} &= T^{-1} \sum_{t=1}^T \mathbf{y}_{t,m+1}^{(k)} \mathbf{y}_{t,m+1}^{(k)\top} = T^{-1} \sum_{t=1}^T \text{mat}_k(\mathcal{X}_t - \bar{\mathcal{X}}) \check{\mathbf{q}}_k^{(m)} \check{\mathbf{q}}_k^{(m)\top} \text{mat}_k^\top(\mathcal{X}_t - \bar{\mathcal{X}}) \\
&= T^{-1} \sum_{t=1}^T (\mathbf{A}_k \text{mat}_k(\mathcal{F}_t - \bar{\mathcal{F}}) \mathbf{A}_{-k}^\top \check{\mathbf{q}}_k^{(m)} + \text{mat}_k(\mathcal{E}_t - \bar{\mathcal{E}}) \check{\mathbf{q}}_k^{(m)})^{\otimes 2} \\
&= T^{-1} \sum_{t=1}^T \left\{ \mathbf{A}_k \text{mat}_k(\mathcal{F}_t - \bar{\mathcal{F}}) \mathbf{A}_{-k}^\top \check{\mathbf{q}}_k^{(m)} + \mathbf{A}_{e,k} \text{mat}_k(\mathcal{F}_{e,t} - \bar{\mathcal{F}}_{e,\cdot}) \mathbf{A}_{e,-k}^\top \check{\mathbf{q}}_k^{(m)} \right. \\
&\quad \left. + \left[ (\boldsymbol{\Sigma}_{\epsilon,1}^{(k)})^{1/2} (\boldsymbol{\epsilon}_{t,1}^{(k)} - \bar{\boldsymbol{\epsilon}}_{\cdot,1}^{(k)}), \dots, (\boldsymbol{\Sigma}_{\epsilon,d_k}^{(k)})^{1/2} (\boldsymbol{\epsilon}_{t,d_k}^{(k)} - \bar{\boldsymbol{\epsilon}}_{\cdot,d_k}^{(k)}) \right] \check{\mathbf{q}}_k^{(m)} \right\}^{\otimes 2} \\
&= \sum_{i=1}^3 \tilde{\mathbf{S}}_{ii,m} + \sum_{i,j=1; i < j}^3 (\tilde{\mathbf{S}}_{ij,m} + \tilde{\mathbf{S}}_{ij,m}^\top), \text{ where} \\
\tilde{\mathbf{S}}_{11,m} &:= \mathbf{A}_k (\mathbf{I}_{r_k} \otimes \check{\mathbf{q}}_k^{(m)\top} \mathbf{A}_{-k}) T^{-1} \tilde{\mathbf{F}}^{(k)} \mathbf{M}_T \tilde{\mathbf{F}}^{(k)\top} (\mathbf{I}_{r_k} \otimes \check{\mathbf{q}}_k^{(m)\top} \mathbf{A}_{-k})^\top \mathbf{A}_k^\top, \\
\tilde{\mathbf{S}}_{12,m} &:= \mathbf{A}_k (\mathbf{I}_{r_k} \otimes \check{\mathbf{q}}_k^{(m)\top} \mathbf{A}_{-k}) T^{-1} \tilde{\mathbf{F}}^{(k)} \mathbf{M}_T \tilde{\boldsymbol{\Theta}}^{(k)\top} \text{diag}^{1/2}(\boldsymbol{\Sigma}_{\epsilon,1}^{(k)}, \dots, \boldsymbol{\Sigma}_{\epsilon,d_k}^{(k)}) (\check{\mathbf{q}}_k^{(m)\top} \otimes \mathbf{I}_{d_k})^\top, \\
\tilde{\mathbf{S}}_{13,m} &:= \mathbf{A}_k (\mathbf{I}_{r_k} \otimes \check{\mathbf{q}}_k^{(m)\top} \mathbf{A}_{-k}) T^{-1} \tilde{\mathbf{F}}^{(k)} \mathbf{M}_T \tilde{\mathbf{F}}_e^{(k)\top} (\mathbf{I}_{r_{e,k}} \otimes \check{\mathbf{q}}_k^{(m)\top} \mathbf{A}_{e,-k})^\top \mathbf{A}_{e,k}^\top, \\
\tilde{\mathbf{S}}_{22,m} &:= (\check{\mathbf{q}}_k^{(m)\top} \otimes \mathbf{I}_{d_k}) \text{diag}^{1/2}(\boldsymbol{\Sigma}_{\epsilon,1}^{(k)}, \dots, \boldsymbol{\Sigma}_{\epsilon,d_k}^{(k)}) T^{-1} \tilde{\boldsymbol{\Theta}}^{(k)} \mathbf{M}_T \tilde{\boldsymbol{\Theta}}^{(k)} \text{diag}^{1/2}(\boldsymbol{\Sigma}_{\epsilon,1}^{(k)}, \dots, \boldsymbol{\Sigma}_{\epsilon,d_k}^{(k)}) (\check{\mathbf{q}}_k^{(m)\top} \otimes \mathbf{I}_{d_k})^\top, \\
\tilde{\mathbf{S}}_{23,m} &:= (\check{\mathbf{q}}_k^{(m)\top} \otimes \mathbf{I}_{d_k}) \text{diag}^{1/2}(\boldsymbol{\Sigma}_{\epsilon,1}^{(k)}, \dots, \boldsymbol{\Sigma}_{\epsilon,d_k}^{(k)}) T^{-1} \tilde{\boldsymbol{\Theta}}^{(k)} \mathbf{M}_T \tilde{\mathbf{F}}_e^{(k)\top} (\mathbf{I}_{r_{e,k}} \otimes \check{\mathbf{q}}_k^{(m)\top} \mathbf{A}_{e,-k})^\top \mathbf{A}_{e,k}^\top, \\
\tilde{\mathbf{S}}_{33,m} &:= \mathbf{A}_{e,k} (\mathbf{I}_{r_{e,k}} \otimes \check{\mathbf{q}}_k^{(m)\top} \mathbf{A}_{e,-k}) T^{-1} \tilde{\mathbf{F}}_e^{(k)} \mathbf{M}_T \tilde{\mathbf{F}}_e^{(k)\top} (\mathbf{I}_{r_{e,k}} \otimes \check{\mathbf{q}}_k^{(m)\top} \mathbf{A}_{e,-k})^\top \mathbf{A}_{e,k}^\top,
\end{aligned}$$

with  $\mathbf{M}_T = \mathbf{I}_T - T^{-1} \mathbf{1}_T \mathbf{1}_T^\top$ , and

$$\begin{aligned}
\tilde{\mathbf{F}}^{(k)} &:= [\text{vec}(\text{mat}_k^\top(\mathcal{F}_1)), \dots, \text{vec}(\text{mat}_k^\top(\mathcal{F}_T))] =: [\mathbf{f}_1^{(k)}, \dots, \mathbf{f}_T^{(k)}], \\
\tilde{\mathbf{F}}_e^{(k)} &:= [\text{vec}(\text{mat}_k^\top(\mathcal{F}_{e,1})), \dots, \text{vec}(\text{mat}_k^\top(\mathcal{F}_{e,T}))] =: [\mathbf{f}_{e,1}^{(k)}, \dots, \mathbf{f}_{e,T}^{(k)}], \text{ where} \\
\mathbf{f}_t^{(k)} &:= (f_{t,1,1}^{(k)}, \dots, f_{t,r_k,1}^{(k)}, \dots, f_{t,1,r_k}^{(k)}, \dots, f_{t,r_k,r_k}^{(k)})^\top, \\
\mathbf{f}_{e,t}^{(k)} &:= (f_{e,t,1,1}^{(k)}, \dots, f_{e,t,r_{e,-k},1}^{(k)}, \dots, f_{e,t,1,r_{e,k}}^{(k)}, \dots, f_{e,t,r_{e,-k},r_{e,k}}^{(k)})^\top, \text{ and} \\
\tilde{\boldsymbol{\Theta}}^{(k)} &:= [\boldsymbol{\epsilon}_1^{(k)}, \dots, \boldsymbol{\epsilon}_T^{(k)}] \text{ with } \boldsymbol{\epsilon}_t^{(k)} := (\boldsymbol{\epsilon}_{t,1}^{(k)\top}, \dots, \boldsymbol{\epsilon}_{t,d_k}^{(k)\top})^\top.
\end{aligned}$$

With the notations in Assumption (E2) and (F1), for  $t \in [T]$ , define

$$\begin{aligned}
\mathbf{z}_{e,t}^{(k)} &:= (z_{e,t,1,1}^{(k)}, \dots, z_{e,t,r_{e,-k},1}^{(k)}, \dots, z_{e,t,1,r_{e,k}}^{(k)}, \dots, z_{e,t,r_{e,-k},r_{e,k}}^{(k)})^\top, \\
\mathbf{z}_{\epsilon,t}^{(k)} &:= (z_{\epsilon,t,1,1}^{(k)}, \dots, z_{\epsilon,t,1,d_k}^{(k)}, \dots, z_{\epsilon,t,d_k,1}^{(k)}, \dots, z_{\epsilon,t,d_k,d_k}^{(k)})^\top, \\
\mathbf{z}_{f,t}^{(k)} &:= (z_{f,t,1,1}^{(k)}, \dots, z_{f,t,r_{-k},1}^{(k)}, \dots, z_{f,t,1,r_k}^{(k)}, \dots, z_{f,t,r_{-k},r_k}^{(k)})^\top.
\end{aligned}$$

We further split, for a fixed integer  $N \geq 1$ ,

$$\begin{aligned}
\tilde{\mathbf{F}}^{(k)} &:= \mathbf{F}^{(k)} + \check{\mathbf{F}}^{(k)}, \quad \tilde{\mathbf{F}}_e^{(k)} := \mathbf{F}_e^{(k)} + \check{\mathbf{F}}_e^{(k)}, \quad \tilde{\boldsymbol{\Theta}}^{(k)} := \boldsymbol{\Theta}^{(k)} + \check{\boldsymbol{\Theta}}^{(k)}, \text{ where} \\
\mathbf{F}^{(k)} &:= \left( \sum_{q=0}^{NT} a_{f,q} \mathbf{z}_{f,1-q}^{(k)}, \dots, \sum_{q=0}^{NT} a_{f,q} \mathbf{z}_{f,T-q}^{(k)} \right) = (\mathbf{z}_{f,1-NT}^{(k)}, \dots, \mathbf{z}_{f,T}^{(k)}) \mathcal{A}_{f,T} =: \mathbf{Z}_f^{(k)} \mathcal{A}_{f,T}, \\
\mathbf{F}_e^{(k)} &:= \left( \sum_{q=0}^{NT} a_{e,q} \mathbf{z}_{e,1-q}^{(k)}, \dots, \sum_{q=0}^{NT} a_{e,q} \mathbf{z}_{e,T-q}^{(k)} \right) = (\mathbf{z}_{e,1-NT}^{(k)}, \dots, \mathbf{z}_{e,T}^{(k)}) \mathcal{A}_{e,T} =: \mathbf{Z}_e^{(k)} \mathcal{A}_{e,T}, \\
\boldsymbol{\Theta}^{(k)} &:= \left( \sum_{q=0}^{NT} a_{\epsilon,q} \mathbf{z}_{\epsilon,1-q}^{(k)}, \dots, \sum_{q=0}^{NT} a_{\epsilon,q} \mathbf{z}_{\epsilon,T-q}^{(k)} \right) = (\mathbf{z}_{\epsilon,1-NT}^{(k)}, \dots, \mathbf{z}_{\epsilon,T}^{(k)}) \mathcal{A}_{\epsilon,T} =: \mathbf{Z}_\epsilon^{(k)} \mathcal{A}_{\epsilon,T},
\end{aligned}$$

with  $\mathcal{A}_{f,T}$ ,  $\mathcal{A}_{e,T}$  and  $\mathcal{A}_{\epsilon,T}$  defined in Assumption (RE1), and  $\check{\mathbf{F}}^{(k)}$ ,  $\check{\mathbf{F}}_e^{(k)}$  and  $\check{\mathbf{\Theta}}^{(k)}$  the remainders of the truncations. Then we define

$$\begin{aligned}\tilde{\mathbf{S}}_{ij,m} &= \mathbf{S}_{ij,m} + \check{\mathbf{S}}_{ij,m}, \quad \text{where} \\ \mathbf{S}_{11,m} &:= \mathbf{A}_k (\mathbf{I}_{r_k} \otimes \check{\mathbf{q}}_k^{(m)\top} \mathbf{A}_{-k}) T^{-1} \mathbf{F}^{(k)} \mathbf{M}_T \mathbf{F}^{(k)\top} (\mathbf{I}_{r_k} \otimes \check{\mathbf{q}}_k^{(m)\top} \mathbf{A}_{-k})^\top \mathbf{A}_k^\top,\end{aligned}\tag{S.45}$$

and similarly for other  $\mathbf{S}_{ij,m}$ 's. We first prove a lemma on how large the quadratic form

$$\hat{g}_{-k} := \check{\mathbf{q}}_k^{(0)\top} \mathbf{A}_{-k} \mathbf{A}_{-k}^\top \check{\mathbf{q}}_k^{(0)}\tag{S.46}$$

is, before showing how small each remainder term  $\check{\mathbf{S}}_{ij,0}$  is in the next lemma.

**Lemma 5** *Let all the assumptions in Theorem 4.1 be satisfied. Then*

$$\hat{g}_{-k} \asymp_P \prod_{j=1; j \neq k}^K d_j^{\alpha_{j,1}}.$$

To put some perspectives into Lemma 5, recall that  $\check{\mathbf{q}}_k^{(0)} = \hat{\mathbf{U}}_{-k,pre} = \hat{\mathbf{U}}_{-k,pre,(1)}$  is estimating the direction  $\mathbf{U}_{-k,(1)}$ . Hence the quadratic form  $\hat{g}_{-k}$  is estimating

$$\mathbf{U}_{-k,(1)}^\top \mathbf{A}_{-k} \mathbf{A}_{-k}^\top \mathbf{U}_{-k,(1)} = \mathbf{U}_{-k,(1)}^\top \mathbf{U}_{-k} \mathbf{G}_{-k} \mathbf{U}_{-k}^\top \mathbf{U}_{-k,(1)} = (\mathbf{G}_{-k})_{11} \asymp \prod_{j=1; j \neq k}^K d_j^{\alpha_{j,1}},$$

where the last rate is the result of (S.6) in Lemma 2. Hence  $\hat{g}_{-k}$  has the same rate as the quadratic form that it is estimating.

*Proof of Lemma 5.*

$$\begin{aligned}\hat{g}_{-k} &= \hat{\mathbf{q}}_{-k,pre}^\top \mathbf{A}_{-k} \mathbf{A}_{-k}^\top \hat{\mathbf{q}}_{-k,pre} = \hat{\mathbf{U}}_{-k,pre,(1)}^\top \mathbf{A}_{-k} \mathbf{A}_{-k}^\top \hat{\mathbf{U}}_{-k,pre,(1)} \\ &= \mathbf{U}_{-k,(1)}^\top \mathbf{A}_{-k} \mathbf{A}_{-k}^\top \mathbf{U}_{-k,(1)} + (\hat{\mathbf{U}}_{-k,pre,(1)} - \mathbf{U}_{-k,(1)})^\top \mathbf{A}_{-k} \mathbf{A}_{-k}^\top (\hat{\mathbf{U}}_{-k,pre,(1)} - \mathbf{U}_{-k,(1)}) \\ &\quad + 2(\hat{\mathbf{U}}_{-k,pre,(1)} - \mathbf{U}_{-k,(1)})^\top \mathbf{A}_{-k} \mathbf{A}_{-k}^\top \mathbf{U}_{-k,(1)} \\ &\geq \mathbf{U}_{-k,(1)}^\top \mathbf{A}_{-k} \mathbf{A}_{-k}^\top \mathbf{U}_{-k,(1)} + 2(\hat{\mathbf{U}}_{-k,pre,(1)} - \mathbf{U}_{-k,(1)})^\top \mathbf{A}_{-k} \mathbf{A}_{-k}^\top \mathbf{U}_{-k,(1)} \\ &= \mathbf{U}_{-k,(1)}^\top \mathbf{U}_{-k} \mathbf{G}_{-k} \mathbf{U}_{-k}^\top \mathbf{U}_{-k,(1)} + 2(\hat{\mathbf{U}}_{-k,pre,(1)} - \mathbf{U}_{-k,(1)})^\top \mathbf{U}_{-k} \mathbf{G}_{-k} \mathbf{U}_{-k}^\top \mathbf{U}_{-k,(1)},\end{aligned}\tag{S.47}$$

where the last line used the singular value decomposition of  $\mathbf{A}_k$  in (3.9), and

$$\begin{aligned}\hat{\mathbf{U}}_{-k,pre,(1)} &:= \mathbf{U}_{K,pre,(1)} \otimes \cdots \otimes \mathbf{U}_{k+1,pre,(1)} \otimes \mathbf{U}_{k-1,pre,(1)} \otimes \cdots \otimes \mathbf{U}_{1,pre,(1)}, \\ \mathbf{U}_{-k} &:= \mathbf{U}_K \otimes \cdots \otimes \mathbf{U}_{k+1} \otimes \mathbf{U}_{k-1} \otimes \cdots \otimes \mathbf{U}_1, \quad \mathbf{G}_{-k} := \mathbf{G}_K \otimes \cdots \otimes \mathbf{G}_{k+1} \otimes \mathbf{G}_{k-1} \otimes \cdots \otimes \mathbf{G}_1, \\ \mathbf{U}_{-k,(1)} &:= \mathbf{U}_{K,(1)} \otimes \cdots \otimes \mathbf{U}_{k+1,(1)} \otimes \mathbf{U}_{k-1,(1)} \otimes \cdots \otimes \mathbf{U}_{1,(1)}.\end{aligned}$$

But (S.6) in Lemma 2 proves that  $\mathbf{G}_k$  has the  $j$ -th diagonal element with order  $d_k^{\alpha_{k,j}}$  for  $j \in [r_k]$  under Assumption (L1), which is the same as those in  $\mathbf{D}_k$ . At the same time, induction (proof omitted) easily gives

$$\|\hat{\mathbf{U}}_{-k,pre,(1)} - \mathbf{U}_{-k,(1)}\| = O_P \left( \sum_{j=1; j \neq k}^K \|\hat{\mathbf{U}}_{j,pre,(1)} - \mathbf{U}_{j,(1)}\| \right) = O_P \left( K \sqrt{\frac{r_{\max}}{T}} + \sum_{j=1; j \neq k}^K d_j^{-\alpha_{j,1}} c_{j,max}^{1/2} \right),\tag{S.48}$$

with  $r_{\max} := \max_{k \in [K]} r_k$ . Hence from the decomposition after (S.47), we then have, assuming the above rate is  $o(1)$ ,

$$\hat{g}_{-k} \asymp (1 + o_P(1)) (\mathbf{G}_{-k})_{11} \asymp_P \prod_{j=1; j \neq k}^K d_j^{\alpha_{j,1}}.$$

**Lemma 6** *Let all the assumptions in Theorem 4.1 be satisfied. Then for each  $k \in [K]$ ,*

$$\sum_{i,j=1; i \leq j}^3 \|\check{\mathbf{S}}_{ij,0}\| = o_P(T^{-1}).$$

*Proof of Lemma 6.* Define

$$\begin{aligned} \mathbf{Q}_{1,m}^{(k)} &:= \mathbf{A}_k(\mathbf{I}_{r_k} \otimes \check{\mathbf{Q}}_k^{(m)\top} \mathbf{A}_{-k}), \\ \mathbf{Q}_{2,m}^{(k)} &:= (\check{\mathbf{Q}}_k^{(m)\top} \otimes \mathbf{I}_{d_k}) \text{diag}^{1/2}(\boldsymbol{\Sigma}_{\epsilon,1}^{(k)}, \dots, \boldsymbol{\Sigma}_{\epsilon,d_{-k}}^{(k)}), \\ \mathbf{Q}_{3,m}^{(k)} &:= \mathbf{A}_{e,k}(\mathbf{I}_{r_{e,k}} \otimes \check{\mathbf{Q}}_k^{(m)\top} \mathbf{A}_{e,-k}). \end{aligned} \quad (\text{S.49})$$

Then

$$\begin{aligned} \|\mathbf{Q}_{1,0}^{(k)}\|^2 &= (\check{\mathbf{Q}}_k^{(0)\top} \mathbf{A}_{-k} \mathbf{A}_{-k}^\top \check{\mathbf{Q}}_k^{(0)}) \|\mathbf{A}_k\|^2 = O_P\left(\prod_{j=1; j \neq k}^K d_j^{\alpha_{j,1}} \cdot d_k^{\alpha_{k,1}}\right) = O_P(g_s), \\ \|\mathbf{Q}_{2,0}^{(k)}\|^2 &= O_P\left(\max_{j \in [d_{-k}]} \|\boldsymbol{\Sigma}_{\epsilon,j}^{(k)}\|\right) = O_P(1), \quad \|\mathbf{Q}_{3,0}^{(k)}\|^2 = (\check{\mathbf{Q}}_k^{(0)\top} \mathbf{A}_{e,-k} \mathbf{A}_{e,-k}^\top \check{\mathbf{Q}}_k^{(0)}) \|\mathbf{A}_{e,k}\|^2 = O_P(1), \end{aligned}$$

where the first line used Lemma 5, and the results for  $\mathbf{Q}_{2,0}^{(k)}$  and  $\mathbf{Q}_{3,0}^{(k)}$  used finiteness of  $K$  and Assumption (E1). We also have, by Assumption (RE1),

$$\begin{aligned} E\|T^{-1/2}\mathbf{F}^{(k)}\|_F^2 &= T^{-1} E \text{tr}(\mathcal{A}_{f,T}^\top \mathbf{Z}_f^{(k)\top} \mathbf{Z}_f^{(k)} \mathcal{A}_{f,T}) = r T^{-1} \text{tr}(\mathcal{A}_{f,T} \mathcal{A}_{f,T}^\top) = r(1 - o(T^{-2}d^{-2})), \\ E\|T^{-1/2}\check{\mathbf{F}}^{(k)}\|_F^2 &= O(rT \cdot T^{-1} \cdot o(T^{-2}d^{-2})) = o(rT^{-2}d^{-2}), \\ E\|T^{-1/2}\boldsymbol{\Theta}^{(k)}\|_F^2 &= d(1 - o(T^{-2}d^{-2})), \quad E\|T^{-1/2}\check{\boldsymbol{\Theta}}^{(k)}\|_F^2 = o(dT^{-2}d^{-2}) = o(T^{-2}d^{-1}), \\ E\|T^{-1/2}\mathbf{F}_e^{(k)}\|_F^2 &= r_e(1 - o(T^{-2}d^{-2})), \quad E\|T^{-1/2}\check{\mathbf{F}}_e^{(k)}\|_F^2 = o(r_e T^{-2}d^{-2}). \end{aligned}$$

Then writing  $\check{\mathbf{S}}_{11,0} = I_1 + I_1^\top + I_2$ , where

$$I_1 := T^{-1} \mathbf{Q}_{1,0}^{(k)} \check{\mathbf{F}}^{(k)} \mathbf{M}_T \mathbf{F}^{(k)} \mathbf{Q}_{1,0}^{(k)\top}, \quad I_2 := T^{-1} \mathbf{Q}_{1,0}^{(k)} \check{\mathbf{F}}^{(k)} \mathbf{M}_T \check{\mathbf{F}}^{(k)} \mathbf{Q}_{1,0}^{(k)\top},$$

we have for  $a > 0$ ,

$$\begin{aligned} P(\|I_1\| \geq a) &\leq \|\mathbf{Q}_{1,0}^{(k)}\|^2 E^{1/2} \|T^{-1/2}\mathbf{F}^{(k)}\|_F^2 \cdot E^{1/2} \|T^{-1/2}\check{\mathbf{F}}^{(k)}\|_F^2 / a \\ &= o(g_s \cdot r^{1/2} \cdot r^{1/2} T^{-2/2} d^{-2/2} / a) = o(r g_s T^{-1} d^{-1} / a) = o(T^{-1}), \end{aligned}$$

showing that  $\|I_1\| = o_P(T^{-1})$ . Similarly (details omitted),  $\|I_2\| = o_P(T^{-2}d^{-1})$ , so that  $\|\check{\mathbf{S}}_{11,0}\| = o_P(T^{-1})$ . Similar to the above arguments, we can show that (details omitted)

$$\begin{aligned} \|\check{\mathbf{S}}_{12,0}\| &= o_P(r^{1/2} g_s^{1/2} T^{-1} d^{-1/2}) = o_P(T^{-1}), \quad \|\check{\mathbf{S}}_{13,0}\| = o_P(g_s^{1/2} r^{1/2} r_e^{1/2} T^{-1} d^{-1}) = o_P(T^{-1}), \\ \|\check{\mathbf{S}}_{22,0}\| &= o_P(d^{1/2} \cdot T^{-1} d^{-1/2}) = o_P(T^{-1}), \quad \|\check{\mathbf{S}}_{23,0}\| = o_P(r_e^{1/2} T^{-1} d^{-1/2}) = o_P(T^{-1}), \\ \|\check{\mathbf{S}}_{33,0}\| &= o_P(r_e^{1/2} \cdot r_e^{1/2} T^{-1} d^{-1}) = o_P(T^{-1}). \end{aligned}$$

This completes the proof of the lemma.  $\square$

Before presenting the next lemma, define, for  $j \in [d_{-k}]$  and  $k \in [K]$ , and  $m$  a non-negative integer,

$$\begin{aligned} \mathbf{S}'_{11,m} &= (\check{\mathbf{Q}}_k^{(m)\top} \mathbf{A}_{-k} \mathbf{A}_{-k}^\top \check{\mathbf{Q}}_k^{(m)}) \mathbf{A}_k \mathbf{A}_k^\top, \\ \mathbf{S}''_{11,m} &= \mathbf{A}_k(\mathbf{I}_{r_k} \otimes \check{\mathbf{Q}}_k^{(m)\top} \mathbf{A}_{-k})(T^{-1} \mathbf{F}^{(k)} \mathbf{M}_T \mathbf{F}^{(k)\top} - \mathbf{I}_r)(\mathbf{I}_{r_k} \otimes \check{\mathbf{Q}}_k^{(m)\top} \mathbf{A}_{-k})^\top \mathbf{A}_k^\top, \end{aligned}$$

so that  $\mathbf{S}_{11,m} = \mathbf{S}'_{11,m} + \mathbf{S}''_{11,m}$ , where  $\mathbf{S}_{11,m}$  is defined in (S.45). Then

$$\begin{aligned}\tilde{\mathbf{S}}_{y,m+1}^{(k)} &= T^{-1} \sum_{t=1}^T \mathbf{y}_{t,m+1}^{(k)} \mathbf{y}_{t,m+1}^{(k)\top} = \mathbf{S}'_{11,m} + \mathbf{S}''_{11,m} + \mathbf{S}_{22,m} + \mathbf{S}_{33,m} + \sum_{i,j=1;i < j}^3 (\mathbf{S}_{ij,m} + \mathbf{S}_{ij,m}^\top) + \sum_{i,j=1}^3 \check{\mathbf{S}}_{ij,m} \\ &=: \mathbf{S}'_{11,m} + \mathbf{E}_m.\end{aligned}\tag{S.50}$$

**Lemma 7** *Let all the assumptions in Theorem 4.1 be satisfied. Then for  $j \in [d_{\cdot k}]$  and  $k \in [K]$ , defining  $\check{g}_{\cdot k}^{(m)} := \check{\mathbf{q}}_{\cdot k}^{(m)\top} \mathbf{A}_{\cdot k} \mathbf{A}_{\cdot k}^\top \check{\mathbf{q}}_{\cdot k}^{(m)}$ , we have*

$$\begin{aligned}\|\mathbf{S}''_{11,m}\| &= O_P\left(\check{g}_{\cdot k}^{(m)} d_k^{\alpha_{k,1}} \sqrt{\frac{r}{T}}\right), \quad \|\mathbf{S}_{12,m}\| = O_P\left\{(\check{g}_{\cdot k}^{(m)})^{1/2} d_k^{\alpha_{k,1}/2} \left(\sqrt{\frac{r d_k}{T}} + \|\check{\mathbf{q}}_{\cdot k}^{(m)} - \mathbf{U}_{\cdot k,(1)}\| \sqrt{\frac{r d}{T}}\right)\right\}, \\ \|\mathbf{S}_{13,m}\| &= O_P\left((\check{g}_{\cdot k}^{(m)})^{1/2} d_k^{\alpha_{k,1}/2} \sqrt{\frac{r r_e}{T}}\right), \\ \|\mathbf{S}_{22,m}\| &= O_P\left(1 + \|\check{\mathbf{q}}_{\cdot k}^{(m)} - \mathbf{U}_{\cdot k,(1)}\|^2 \frac{d}{T} + \frac{d_k}{\sqrt{T}} + \|\check{\mathbf{q}}_{\cdot k}^{(m)} - \mathbf{U}_{\cdot k,(1)}\| \sqrt{\frac{d_k d}{T}}\right), \\ \|\mathbf{S}_{23,m}\| &= O_P\left\{\sqrt{\frac{r_e d_k}{T}} + \|\check{\mathbf{q}}_{\cdot k}^{(m)} - \mathbf{U}_{\cdot k,(1)}\| \sqrt{\frac{r_e d}{T}}\right\}, \quad \|\mathbf{S}_{33,m}\| = O_P(1),\end{aligned}$$

where  $\mathbf{U}_{\cdot k,(1)} := \mathbf{U}_{K,(1)} \otimes \cdots \otimes \mathbf{U}_{k+1,(1)} \otimes \mathbf{U}_{k-1,(1)} \otimes \cdots \otimes \mathbf{U}_{1,(1)}$ .

*Proof of Lemma 7.* Using (3.9) and the fact that (S.6) proves that  $\lambda_j(\mathbf{G}_k) \asymp d_k^{\alpha_{k,j}}$  for  $j \in [r_k]$ , we have

$$\begin{aligned}\|\mathbf{S}''_{11,m}\| &\leq (\check{\mathbf{q}}_{\cdot k}^{(m)\top} \mathbf{A}_{\cdot k} \mathbf{A}_{\cdot k}^\top \check{\mathbf{q}}_{\cdot k}^{(m)}) \|\mathbf{A}_{\cdot k}\|^2 \cdot \|T^{-1} \mathbf{Z}_f^{(k)} \mathcal{A}_{f,T} \mathbf{M}_T \mathcal{A}_{f,T}^\top \mathbf{Z}_f^{(k)\top} - \mathbf{I}_r\| \\ &= \check{g}_{\cdot k}^{(m)} \|\mathbf{G}_k\| \cdot \|T^{-1} \mathbf{Z}_f^{(k)} \mathcal{A}_{f,T} \mathbf{M}_T \mathcal{A}_{f,T}^\top \mathbf{Z}_f^{(k)\top} - \mathbf{I}_r\| = O_P\left(\check{g}_{\cdot k}^{(m)} d_k^{\alpha_{k,1}} \sqrt{\frac{r}{T}}\right),\end{aligned}$$

where the last equality used Theorem 2.8 of Wang and Paul (2014), which can be applied since we have  $r = o(T^{1/3})$  and with fourth order moments exist for the elements in  $\mathbf{Z}_f^{(k)}$  from Assumption (L1) and (R1) respectively, and that by Assumption (RE1),

$$\begin{aligned}\|\mathcal{A}_{f,T} \mathbf{M}_T \mathcal{A}_{f,T}^\top\| &\leq \|\mathbf{A}_{f,T}\|^2 < \infty, \\ \frac{1}{(N+1)T} \text{tr}(\mathcal{A}_{f,T} \mathbf{M}_T \mathcal{A}_{f,T}^\top) &= \frac{1}{(N+1)T} (\text{tr}(\mathcal{A}_{f,T}^\top \mathcal{A}_{f,T}) - T^{-1} \mathbf{1}_T^\top \mathcal{A}_{f,T}^\top \mathcal{A}_{f,T} \mathbf{1}_T) \rightarrow \frac{1}{N+1}, \\ \frac{1}{(N+1)T} \text{tr}(\mathcal{A}_{f,T} \mathbf{M}_T \mathcal{A}_{f,T}^\top)^2 &= \frac{1}{(N+1)T} \{\text{tr}(\mathcal{A}_{f,T}^\top \mathcal{A}_{f,T})^2 - 2T^{-1} \mathbf{1}_T^\top (\mathcal{A}_{f,T}^\top \mathcal{A}_{f,T})^2 \mathbf{1}_T \\ &\quad + T^{-2} (\mathbf{1}_T^\top \mathcal{A}_{f,T}^\top \mathcal{A}_{f,T} \mathbf{1}_T)^2\} \rightarrow \frac{a_1 - 2a_2 + a_3^2}{N+1}.\end{aligned}$$

For  $\|\mathbf{S}_{12,m}\|$ , define the notation  $\mathbf{U}_S$  to be a sub-matrix of  $\mathbf{U}$  restricted to the rows indexed by  $S$ . Let  $\mathbf{Z}_j := (\mathbf{Z}_\epsilon^{(k)})_{B_j} \in \mathbb{R}^{d_k \times (M+1)T}$ , where  $B_j := \{(j-1)d_k + 1, \dots, j d_k\}$ ,  $j = 1, \dots, d_{\cdot k}$ . Let also  $\mathbf{a}_{\ell,j} := (\boldsymbol{\Sigma}_{\epsilon,j}^{(k)})_\ell^\top$ ,  $\ell \in [d_k]$ . Using the notation in (S.49), for  $h \in [r]$  and  $\ell \in [d_k]$ , the  $(h, \ell)$ -th element of  $T^{-1} \mathbf{Z}_f^{(k)} \mathcal{A}_{f,T} \mathbf{M}_T \mathcal{A}_{f,T}^\top \mathbf{Z}_\epsilon^{(k)\top} \mathbf{Q}_{2,m}^{(k)\top}$  is  $J_1 + J_2$ , where

$$\begin{aligned}J_1 &:= T^{-1} \sum_{j=1}^{d_{\cdot k}} (\mathbf{U}_{\cdot k,(1)})_j \mathbf{a}_{\ell,j}^\top \mathbf{Z}_j \mathcal{A}_{\epsilon,T} \mathbf{M}_T \mathcal{A}_{f,T}^\top (\mathbf{Z}_f^{(k)})_h, \\ J_2 &:= T^{-1} \sum_{j=1}^{d_{\cdot k}} (\check{\mathbf{q}}_{\cdot k}^{(m)} - \mathbf{U}_{\cdot k,(1)})_j \mathbf{a}_{\ell,j}^\top \mathbf{Z}_j \mathcal{A}_{\epsilon,T} \mathbf{M}_T \mathcal{A}_{f,T}^\top (\mathbf{Z}_f^{(k)})_h.\end{aligned}$$



We have  $EJ_1 = 0$ , and by Assumption (E1),

$$E|J_1|^2 = T^{-2} \sum_{j=1}^{d_k} (\mathbf{U}_{-k,(1)})_j^2 \|\mathbf{a}_{\ell,j}\|^2 \|\mathcal{A}_{f,T} \mathbf{M}_T \mathcal{A}_{\epsilon,T}^\top\|_F^2 \leq O(T^{-2}) \cdot \|\mathcal{A}_{f,T}\| \|\mathbf{M}_T\| \cdot \|\mathcal{A}_{\epsilon,T}\|_F^2 = O(T^{-1}),$$

where the last equality is from Assumption (RE1). It means that  $J_1 = O_P(T^{-1/2})$ . For  $J_2$ , using the Cauchy-Schwarz inequality and that  $\mathbf{a}_{\ell,j}^\top \mathbf{Z}_j \mathcal{A}_{\epsilon,T} \mathbf{M}_T \mathcal{A}_{f,T}^\top (\mathbf{Z}_f^{(k)})_h = O_P(T^{1/2})$  from the analysis of  $J_1$  above,

$$\begin{aligned} J_2 &\leq \|\check{\mathbf{q}}_{\mathbf{L}_k}^{(m)} - \mathbf{U}_{-k,(1)}\| \cdot T^{-1} \left( \sum_{j=1}^{d_k} (\mathbf{a}_{\ell,j}^\top \mathbf{Z}_j \mathcal{A}_{f,T} \mathbf{M}_T \mathcal{A}_{\epsilon,T}^\top (\mathbf{Z}_f^{(k)})_h)^2 \right)^{1/2} \\ &= \|\check{\mathbf{q}}_{\mathbf{L}_k}^{(m)} - \mathbf{U}_{-k,(1)}\| \cdot T^{-1} O_P(d_k T)^{1/2} = O_P \left( \|\check{\mathbf{q}}_{\mathbf{L}_k}^{(m)} - \mathbf{U}_{-k,(1)}\| \sqrt{\frac{d_k}{T}} \right). \end{aligned}$$

With the above, we then have

$$\begin{aligned} \|\mathbf{S}_{12,m}\| &= \|\mathbf{Q}_{1,m}^{(k)} T^{-1} \mathbf{Z}_f^{(k)} \mathcal{A}_{f,T} \mathbf{M}_T \mathcal{A}_{\epsilon,T}^\top \mathbf{Z}_\epsilon^{(k)\top} \mathbf{Q}_{2,m}^{(k)\top}\| \leq \|\mathbf{Q}_{1,m}^{(k)}\| \cdot \sqrt{rd_k} \cdot O_P(J_1 + J_2) \\ &= O_P \left\{ (\check{g}_{-k}^{(m)})^{1/2} d_k^{\alpha_{k,1}/2} \left( \sqrt{\frac{rd_k}{T}} + \|\check{\mathbf{q}}_{\mathbf{L}_k}^{(m)} - \mathbf{U}_{-k,(1)}\| \sqrt{\frac{rd_k}{T}} \right) \right\}. \end{aligned}$$

For  $\mathbf{S}_{13,m}$ , using the definitions in (S.49) and the bounds for  $\|\mathbf{Q}_{3,m}^{(k)}\|$  in Lemma 6,

$$\begin{aligned} \|\mathbf{S}_{13,m}\| &\leq \|\mathbf{Q}_{1,m}^{(k)}\| \cdot \|\mathbf{Q}_{3,m}^{(k)}\| \cdot \|T^{-1} \mathbf{Z}_f^{(k)} \mathcal{A}_{f,T} \mathbf{M}_T \mathcal{A}_{e,T}^\top \mathbf{Z}_e^{(k)\top}\| \\ &\leq (\check{g}_{-k}^{(m)})^{1/2} d_k^{\alpha_{k,1}/2} \cdot 1 \cdot O_P(T^{-1} \sqrt{rr_e} \|\mathcal{A}_{f,T} \mathbf{M}_T \mathcal{A}_{e,T}^\top\|_F) = O_P \left( (\check{g}_{-k}^{(m)})^{1/2} d_k^{\alpha_{k,1}/2} \sqrt{\frac{rr_e}{T}} \right), \end{aligned}$$

where the last line used the fact that an element in  $\mathbf{Z}_f^{(k)} \mathcal{A}_{f,T} \mathbf{M}_T \mathcal{A}_{e,T}^\top \mathbf{Z}_e^{(k)\top}$  is  $O_P(\|\mathcal{A}_{f,T} \mathbf{M}_T \mathcal{A}_{e,T}^\top\|_F) = O_P(T^{1/2})$  by a similar calculation for treating  $J_1$  above.

For  $\mathbf{S}_{22,m}$ , decompose  $\mathbf{Q}_{2,m}^{(k)} = \mathbf{Q}_{2,m,0}^{(k)} + \mathbf{Q}_{2,m,e}^{(k)}$ , where

$$\mathbf{Q}_{2,m,0}^{(k)} := (\mathbf{U}_{-k,(1)}^\top \otimes I_{d_k}) \text{diag}^{1/2}(\boldsymbol{\Sigma}_{\epsilon,1}^{(k)}, \dots, \boldsymbol{\Sigma}_{\epsilon,d_k}^{(k)}), \quad \mathbf{Q}_{2,m,e}^{(k)} := ((\check{\mathbf{q}}_{\mathbf{L}_k}^{(m)} - \mathbf{U}_{-k,(1)})^\top \otimes I_{d_k}) \text{diag}^{1/2}(\boldsymbol{\Sigma}_{\epsilon,1}^{(k)}, \dots, \boldsymbol{\Sigma}_{\epsilon,d_k}^{(k)}).$$

Then  $\mathbf{S}_{22,m} = I_0 + I_1 + I_2 + I_3$ , where

$$\begin{aligned} I_0 &:= \mathbf{Q}_{2,m}^{(k)} \mathbf{Q}_{2,m}^{(k)\top}, \\ I_1 &:= \mathbf{Q}_{2,m,0}^{(k)} (T^{-1} \mathbf{Z}_\epsilon^{(k)} \mathcal{A}_{\epsilon,T} \mathbf{M}_T \mathcal{A}_{\epsilon,T}^\top \mathbf{Z}_\epsilon^{(k)\top} - \mathbf{I}_d) \mathbf{Q}_{2,m,0}^{(k)\top}, \\ I_2 &:= \mathbf{Q}_{2,m,e}^{(k)} (T^{-1} \mathbf{Z}_\epsilon^{(k)} \mathcal{A}_{\epsilon,T} \mathbf{M}_T \mathcal{A}_{\epsilon,T}^\top \mathbf{Z}_\epsilon^{(k)\top} - \mathbf{I}_d) \mathbf{Q}_{2,m,0}^{(k)\top}, \\ I_3 &:= \mathbf{Q}_{2,m,e}^{(k)} (T^{-1} \mathbf{Z}_\epsilon^{(k)} \mathcal{A}_{\epsilon,T} \mathbf{M}_T \mathcal{A}_{\epsilon,T}^\top \mathbf{Z}_\epsilon^{(k)\top} - \mathbf{I}_d) \mathbf{Q}_{2,m,e}^{(k)\top}. \end{aligned}$$

Firstly,

$$\|I_0\| = \|\mathbf{Q}_{2,m}^{(k)}\|^2 \leq \max_{j \in [d_k]} \|\boldsymbol{\Sigma}_{\epsilon,j}^{(k)}\| = O(1).$$

Using the same notations as in the treatment of  $\mathbf{S}_{12,m}$  before within this proof, for  $\ell, h \in [d_k]$ , the  $(\ell, h)$  entry of  $I_1$  is given by

$$(I_1)_{\ell,h} = T^{-1} \left( \sum_{j=1}^{d_k} (\mathbf{U}_{-k,(1)})_j \mathbf{a}_{\ell,j}^\top \mathbf{Z}_j \right) \mathcal{A}_{\epsilon,T} \mathbf{M}_T \mathcal{A}_{\epsilon,T}^\top \left( \sum_{j=1}^{d_k} (\mathbf{U}_{-k,(1)})_j \mathbf{a}_{h,j}^\top \mathbf{Z}_j \right)^\top - \sum_{j=1}^{d_k} (\mathbf{U}_{-k,(1)})_j^2 \mathbf{a}_{\ell,j}^\top \mathbf{a}_{h,j}.$$

Hence by Assumption (RE1) and (E1), writing  $\mathbf{G} := \mathcal{A}_{\epsilon,T} \mathbf{M}_T \mathcal{A}_{\epsilon,T}^\top$ ,

$$E(I_1)_{\ell,h} = \sum_{j=1}^{d_k} (\mathbf{U}_{-k,(1)})_j^2 \mathbf{a}_{\ell,j}^\top \mathbf{a}_{h,j} \{T^{-1} \text{tr}(\mathbf{G}) - 1\} = O(T^{-1/2}).$$

Also,

$$\begin{aligned}
E(I_1)_{\ell,h}^2 &= T^{-2} \sum_{i,j=1}^{d_k} (\mathbf{U}_{-k,(1)})_i^2 (\mathbf{U}_{-k,(1)})_j^2 E\{(\mathbf{a}_{\ell,i}^\top \mathbf{Z}_i \mathbf{G} \mathbf{Z}_i^\top \mathbf{a}_{h,i})(\mathbf{a}_{\ell,j}^\top \mathbf{Z}_j \mathbf{G} \mathbf{Z}_j^\top \mathbf{a}_{h,j})\} \\
&\quad + T^{-2} \sum_{i \neq j} (\mathbf{U}_{-k,(1)})_i^2 (\mathbf{U}_{-k,(1)})_j^2 \{E(\mathbf{a}_{\ell,i}^\top \mathbf{Z}_i \mathbf{G} \mathbf{Z}_j^\top \mathbf{a}_{h,j})^2 + E\{\mathbf{a}_{\ell,i}^\top \mathbf{Z}_i \mathbf{G} \mathbf{Z}_j^\top \mathbf{a}_{h,j} \mathbf{a}_{\ell,j}^\top \mathbf{Z}_j \mathbf{G} \mathbf{Z}_i^\top \mathbf{a}_{h,i}\}\} \\
&\quad - 2T^{-1} \text{tr}(\mathbf{G}) \left( \sum_{j=1}^{d_k} (\mathbf{U}_{-k,(1)})_j^2 \mathbf{a}_{\ell,j}^\top \mathbf{a}_{h,j} \right)^2 + \left( \sum_{j=1}^{d_k} (\mathbf{U}_{-k,(1)})_j^2 \mathbf{a}_{\ell,j}^\top \mathbf{a}_{h,j} \right)^2 \\
&= T^{-2} \sum_{i=1}^{d_k} (\mathbf{U}_{-k,(1)})_i^4 E(\mathbf{a}_{\ell,i}^\top \mathbf{Z}_i \mathbf{G} \mathbf{Z}_i^\top \mathbf{a}_{h,i})^2 + T^{-2} \sum_{i \neq j} (\mathbf{U}_{-k,(1)})_i^2 (\mathbf{U}_{-k,(1)})_j^2 (\mathbf{a}_{\ell,i}^\top \mathbf{a}_{h,i})(\mathbf{a}_{\ell,j}^\top \mathbf{a}_{h,j}) \text{tr}^2(\mathbf{G}) \\
&\quad + T^{-2} \sum_{i \neq j} (\mathbf{U}_{-k,(1)})_i^2 (\mathbf{U}_{-k,(1)})_j^2 (\|\mathbf{a}_{\ell,i}\|^2 \|\mathbf{a}_{h,j}\|^2 + (\mathbf{a}_{\ell,i}^\top \mathbf{a}_{h,i})(\mathbf{a}_{\ell,j}^\top \mathbf{a}_{h,j})) \text{tr}(\mathbf{G}^2) \\
&\quad - (2T^{-1} \text{tr}(\mathbf{G}) - 1) \left( \sum_{j=1}^{d_k} (\mathbf{U}_{-k,(1)})_j^2 \mathbf{a}_{\ell,j}^\top \mathbf{a}_{h,j} \right)^2. \tag{S.51}
\end{aligned}$$

Now define  $\mathbf{w}_\ell := \mathbf{Z}_i^\top \mathbf{a}_{\ell,i}$ . Then  $E(\mathbf{w}_\ell) = \mathbf{0}$  and  $\text{cov}(\mathbf{w}_\ell, \mathbf{w}_h) = \mathbf{a}_{\ell,i}^\top \mathbf{a}_{h,i} \mathbf{I}_{(M+1)T}$ , with elements in  $\mathbf{w}_\ell$  independent of each other for each  $\ell \in [d_k]$  and  $i \in [d_k]$ . Hence

$$\begin{aligned}
E(\mathbf{a}_{\ell,i}^\top \mathbf{Z}_i \mathbf{G} \mathbf{Z}_i^\top \mathbf{a}_{h,i})^2 &= E(\mathbf{w}_\ell^\top \mathbf{G} \mathbf{w}_h) = E\left( \sum_{j=1}^{(N+1)T} (\mathbf{G})_{jj} (\mathbf{w}_\ell)_j (\mathbf{w}_h)_j + \sum_{j_1 \neq j_2} (\mathbf{G})_{j_1 j_2} (\mathbf{w}_\ell)_{j_1} (\mathbf{w}_h)_{j_2} \right)^2 \\
&= \sum_{j=1}^{(N+1)T} (\mathbf{G})_{jj}^2 E[(\mathbf{w}_\ell)_j^2 (\mathbf{w}_h)_j^2] + \sum_{j_1 \neq j_2} (\mathbf{G})_{j_1 j_1} (\mathbf{G})_{j_2 j_2} (\mathbf{a}_{\ell,i}^\top \mathbf{a}_{h,i})^2 \\
&\quad + \sum_{j_1 \neq j_2} (\mathbf{G})_{j_1 j_2}^2 (\|\mathbf{a}_{\ell,i}\|^2 \|\mathbf{a}_{h,i}\|^2 + (\mathbf{a}_{\ell,i}^\top \mathbf{a}_{h,i})^2) \\
&= \sum_{j=1}^{(N+1)T} (\mathbf{G})_{jj}^2 \left\{ \|\mathbf{a}_{h,i}\|^2 \|\mathbf{a}_{\ell,i}\|^2 + (\nu_4 - 3) \mathbf{a}_{h,i}^\top \text{diag}(\mathbf{a}_{\ell,i} \mathbf{a}_{\ell,i}^\top) \mathbf{a}_{h,i} + 2(\mathbf{a}_{\ell,i}^\top \mathbf{a}_{h,i})^2 \right\} \\
&\quad + \sum_{j_1 \neq j_2} (\mathbf{G})_{j_1 j_1} (\mathbf{G})_{j_2 j_2} (\mathbf{a}_{\ell,i}^\top \mathbf{a}_{h,i})^2 + \sum_{j_1 \neq j_2} (\mathbf{G})_{j_1 j_2}^2 (\|\mathbf{a}_{\ell,i}\|^2 \|\mathbf{a}_{h,i}\|^2 + (\mathbf{a}_{\ell,i}^\top \mathbf{a}_{h,i})^2) \\
&= \|\mathbf{a}_{h,i}\|^2 \|\mathbf{a}_{\ell,i}\|^2 \text{tr}(\mathbf{G}^2) + (\mathbf{a}_{\ell,i}^\top \mathbf{a}_{h,i})^2 (\text{tr}^2(\mathbf{G}) + \text{tr}(\mathbf{G}^2)) \\
&\quad + (\nu_4 - 3) \text{tr}(\text{diag}^2(\mathbf{G})) \mathbf{a}_{h,i}^\top \text{diag}(\mathbf{a}_{\ell,i} \mathbf{a}_{\ell,i}^\top) \mathbf{a}_{h,i}, \tag{S.52}
\end{aligned}$$

where  $\nu_4 := E(\mathbf{Z}_i)_{11}^4 < \infty$  by Assumption (R1), and we used Lemma (A.2) of Li et al. (2019). Substitute this back to (S.51), we have  $E(I_1)_{\ell,h}^2 = H_1 + H_2 + H_3 + H_4$ , where

$$\begin{aligned}
H_1 &:= (T^{-2} \text{tr}^2(\mathbf{G}) - 2T^{-1} \text{tr}(\mathbf{G}) + 1) \left( \sum_{j=1}^{d_k} (\mathbf{U}_{-k,(1)})_j^2 \mathbf{a}_{\ell,j}^\top \mathbf{a}_{h,j} \right)^2, \\
H_2 &:= T^{-2} \text{tr}(\mathbf{G}^2) \left( \sum_{j=1}^{d_k} (\mathbf{U}_{-k,(1)})_j^2 \mathbf{a}_{\ell,j}^\top \mathbf{a}_{h,j} \right)^2, \\
H_3 &:= T^{-2} \text{tr}(\mathbf{G}^2) \left( \sum_{j=1}^{d_k} (\mathbf{U}_{-k,(1)})_j^2 \mathbf{a}_{\ell,j}^\top \mathbf{a}_{h,j} \right) \left( \sum_{j=1}^{d_k} (\mathbf{U}_{-k,(1)})_j^2 \mathbf{a}_{h,j}^\top \mathbf{a}_{\ell,j} \right), \\
H_4 &:= T^{-2} \text{tr}(\text{diag}^2(\mathbf{G})) \sum_{j=1}^{d_k} (\mathbf{U}_{-k,(1)})_j^4 (\nu_4 - 3) \mathbf{a}_{h,j}^\top \text{diag}(\mathbf{a}_{\ell,j} \mathbf{a}_{\ell,j}^\top) \mathbf{a}_{h,j}.
\end{aligned}$$

By Assumption (RE1), we have

$$\begin{aligned} |H_1|, |H_2|, |H_3| &= O(T^{-1}) \cdot O\left(\max_{j \in [d_k]} \|\Sigma_{\epsilon, j}^{(k)}\|_{\max}^2\right) = O(T^{-1}), \\ |H_4| &\leq T^{-2} \text{tr}(\mathbf{G}^2)(\nu_4 - 3) \max_{j \in [d_k]} \|\Sigma_{\epsilon, j}^{(k)}\|_{\max}^2 = O(T^{-1}). \end{aligned}$$

Hence we can conclude that  $\|I_1\| = O_P(d_k T^{-1/2})$ .

For  $I_2$ , define  $\hat{e}_j := (\check{\mathbf{q}}_k^{(m)} - \mathbf{U}_{-k, (1)})_j$ . Then for  $\ell, h \in [d_k]$ , the  $(\ell, h)$  entry of  $I_2$  is given by

$$(I_2)_{\ell, h} = \sum_{j_1=1}^{d_k} \hat{e}_{j_1} \left( T^{-1} \mathbf{a}_{\ell, j_1}^T \mathbf{Z}_{j_1} \mathbf{G} \sum_{j_2=1}^{d_k} (\mathbf{U}_{-k, (1)})_{j_2} \mathbf{Z}_{j_2}^T \mathbf{a}_{h, j_2} - (\mathbf{U}_{-k, (1)})_{j_1} \mathbf{a}_{\ell, j_1}^T \mathbf{a}_{h, j_1} \right) =: \sum_{j_1=1}^{d_k} \hat{e}_{j_1} g_{j_1, \ell, h}.$$

By Assumption (RE1) and (E1),

$$E(g_{j_1, h, \ell}) = (T^{-1} \text{tr}(\mathbf{G}) - 1) (\mathbf{U}_{-k, (1)})_{j_1} \mathbf{a}_{\ell, j_1}^T \mathbf{a}_{h, j_1} = O(T^{-1/2}).$$

Also, similar to the treatment of  $I_1$  and using (S.52),

$$\begin{aligned} E(g_{j_1, h, \ell}^2) &= T^{-2} \sum_{j=1}^{d_k} (\mathbf{U}_{-k, (1)})_j^2 E(\mathbf{a}_{\ell, j_1}^T \mathbf{Z}_{j_1} \mathbf{G} \mathbf{Z}_{j_1}^T \mathbf{a}_{h, j_1})^2 \\ &\quad - 2T^{-1} \text{tr}(\mathbf{G}) (\mathbf{U}_{-k, (1)})_{j_1}^2 (\mathbf{a}_{\ell, j_1}^T \mathbf{a}_{h, j_1})^2 + (\mathbf{U}_{-k, (1)})_{j_1}^2 (\mathbf{a}_{\ell, j_1}^T \mathbf{a}_{h, j_1})^2 \\ &= \sum_{j=1}^{d_k} (\mathbf{U}_{-k, (1)})_j^2 T^{-2} \text{tr}(\mathbf{G}^2) \|\mathbf{a}_{h, j}\|^2 \|\mathbf{a}_{\ell, j_1}\|^2 \\ &\quad + (T^{-1} \text{tr}(\mathbf{G}) - 1)^2 (\mathbf{U}_{-k, (1)})_{j_1}^2 (\mathbf{a}_{\ell, j_1}^T \mathbf{a}_{h, j_1})^2 + T^{-2} \text{tr}(\mathbf{G}^2) (\mathbf{U}_{-k, (1)})_{j_1}^2 (\mathbf{a}_{\ell, j_1}^T \mathbf{a}_{h, j_1})^2 \\ &\quad + T^{-2} (\mathbf{U}_{-k, (1)})_{j_1}^2 (\nu_4 - 3) \text{tr}(\text{diag}^2(\mathbf{G})) \mathbf{a}_{h, j_1}^T \text{diag}(\mathbf{a}_{\ell, j_1} \mathbf{a}_{\ell, j_1}^T) \mathbf{a}_{h, j_1} = O(T^{-1}), \end{aligned}$$

so that we can conclude that  $\|I_2\| = O_P\left(d_k \|\check{\mathbf{q}}_k^{(m)} - \mathbf{U}_{-k, (1)}\| \sqrt{\frac{d_k}{T}}\right) = O_P\left(d_k^{1/2} \|\check{\mathbf{q}}_k^{(m)} - \mathbf{U}_{-k, (1)}\| \sqrt{\frac{d}{T}}\right)$ .

For  $I_3$ , using Theorem 2 of Lataa (2004),

$$\begin{aligned} \|I_3\| &\leq \|\mathbf{Q}_{2, m, \epsilon}^{(k)}\|^2 \|T^{-1} \mathbf{Z}_{\epsilon}^{(k)} \mathbf{G} \mathbf{Z}_{\epsilon}^{(k)T} - \mathbf{I}_d\| \leq \|\check{\mathbf{q}}_k^{(m)} - \mathbf{U}_{-k, (1)}\|^2 \max_{j \in [d_k]} \|\Sigma_{\epsilon, j}^{(k)}\| \cdot O_P(1 + d/T) \\ &= O_P(\|\check{\mathbf{q}}_k^{(m)} - \mathbf{U}_{-k, (1)}\|^2 (1 + d/T)). \end{aligned}$$

Hence

$$\begin{aligned} \|\mathbf{S}_{22, m}\| &= \|I_0\| + \|I_1\| + 2\|I_2\| + \|I_3\| \\ &= O_P\left(1 + \frac{d_k}{\sqrt{T}} + \|\check{\mathbf{q}}_k^{(m)} - \mathbf{U}_{-k, (1)}\| \sqrt{\frac{d_k d}{T}} + \|\check{\mathbf{q}}_k^{(m)} - \mathbf{U}_{-k, (1)}\|^2 \left(\frac{d}{T} + 1\right)\right) \\ &= O_P\left(1 + \|\check{\mathbf{q}}_k^{(m)} - \mathbf{U}_{-k, (1)}\|^2 \frac{d}{T} + \frac{d_k}{\sqrt{T}} + \|\check{\mathbf{q}}_k^{(m)} - \mathbf{U}_{-k, (1)}\| \sqrt{\frac{d_k d}{T}}\right). \end{aligned}$$

For  $\mathbf{S}_{23, m}$ , the treatment is exactly parallel to the treatment of  $\mathbf{S}_{12, m}$ , so that

$$\begin{aligned} \|\mathbf{S}_{23, m}\| &= \|\mathbf{Q}_{3, m}^{(k)}\| \cdot \sqrt{r_e d_k} \cdot O_P(J_1 + J_2) \\ &= O_P\left\{\sqrt{\frac{r_e d_k}{T}} + \|\check{\mathbf{q}}_k^{(m)} - \mathbf{U}_{-k, (1)}\| \sqrt{\frac{r_e d}{T}}\right\}. \end{aligned}$$

Finally, by Assumption (RE1) and Theorem 2 of Lataa (2004) and that  $r_e = o(T)$ ,

$$\|\mathbf{S}_{33, m}\| \leq \|\mathbf{Q}_{3, m}^{(k)}\|^2 O_P\left(1 + \sqrt{\frac{r_e}{T}}\right) = O_P(1).$$

This completes the proof of the lemma.  $\square$

*Proof of Theorem 4.1.* Firstly, (S.6) in Lemma 2 proves that  $\mathbf{G}_k$  has the  $j$ -th diagonal element with order  $d_k^{\alpha_{k,j}}$  for  $j \in [r_k]$  under Assumption (L1), which is the same as those in  $\mathbf{D}_k$ . Define  $\mathbf{B}_k \in \mathbb{R}^{d_k \times d_k - r_k}$  to be an orthogonal compliment of  $\mathbf{U}_k = (\mathbf{U}_{k,(1)}, \mathbf{U}_{k,(2:r_k)})$ , in the sense that  $\mathbf{V}_k := (\mathbf{U}_k, \mathbf{B}_k)$  has  $\mathbf{V}_k \mathbf{V}_k^T = \mathbf{V}_k^T \mathbf{V}_k = \mathbf{I}_{d_k}$ . Then using the SVD of  $\mathbf{A}_k$  in (3.9),

$$\mathbf{V}_k^T \mathbf{S}'_{11,m} \mathbf{V}_k = \begin{pmatrix} \check{g}_{-k}^{(m)} \mathbf{G}_k & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix},$$

with the  $(j, j)$  element of  $\check{g}_{-k}^{(m)} \mathbf{G}_k$  of order  $\check{g}_{-k}^{(m)} d_k^{\alpha_{k,j}}$ ,  $j \in [r_k]$ . By Assumption (L1'), we then have

$$\text{sep}(\mathbf{U}_{k,(1)}^T \mathbf{S}'_{11,m} \mathbf{U}_{k,(1)}, (\mathbf{U}_{k,(2:r_k)}, \mathbf{B}_k)^T \mathbf{S}'_{11,m} (\mathbf{U}_{k,(2:r_k)}, \mathbf{B}_k)) \asymp \check{g}_{-k}^{(m)} d_k^{\alpha_{k,1}},$$

where

$$\text{sep}(\mathbf{D}_1, \mathbf{D}_2) := \min_{\lambda \in \lambda(\mathbf{D}_1), \mu \in \lambda(\mathbf{D}_2)} |\lambda - \mu|.$$

Then by Lemma 3 of Lam et al. (2011), which is Theorem 8.1.10 in Golub and Van Loan (1996), since  $\mathbf{U}_{k,(1)}$  is an eigenvector of  $\mathbf{S}'_{11,m}$  and hence the span of  $\mathbf{U}_{k,(1)}$  is an invariant subspace for  $\mathbf{S}'_{11,m}$ , for the matrix  $\mathbf{E}_m$  in (S.50), if

$$\|\mathbf{E}_m\| = o_P(\check{g}_{-k}^{(m)} d_k^{\alpha_{k,1}}),$$

then there exists  $\check{\mathbf{q}}_k^{(m+1)}$  which is an eigenvector of  $\tilde{\Sigma}_{y,m+1}^{(k)}$  such that

$$\|\check{\mathbf{q}}_k^{(m+1)} - \mathbf{U}_{k,(1)}\| = O(\|\mathbf{E}_m\| / (\check{g}_{-k}^{(m)} d_k^{\alpha_{k,1}})). \quad (\text{S.53})$$

We prove by induction on the integer  $m$  for the following statements:

I(m):  $\check{g}_{-k}^{(m)} d_k^{\alpha_{k,1}} \asymp_P g_s$  for each  $k \in [K]$ .

II(m): For each  $k \in [K]$ ,

$$\|\check{\mathbf{q}}_k^{(m+1)} - \mathbf{U}_{k,(1)}\| = O_P \left\{ \sqrt{\frac{r}{T}} + g_s^{-1/2} \|\check{\mathbf{q}}_k^{(m)} - \mathbf{U}_{k,(1)}\| \sqrt{\frac{rd}{T}} \right\} = o_P(1).$$

The statement I(0) is proved exactly by Lemma 5. For statement II(0), using Lemma 6 and Lemma 7 at  $m = 0$  and combine with the result that  $\check{g}_{-k}^{(0)} = \hat{g}_{-k} \asymp_P g_s / d_k^{\alpha_{k,1}}$  from I(0), we have from (S.50) that

$$\begin{aligned} \|\mathbf{E}_0\| &= O_P \left( \|\mathbf{S}''_{11,0}\| + \|\mathbf{S}_{22,0}\| + \|\mathbf{S}_{33,0}\| + \sum_{i < j=1}^3 (\|\mathbf{S}_{ij,0}\| + \|\mathbf{S}_{ij,0}^T\|) + \sum_{i,j=1}^3 \|\check{\mathbf{S}}_{ij,0}\| \right) \\ &= O_P \left( g_s \sqrt{\frac{r}{T}} + g_s^{1/2} \|\check{\mathbf{q}}_k^{(0)} - \mathbf{U}_{k,(1)}\| \sqrt{\frac{rd}{T}} + \|\check{\mathbf{q}}_k^{(0)} - \mathbf{U}_{k,(1)}\|^2 \frac{d}{T} \right) \\ &= O_P \left( g_s \sqrt{\frac{r}{T}} + g_s^{1/2} b_k \sqrt{\frac{rd}{T}} + b_k^2 \frac{d}{T} \right), \end{aligned}$$

where the dominating rates in the second line are the results of the explicit rate assumptions presented at the beginning of Theorem 4.1, and the last line used (S.48) and the notation  $b_k$  used in Theorem 4.1. Hence from (S.53),

$$\begin{aligned} \|\check{\mathbf{q}}_k^{(1)} - \mathbf{U}_{k,(1)}\| &= O_P(\|\mathbf{E}_0\| / (\check{g}_{-k}^{(0)} d_k^{\alpha_{k,1}})) = O_P(\|\mathbf{E}_0\| / g_s) \\ &= O_P \left( \sqrt{\frac{r}{T}} + g_s^{-1/2} \|\check{\mathbf{q}}_k^{(0)} - \mathbf{U}_{k,(1)}\| \sqrt{\frac{rd}{T}} + g_s^{-1} \|\check{\mathbf{q}}_k^{(0)} - \mathbf{U}_{k,(1)}\|^2 \frac{d}{T} \right) \\ &= O_P \left( \sqrt{\frac{r}{T}} + g_s^{-1/2} b_k \sqrt{\frac{rd}{T}} + g_s^{-1} b_k^2 \frac{d}{T} \right) = O_P \left( \sqrt{\frac{r}{T}} + g_s^{-1/2} b_k \sqrt{\frac{rd}{T}} \right) = o_P(1), \end{aligned}$$

which is statement II(0), and the first statement of Theorem 4.1 is proved. Hence both I( $m$ ) and II( $m$ ) are true at  $m = 0$ .

Assume that both statements are true at a non-negative integer  $m$ . Then with II( $m$ ), like (S.47),

$$\begin{aligned}\check{g}_{-k}^{(m+1)} &= \check{\mathbf{q}}_k^{(m+1)\top} \mathbf{A}_{-k} \mathbf{A}_{-k}^\top \check{\mathbf{q}}_k^{(m+1)} \\ &\geq \mathbf{U}_{-k,(1)}^\top \mathbf{U}_{-k} \mathbf{G}_{-k} \mathbf{U}_{-k}^\top \mathbf{U}_{-k,(1)} + 2(\check{\mathbf{q}}_k^{(m+1)} - \mathbf{U}_{-k,(1)})^\top \mathbf{U}_{-k} \mathbf{G}_{-k} \mathbf{U}_{-k}^\top \mathbf{U}_{-k,(1)} \\ &= (1 + o_P(1))(\mathbf{G}_{-k})_{11} \asymp_P \prod_{j=1; j \neq k}^K d_j^{\alpha_{j,1}} = g_s d_k^{-\alpha_{k,1}},\end{aligned}$$

which proves statement I( $m+1$ ). With this, then using the definitions in (S.49), we have

$$\|\mathbf{Q}_{1,m+1}^{(k)}\| = (\check{\mathbf{q}}_k^{(m+1)\top} \mathbf{A}_{-k} \mathbf{A}_{-k}^\top \check{\mathbf{q}}_k^{(m+1)}) \|\mathbf{A}_k\|^2 = \check{g}_{-k}^{(m+1)} \cdot d_k^{\alpha_{k,1}} = O_P(g_s), \quad \|\mathbf{Q}_{2,m+1}^{(k)}\| = O(1) = \|\mathbf{Q}_{3,m+1}^{(k)}\|.$$

Then the rest of the proofs of Lemma 6 follow through and we can conclude that

$$\sum_{i < j=1}^3 \|\check{\mathbf{S}}_{ij,m+1}\| = o_P(T^{-1}).$$

With the above, and Lemma 7 at  $m+1$ , we have from (S.50) that

$$\begin{aligned}\|\mathbf{E}_{m+1}\| &= O_P\left(\|\mathbf{S}_{11,m+1}''\| + \|\mathbf{S}_{22,m+1}\| + \|\mathbf{S}_{33,m+1}\| + \sum_{i < j=1}^3 (\|\mathbf{S}_{ij,m+1}\| + \|\mathbf{S}_{ij,m+1}^\top\|) + \sum_{i,j=1}^3 \|\check{\mathbf{S}}_{ij,m+1}\|\right) \\ &= O_P\left(g_s \sqrt{\frac{r}{T}} + g_s^{1/2} \|\check{\mathbf{q}}_k^{(m+1)} - \mathbf{U}_{-k,(1)}\| \sqrt{\frac{rd}{T}} + \|\check{\mathbf{q}}_k^{(m+1)} - \mathbf{U}_{-k,(1)}\|^2 \frac{d}{T}\right),\end{aligned}$$

so that by (S.53) and the proved I( $m+1$ ),

$$\begin{aligned}\|\check{\mathbf{q}}_k^{(m+2)} - \mathbf{U}_{k,(1)}\| &= O_P(\|\mathbf{E}_{m+1}\| / (\check{g}_{-k}^{(m+1)} d_k^{\alpha_{k,1}})) = O_P(\|\mathbf{E}_{m+1}\| / g_s) \\ &= O_P\left(\sqrt{\frac{r}{T}} + g_s^{-1/2} \|\check{\mathbf{q}}_k^{(m+1)} - \mathbf{U}_{-k,(1)}\| \sqrt{\frac{rd}{T}} + g_s^{-1} \|\check{\mathbf{q}}_k^{(m+1)} - \mathbf{U}_{-k,(1)}\|^2 \frac{d}{T}\right) \\ &= O_P\left(\sqrt{\frac{r}{T}} + g_s^{-1/2} \|\check{\mathbf{q}}_k^{(m+1)} - \mathbf{U}_{-k,(1)}\| \sqrt{\frac{rd}{T}}\right),\end{aligned}$$

which is  $o_P(1)$  since  $\|\check{\mathbf{q}}_k^{(m+1)} - \mathbf{U}_{-k,(1)}\| = o_P(1)$  by assumption II( $m$ ), and that  $rdg_s^{-1} = o(T)$ . This proves statement II( $m+1$ ), and hence we have proved the statements I( $m$ ) and II( $m$ ) for any non-negative integers  $m$  by induction.

To prove the second part of Theorem 4.1, define  $e_{m,k} := \|\check{\mathbf{q}}_k^{(m)} - \mathbf{U}_{k,(1)}\|$  and  $e_{m,-k} := \|\check{\mathbf{q}}_k^{(m)} - \mathbf{U}_{-k,(1)}\|$ . We have from an argument similar to the one in (S.48) that

$$e_{m,-k} = O_P\left(\sum_{j=1; j \neq k}^K e_{m,j}\right). \quad (\text{S.54})$$

We see from statement II( $m$ ) that the coefficient rate of  $\|\check{\mathbf{q}}_k^{(m)} - \mathbf{U}_{-k,(1)}\|$  is, for each  $k \in [K]$ ,  $u_k := g_s^{-1/2} \sqrt{rd/T}$ , which is  $o(1)$  by the assumption  $rdg_s^{-1} = o(T)$  in Theorem 4.1. The statement II( $m+n-1$ ) then implies that

$$e_{m+n,k} = O_P\left(\sqrt{\frac{r}{T}} + u_k e_{m+n-1,-k}\right) = O_P\left(\sqrt{\frac{r}{T}} + u_k \sum_{j=1; j \neq k}^K e_{m+n-1,j}\right).$$

Defining  $\mathbf{e}_m := (e_{m,1}, \dots, e_{m,K})^\top$ , the above becomes

$$\mathbf{e}_{m+n} = O_P\left(\sqrt{\frac{r}{T}}\mathbf{1}_K + \mathbf{W}\mathbf{e}_{m+n-1}\right), \quad \text{where } \mathbf{W} := \begin{pmatrix} 0 & u_1 & u_1 & \cdots & u_1 \\ u_2 & 0 & u_2 & \cdots & u_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ u_K & u_K & u_K & \cdots & 0 \end{pmatrix},$$

with  $\|\mathbf{W}\|_\infty \leq K \max_{k \in [K]} u_k = o(1)$  since  $K$  is assumed finite. Hence iterating the above  $n-1$  more times, we have

$$\begin{aligned} \mathbf{e}_{m+n} &= O_P\left(\sqrt{\frac{r}{T}}(\mathbf{I}_K + \mathbf{W} + \cdots + \mathbf{W}^{n-1})\mathbf{1}_K + \mathbf{W}^n\mathbf{e}_m\right), \quad \text{implying} \\ \|\mathbf{e}_{m+n}\|_{\max} &= O_P\left(\sqrt{\frac{r}{T}}(1 + \|\mathbf{W}\|_\infty + \cdots + \|\mathbf{W}\|_\infty^{n-1}) + \|\mathbf{W}\|_\infty^n \|\mathbf{e}_m\|_{\max}\right) \\ &= O_P\left(\sqrt{\frac{r}{T}} + \|\mathbf{W}\|_\infty^n \|\mathbf{e}_m\|_{\max}\right) = O_P\left(\sqrt{\frac{r}{T}}\right) \end{aligned}$$

for  $n$  large enough. This completes the proof of the theorem.  $\square$

*Proof of Theorem 4.2.* Using the notations in (S.45) and in Lemma 7, since  $\mathbf{S}_{11,m}$  is sandwiched by  $\mathbf{A}_k$  and  $\mathbf{A}_k^\top$ , implying that it is sandwiched by  $\mathbf{U}_k$  and  $\mathbf{U}_k^\top$ , the span of the columns of  $\mathbf{U}_k$  forms an invariant subspace for  $\mathbf{S}_{11,m}$ . Now Let  $\mathbf{B}_k$  be the orthogonal complement of  $\mathbf{U}_k$ , in the sense that  $\mathbf{U} := (\mathbf{U}_k, \mathbf{B}_k)$  is an orthogonal matrix. Then we have

$$\begin{aligned} \mathbf{U}^\top \mathbf{S}_{11,m} \mathbf{U} &= \begin{pmatrix} \mathbf{U}_k^\top \mathbf{S}_{11,m} \mathbf{U}_k & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}, \quad \text{where } \text{sep}(\mathbf{U}_k^\top \mathbf{S}_{11,m} \mathbf{U}_k, \mathbf{0}) = \lambda_{r_k}(\mathbf{U}_k^\top \mathbf{S}_{11,m} \mathbf{U}_k), \quad \text{and} \\ \lambda_{r_k}(\mathbf{U}_k^\top \mathbf{S}_{11,m} \mathbf{U}_k) &= \lambda_{r_k}(\mathbf{V}_k \mathbf{G}_k \mathbf{V}_k^\top (\mathbf{I}_{r_k} \otimes \check{\mathbf{q}}_k^{(m)\top} \mathbf{A}_{-k}) T^{-1} \mathbf{Z}_f^{(k)} \mathcal{A}_{f,T} \mathbf{M}_T \mathcal{A}_{f,T}^\top \mathbf{Z}_f^{(k)\top} (\mathbf{I}_{r_k} \otimes \check{\mathbf{q}}_k^{(m)\top} \mathbf{A}_{-k})^\top) \\ &\geq \lambda_{r_k}(\mathbf{G}_k) \lambda_{r_k}(\check{g}_{-k}^{(m)} \mathbf{I}_{r_k}) \lambda_r(T^{-1} \mathbf{Z}_f^{(k)} \mathcal{A}_{f,T} \mathbf{M}_T \mathcal{A}_{f,T}^\top \mathbf{Z}_f^{(k)\top}) \asymp_P (\check{g}_{-k}^{(m)} d_k^{\alpha_k, r_k}), \end{aligned}$$

where we used Theorem 2.8 of Wang and Paul (2014) (applicable since  $r_k = o(T^{1/3})$  by Assumption (L1) and all variables are of bounded fourth moments by Assumption (R1)) to conclude that

$$\lambda_r(T^{-1} \mathbf{Z}_f^{(k)} \mathcal{A}_{f,T} \mathbf{M}_T \mathcal{A}_{f,T}^\top \mathbf{Z}_f^{(k)\top}) \asymp_P (N+1) \left( \frac{\text{tr}(\mathcal{A}_{f,T} \mathbf{M}_T \mathcal{A}_{f,T}^\top)}{(N+1)T} + \sqrt{\frac{r}{(N+1)T}} \right) = 1 + O(T^{-1/2}), \quad (\text{S.55})$$

and the last equality used Assumption (RE1). Hence Lemma 3 of Lam et al. (2011) implies that there exists  $\check{\mathbf{U}}_k$  with  $\check{\mathbf{U}}_k^\top \check{\mathbf{U}}_k = \mathbf{I}_{r_k}$  and

$$\|\check{\mathbf{U}}_k - \mathbf{U}_k\| = O_P\left\{\left(\|\mathbf{S}_{12,m}\| + \|\mathbf{S}_{13,m}\| + \|\mathbf{S}_{22,m}\| + \|\mathbf{S}_{23,m}\| + \|\mathbf{S}_{33,m}\| + \sum_{i,j=1}^3 \|\check{\mathbf{S}}_{ij,m}\|\right) / \text{sep}(\mathbf{U}_k^\top \mathbf{S}_{11,m} \mathbf{U}_k, \mathbf{0})\right\},$$

such that  $\check{\mathbf{U}}_k$  equals the  $r_k$  eigenvectors corresponding to the first  $r_k$  largest eigenvalues of  $\widetilde{\Sigma}_{y,m+1}^{(k)}$  multiplied with an orthogonal matrix, if we can show that the above rate is  $o_P(1)$ . To this end, statement I(m) in the proof of Theorem 4.1 shows that  $\check{g}_{-k}^{(m)} \asymp_P g_s / d_k^{\alpha_k, 1}$ . And from the rate assumptions in Theorem 4.1 and the results of Lemma 7, we have

$$\begin{aligned} \|\check{\mathbf{U}}_k - \mathbf{U}_k\| &= O_P\left\{\left[1 + \sqrt{\frac{r g_s}{T}}(r_e^{1/2} + d_k^{1/2} + \|\check{\mathbf{q}}_k^{(m)} - \mathbf{U}_{-k,(1)}\| d_k^{1/2})\right] / (g_s d_k^{\alpha_k, r_k - \alpha_k, 1})\right\} \\ &= O_P\left\{d_k^{\alpha_k, 1 - \alpha_k, r_k} \left[g_s^{-1} + \sqrt{\frac{r}{T g_s}}(r_e^{1/2} + d_k^{1/2} + \|\check{\mathbf{q}}_k^{(m)} - \mathbf{U}_{-k,(1)}\| d_k^{1/2})\right]\right\} \quad (\text{S.56}) \end{aligned}$$

If  $m$  is large enough, then following Theorem 4.1, we have by (S.54) that

$$\|\check{\mathbf{q}}_k^{(m)} - \mathbf{U}_{-k,(1)}\| = O_P\left(\sum_{j=1; j \neq k}^K \|\check{\mathbf{q}}_j^{(m)} - \mathbf{U}_{j,(1)}\|\right) = O_P\left(K\sqrt{\frac{r}{T}}\right).$$

Substituting the above rate into (S.56) completes the proof.  $\square$

*Proof of Theorem 5.1.* First, it is easy to see that (RE2) implies that

$$\text{diag}(\boldsymbol{\Sigma}_{y,m+1}^{(k)}) = \check{\mathbf{q}}_k^{(m)\top} \mathbf{A}_{-k} \mathbf{A}_{-k}^\top \check{\mathbf{q}}_k^{(m)} \text{diag}(\mathbf{A}_k \mathbf{A}_k^\top)(1 + o(1)). \quad (\text{S.57})$$

Define

$$\begin{aligned} \mathbf{Qm}_1^{(k)} &:= \text{diag}^{-1/2}(\boldsymbol{\Sigma}_{y,m+1}^{(k)}) \mathbf{A}_k (\mathbf{I}_{r_k} \otimes \check{\mathbf{q}}_k^{(m)\top} \mathbf{A}_{-k}), \\ \mathbf{Qm}_2^{(k)} &:= \text{diag}^{-1/2}(\boldsymbol{\Sigma}_{y,m+1}^{(k)}) (\check{\mathbf{q}}_k^{(m)\top} \otimes \mathbf{I}_{d_k}) \text{diag}^{1/2}(\boldsymbol{\Sigma}_{\epsilon,1}^{(k)}, \dots, \boldsymbol{\Sigma}_{\epsilon,d_k}^{(k)}), \\ \mathbf{Qm}_3^{(k)} &:= \text{diag}^{-1/2}(\boldsymbol{\Sigma}_{y,m+1}^{(k)}) \mathbf{A}_{e,k} (\mathbf{I}_{r_{e,k}} \otimes \check{\mathbf{q}}_k^{(m)\top} \mathbf{A}_{e,-k}), \end{aligned} \quad (\text{S.58})$$

where  $\boldsymbol{\Sigma}_{y,m+1}^{(k)}$  is defined in (5.3). Then we have

$$\mathbf{R}_{y,m+1}^{(k)} = \sum_{j=1}^3 \mathbf{Qm}_j^{(k)} \mathbf{Qm}_j^{(k)\top}.$$

Consider

$$\begin{aligned} \text{tr}(\mathbf{Qm}_1^{(k)} \mathbf{Qm}_1^{(k)\top}) &= \text{tr}(\text{diag}^{-1/2}(\boldsymbol{\Sigma}_{y,m+1}^{(k)}) \cdot \check{g}_{-k}^{(m)} \mathbf{A}_k \mathbf{A}_k^\top \cdot \text{diag}^{-1/2}(\boldsymbol{\Sigma}_{y,m+1}^{(k)})) \\ &= \text{tr}[(\check{g}_{-k}^{(m)})^{-1/2} \text{diag}^{-1/2}(\mathbf{A}_k \mathbf{A}_k^\top)(1 + o(1))^{-1} \cdot \check{g}_{-k}^{(m)} \mathbf{A}_k \mathbf{A}_k^\top \\ &\quad \cdot (\check{g}_{-k}^{(m)})^{-1/2} \text{diag}^{-1/2}(\mathbf{A}_k \mathbf{A}_k^\top)(1 + o(1))^{-1}] \\ &= d_k(1 + o(1)), \end{aligned} \quad (\text{S.59})$$

where the second equality used (S.57). At the same time, for  $j \in [r_k]$ ,

$$\begin{aligned} \lambda_j(\mathbf{Qm}_1^{(k)} \mathbf{Qm}_1^{(k)\top}) &= \lambda_j(\text{diag}^{-1/2}(\mathbf{A}_k \mathbf{A}_k^\top)(1 + o(1)) \mathbf{A}_k \mathbf{A}_k^\top \text{diag}^{-1/2}(\mathbf{A}_k \mathbf{A}_k^\top)(1 + o(1))) \\ &= \lambda_j(\mathbf{U}_k^\top \text{diag}^{-1}(\mathbf{A}_k \mathbf{A}_k^\top) \mathbf{U}_k \mathbf{G}_k(1 + o(1))), \quad \text{with} \\ \lambda_j(\mathbf{G}_k) \lambda_{\min}(\text{diag}^{-1}(\mathbf{A}_k \mathbf{A}_k^\top)(1 + o(1))) &\leq \lambda_j(\mathbf{U}_k^\top \text{diag}^{-1}(\mathbf{A}_k \mathbf{A}_k^\top) \mathbf{U}_k \mathbf{G}_k(1 + o(1))) \\ &\leq \lambda_j(\mathbf{G}_k) \lambda_{\max}(\text{diag}^{-1}(\mathbf{A}_k \mathbf{A}_k^\top)(1 + o(1))), \end{aligned}$$

so that by Assumption (RE2), there are generic constants  $c, C > 0$  such that

$$d_k^{\alpha_{k,j}} \asymp_P c \lambda_j(\mathbf{G}_k) \leq \lambda_j(\mathbf{Qm}_1^{(k)} \mathbf{Qm}_1^{(k)\top}) \leq C \lambda_j(\mathbf{G}_k) \asymp_P d_k^{\alpha_{k,j}} \quad (\text{S.60})$$

in probability, where  $\lambda_j(\mathbf{G}_k)$  being asymptotic in probability to  $d_k^{\alpha_{k,j}}$  is given by (S.6) in Lemma 2. Using (S.60), there exists a constant  $C > 0$  (generic, different from the above) so that for  $j \in [r_k]$ ,

$$\frac{\lambda_1(\mathbf{Qm}_1^{(k)} \mathbf{Qm}_1^{(k)\top})}{\lambda_j(\mathbf{Qm}_1^{(k)} \mathbf{Qm}_1^{(k)\top})} \leq C d_k^{\alpha_{k,1} - \alpha_{k,j}}.$$

in probability. Hence using (S.59), in probability, we have

$$d_k(1 + o(1)) \leq \text{tr}(\mathbf{Qm}_1^{(k)} \mathbf{Qm}_1^{(k)\top}) \leq r_k \lambda_1(\mathbf{Qm}_1^{(k)} \mathbf{Qm}_1^{(k)\top}) \leq r_k \cdot C d_k^{\alpha_{k,1} - \alpha_{k,j}} \lambda_j(\mathbf{Qm}_1^{(k)} \mathbf{Qm}_1^{(k)\top}),$$

implying that, in probability, there exists a constant  $C > 0$  such that as  $T, d_k$  are large enough,

$$\lambda_j(\mathbf{R}_{y,m+1}^{(k)}) \geq \lambda_j(\mathbf{Qm}_1^{(k)} \mathbf{Qm}_1^{(k)\top}) \geq C d_k^{1-\alpha_{k,1}+\alpha_{k,j}}/r_k > 1, \quad (\text{S.61})$$

since  $r_k = o(d_k^{1-\alpha_{k,1}+\alpha_{k,j}})$  by (RE2) for  $j \in [r_k]$ . For  $j \in [d_k]/[r_k]$ ,

$$\begin{aligned} \lambda_j(\mathbf{R}_{y,m+1}^{(k)}) &\leq \lambda_j(\mathbf{Qm}_1^{(k)} \mathbf{Qm}_1^{(k)\top}) + \lambda_1(\mathbf{Qm}_2^{(k)} \mathbf{Qm}_2^{(k)\top}) + \lambda_1(\mathbf{Qm}_3^{(k)} \mathbf{Qm}_3^{(k)\top}) \\ &\leq 0 + \|\text{diag}^{-1}(\boldsymbol{\Sigma}_{y,m+1}^{(k)})\| \left( \sum_{j=1}^{d_k} (\check{\mathbf{q}}_k^{(m)})_j^2 \|\boldsymbol{\Sigma}_{\epsilon,j}^{(k)}\| + \check{\mathbf{q}}_k^{(m)\top} \mathbf{A}_{e,-k} \mathbf{A}_{e,-k}^\top \check{\mathbf{q}}_k^{(m)} \|\mathbf{A}_{e,k}\|^2 \right) \\ &\leq (\check{g}_k^{(m)})^{-1} \|\text{diag}^{-1}(\mathbf{A}_k \mathbf{A}_k^\top)(1 + o(1))\| \cdot O(1) \\ &= O_P(d_k^{\alpha_{k,1}} g_s^{-1}) = o_P(1), \end{aligned} \quad (\text{S.62})$$

where the third inequality is in probability after using (S.57), and the last line used statement I(m) in the proof of Theorem 4.1, that  $\check{g}_k^{(m)} \asymp_P g_s d_k^{-\alpha_{k,1}}$ . This completes the proof of the theorem.  $\square$

Before proving Theorem 5.2, define

$$\mathbf{S}_{y,m+1}^{(k)} := \text{diag}^{-1/2}(\boldsymbol{\Sigma}_{y,m+1}^{(k)}) \tilde{\boldsymbol{\Sigma}}_{y,m+1}^{(k)} \text{diag}^{-1/2}(\boldsymbol{\Sigma}_{y,m+1}^{(k)}).$$

It is then easy to see that

$$\widehat{\mathbf{R}}_{y,m+1}^{(k)} := \text{diag}^{-1/2}(\mathbf{S}_{y,m+1}^{(k)}) \mathbf{S}_{y,m+1}^{(k)} \text{diag}^{-1/2}(\mathbf{S}_{y,m+1}^{(k)}).$$

We state and prove the following lemma.

**Lemma 8** *Assume all the assumptions in Theorem 5.2 hold. Then for  $k \in [K]$ ,*

$$\begin{aligned} \max_{j \in [r_k]} \left| \frac{\lambda_j(\mathbf{S}_{y,m+1}^{(k)})}{\lambda_j(\mathbf{R}_{y,m+1}^{(k)})} - 1 \right| &= O_P \left( r_k d_k^{2\alpha_{k,1}-\alpha_{k,j}-1} \sqrt{\frac{r}{T}} \left( 1 + K \sqrt{\frac{rd}{Tg_s}} + \frac{K^2 r^{1/2} d}{T^{3/2} g_s} \right) \right), \\ \max_{j \in [d_k]} \left| \frac{\lambda_j(\mathbf{S}_{y,m+1}^{(k)})}{\lambda_j(\widehat{\mathbf{R}}_{y,m+1}^{(k)})} - 1 \right| &= O_P \left( \sqrt{\frac{r}{T}} \left[ 1 + d_k^{\alpha_{k,1}/2} g_s^{-1/2} \left( r_e^{1/2} + d_k^{1/2} + K \sqrt{\frac{rd}{T}} \right) + d_k^{\alpha_{k,1}} g_s^{-1} \frac{K^2 r^{1/2} d}{T^{3/2}} \right] \right), \\ \max_{j \in [d_k]/[r_k]} |\lambda_j(\mathbf{S}_{y,m+1}^{(k)}) - \lambda_j(\mathbf{R}_{y,m+1}^{(k)})| &= O_P \left( d_k^{\alpha_{k,1}} g_s^{-1} \left\{ \sqrt{\frac{(r_e + d_k)d_k}{T}} + \frac{K \sqrt{r(r_e + d_k)d}}{T} + \frac{K^2 r d}{T^2} \right\} \right). \end{aligned}$$

*Proof of Lemma 8.* Define, using the definition of  $\mathbf{S}_{ij,m}$  defined in Lemma 7,

$$\mathbf{S}_{ij,r} := \text{diag}^{-1/2}(\boldsymbol{\Sigma}_{y,m+1}^{(k)}) \mathbf{S}_{ij,m} \text{diag}^{-1/2}(\boldsymbol{\Sigma}_{y,m+1}^{(k)}), \quad i, j = 1, 2, 3.$$

Then using (S.58), for  $j \in [r_k]$ , we can decompose

$$\left| \frac{\lambda_j(\mathbf{S}_{y,m+1}^{(k)})}{\lambda_j(\mathbf{R}_{y,m+1}^{(k)})} - 1 \right| \leq \frac{1}{\lambda_j(\mathbf{R}_{y,m+1}^{(k)})} \left( \sum_{i=1}^3 \|\mathbf{S}_{ii,r} - \mathbf{Qm}_i^{(k)} \mathbf{Qm}_i^{(k)\top}\| + 2 \sum_{i < j} \|\mathbf{S}_{ij,r}\| + \sum_{i \neq j} \|\check{\mathbf{S}}_{ij} \text{diag}^{-1}(\boldsymbol{\Sigma}_{y,m+1}^{(k)})\| \right).$$

From (S.57) and Assumption (RE2), we have that

$$\lambda_1(\text{diag}^{-1}(\boldsymbol{\Sigma}_{y,m+1}^{(k)})), \quad \lambda_{d_k}(\text{diag}^{-1}(\boldsymbol{\Sigma}_{y,m+1}^{(k)})) \asymp (\check{g}_k^{(m)})^{-1} \asymp_P d_k^{\alpha_{k,1}} g_s^{-1},$$

where the last order is from statement I(m) in the proof of Theorem 4.1, that  $\check{g}_k^{(m)} d_k^{\alpha_{k,1}} \asymp_P g_s$ . Hence coupled with the results from Lemma 7 and (S.61), and the fact that from an argument similar to (S.48) and the result of Theorem 4.1 that for large enough  $m$ ,

$$\|\check{\mathbf{q}}_k^{(m)} - \mathbf{U}_{-k,(1)}\| = O_P \left( K \sqrt{\frac{r}{T}} \right),$$



we have, using the notations in Lemma 7,

$$\begin{aligned}
\frac{\|\mathbf{S}_{11,r} - \mathbf{Q}\mathbf{m}_1^{(k)}\mathbf{Q}\mathbf{m}_1^{(k)\top}\|}{\lambda_j(\mathbf{R}_{y,m+1}^{(k)})} &\leq \frac{\|\text{diag}^{-1}(\boldsymbol{\Sigma}_{y,m+1}^{(k)})\| \|\mathbf{S}_{11,m}''\|}{\lambda_j(\mathbf{R}_{y,m+1}^{(k)})} = O_P\left(r_k d_k^{\alpha_{k,1}-\alpha_{k,j}-1} (\check{g}_{-k}^{(m)})^{-1} \cdot \check{g}_{-k}^{(m)} d_k^{\alpha_{k,1}} \sqrt{\frac{r}{T}}\right) \\
&= O_P\left(r_k d_k^{2\alpha_{k,1}-\alpha_{k,j}-1} \sqrt{\frac{r}{T}}\right), \\
\frac{\|\mathbf{S}_{22,r} - \mathbf{Q}\mathbf{m}_2^{(k)}\mathbf{Q}\mathbf{m}_2^{(k)\top}\|}{\lambda_j(\mathbf{R}_{y,m+1}^{(k)})} &\leq \frac{\|\text{diag}^{-1}(\boldsymbol{\Sigma}_{y,m+1}^{(k)})\| \|\mathbf{S}_{22,m} - \mathbf{Q}\mathbf{m}_2^{(k)}\mathbf{Q}\mathbf{m}_2^{(k)\top}\|}{\lambda_j(\mathbf{R}_{y,m+1}^{(k)})} \\
&= O_P\left(r_k d_k^{2\alpha_{k,1}-\alpha_{k,j}-1} g_s^{-1} \cdot \left\{ \frac{K^2 r d}{T^2} + \frac{d_k}{\sqrt{T}} + \frac{K \sqrt{r d_k d}}{T} \right\}\right), \\
\frac{\|\mathbf{S}_{33,r} - \mathbf{Q}\mathbf{m}_3^{(k)}\mathbf{Q}\mathbf{m}_3^{(k)\top}\|}{\lambda_j(\mathbf{R}_{y,m+1}^{(k)})} &= O_P\left(r_k d_k^{2\alpha_{k,1}-\alpha_{k,j}-1} g_s^{-1} \cdot \sqrt{\frac{r_e}{T}}\right), \\
\frac{\|\mathbf{S}_{12,r}\|}{\lambda_j(\mathbf{R}_{y,m+1}^{(k)})} &= O_P\left(r_k d_k^{2\alpha_{k,1}-\alpha_{k,j}-1} g_s^{-1/2} \left(\sqrt{\frac{r d_k}{T}} + \frac{K r \sqrt{d}}{T}\right)\right), \\
\frac{\|\mathbf{S}_{13,r}\|}{\lambda_j(\mathbf{R}_{y,m+1}^{(k)})} &= O_P\left(r_k d_k^{2\alpha_{k,1}-\alpha_{k,j}-1} g_s^{-1/2} \sqrt{\frac{r r_e}{T}}\right), \\
\frac{\|\mathbf{S}_{23,r}\|}{\lambda_j(\mathbf{R}_{y,m+1}^{(k)})} &= O_P\left(r_k d_k^{2\alpha_{k,1}-\alpha_{k,j}-1} g_s^{-1} \left(\sqrt{\frac{r_e d_k}{T}} + \frac{K \sqrt{r r_e d}}{T}\right)\right), \\
\frac{\|\check{\mathbf{S}}_{ij} \text{diag}^{-1}(\boldsymbol{\Sigma}_{y,m+1}^{(k)})\|}{\lambda_j(\mathbf{R}_{y,m+1}^{(k)})} &= O_P\left(r_k d_k^{2\alpha_{k,1}-\alpha_{k,j}-1} g_s^{-1} T^{-1}\right).
\end{aligned}$$

These implies that

$$\max_{j \in [r_k]} \left| \frac{\lambda_j(\mathbf{S}_{y,m+1}^{(k)})}{\lambda_j(\mathbf{R}_{y,m+1}^{(k)})} - 1 \right| = O_P\left(r_k d_k^{2\alpha_{k,1}-\alpha_{k,j}-1} \sqrt{\frac{r}{T}} \left(1 + K \sqrt{\frac{r d}{T g_s}} + \frac{K^2 r^{1/2} d}{T^{3/2} g_s}\right)\right),$$

which is the first statement in the lemma.

For the second statement, for  $j \in [d_k]$  and  $k \in [K]$ , defining  $\mathbf{u}_j$  to be a unit vector with the  $j$ -th position being 1 and 0 elsewhere,

$$\begin{aligned}
\left| \frac{\lambda_j(\mathbf{S}_{y,m+1}^{(k)})}{\lambda_j(\widehat{\mathbf{R}}_{y,m+1}^{(k)})} - 1 \right| &= \left| \frac{\lambda_j(\mathbf{S}_{y,m+1}^{(k)})}{\lambda_j(\mathbf{S}_{y,m+1}^{(k)} \text{diag}^{-1}(\mathbf{S}_{y,m+1}^{(k)}))} - 1 \right| \leq \max_{j \in [d_k]} |\lambda_j(\text{diag}(\mathbf{S}_{y,m+1}^{(k)})) - 1| \\
&= \max_{j \in [d_k]} |\lambda_j(\text{diag}(\mathbf{S}_{y,m+1}^{(k)} - \mathbf{R}_{y,m+1}^{(k)}))| = \max_{j \in [d_k]} |\mathbf{u}_j^\top (\mathbf{S}_{y,m+1}^{(k)} - \mathbf{R}_{y,m+1}^{(k)}) \mathbf{u}_j| \\
&\leq \max_{j \in [d_k]} \left| \mathbf{u}_j^\top \left( \mathbf{S}_{11,r}^{(k)} - \mathbf{Q}\mathbf{m}_1^{(k)}\mathbf{Q}\mathbf{m}_1^{(k)\top} \right) \mathbf{u}_j \right| + 2 \max_{j \in [d_k]} |\mathbf{u}_j^\top \mathbf{S}_{12,r}^{(k)} \mathbf{u}_j| + 2 \max_{j \in [d_k]} |\mathbf{u}_j^\top \mathbf{S}_{13,r}^{(k)} \mathbf{u}_j| \\
&\quad + \sum_{i=2}^3 |\mathbf{u}_j^\top (\mathbf{S}_{ii,r}^{(k)} - \mathbf{Q}\mathbf{m}_i^{(k)}\mathbf{Q}\mathbf{m}_i^{(k)\top}) \mathbf{u}_j| + 2 |\mathbf{u}_j^\top \mathbf{S}_{23,r}^{(k)} \mathbf{u}_j| + \sum_{i \neq j} \|\text{diag}^{-1}(\boldsymbol{\Sigma}_{y,m+1}^{(k)}) \check{\mathbf{S}}_{ij}\|.
\end{aligned}$$

But similar to the above,

$$\begin{aligned}
\max_{j \in [d_k]} |\mathbf{u}_j^\top (\mathbf{S}_{11,r} - \mathbf{Q}\mathbf{m}_1^{(k)}\mathbf{Q}\mathbf{m}_1^{(k)\top}) \mathbf{u}_j| &= O_P(\|T^{-1}\mathbf{Z}_f^{(k)}\mathcal{A}_{f,T}\mathbf{M}_T\mathcal{A}_{f,T}^\top\mathbf{Z}_f^{(k)\top} - \mathbf{I}_r\|) = O_P\left(\sqrt{\frac{r}{T}}\right), \\
\max_{j \in [d_k]} |\mathbf{u}_j^\top \mathbf{S}_{12,r}^{(k)} \mathbf{u}_j| &= O_P\left((\check{g}_{-k}^{(m)})^{-1/2} \left(\sqrt{\frac{rd_k}{T}} + \frac{Kr\sqrt{d}}{T}\right)\right), \quad \max_{j \in [d_k]} |\mathbf{u}_j^\top \mathbf{S}_{13,r} \mathbf{u}_j| = O_P\left((\check{g}_{-k}^{(m)})^{-1/2} \sqrt{\frac{rr_e}{T}}\right), \\
\max_{j \in [d_k]} |\mathbf{u}_j^\top (\mathbf{S}_{22,r} - \mathbf{Q}\mathbf{m}_2^{(k)}\mathbf{Q}\mathbf{m}_2^{(k)\top}) \mathbf{u}_j| &= O_P\left((\check{g}_{-k}^{(m)})^{-1} \left\{\frac{K^2rd}{T^2} + \frac{d_k}{\sqrt{T}} + \frac{K\sqrt{rd_kd}}{T}\right\}\right), \\
\max_{j \in [d_k]} |\mathbf{u}_j^\top \mathbf{S}_{23,r}^{(k)} \mathbf{u}_j| &= O_P\left((\check{g}_{-k}^{(m)})^{-1} \left(\sqrt{\frac{r_e d_k}{T}} + \frac{K\sqrt{rr_e d}}{T}\right)\right), \\
\max_{j \in [d_k]} |\mathbf{u}_j^\top (\mathbf{S}_{33,r} - \mathbf{Q}\mathbf{m}_3^{(k)}\mathbf{Q}\mathbf{m}_3^{(k)\top}) \mathbf{u}_j| &= O_P\left((\check{g}_{-k}^{(m)})^{-1} \sqrt{\frac{r_e}{T}}\right), \quad \|\text{diag}^{-1}(\boldsymbol{\Sigma}_{y,m+1}^{(k)})\check{\mathbf{S}}_{ij}\| = O_P((\check{g}_{-k}^{(m)})^{-1}T^{-1}).
\end{aligned}$$

Hence the above implies that

$$\begin{aligned}
\max_{j \in [d_k]} \left| \frac{\lambda_j(\mathbf{S}_{y,m+1}^{(k)})}{\lambda_j(\widehat{\mathbf{R}}_{y,m+1}^{(k)})} - 1 \right| &= O_P\left(\sqrt{\frac{r}{T}} \left[1 + (\check{g}_{-k}^{(m)})^{-1/2} \left(r_e^{1/2} + d_k^{1/2} + K\sqrt{\frac{rd}{T}}\right) + (\check{g}_{-k}^{(m)})^{-1} \frac{K^2 r^{1/2} d}{T^{3/2}}\right]\right) \\
&= O_P\left(\sqrt{\frac{r}{T}} \left[1 + d_k^{\alpha_{k,1}/2} g_s^{-1/2} \left(r_e^{1/2} + d_k^{1/2} + K\sqrt{\frac{rd}{T}}\right) + d_k^{\alpha_{k,1}} g_s^{-1} \frac{K^2 r^{1/2} d}{T^{3/2}}\right]\right),
\end{aligned}$$

which is the second statement of the lemma.

Finally, consider for  $j \in [d_k]/[r_k]$ ,

$$\begin{aligned}
|\lambda_j(\mathbf{S}_{y,m+1}^{(k)}) - \lambda_j(\mathbf{R}_{y,m+1}^{(k)})| &\leq \left| \lambda_j\left(\mathbf{S}_{11,r} + \sum_{j=2}^3 (\mathbf{S}_{1j,r} + \mathbf{S}_{j1,r} + \mathbf{Q}\mathbf{m}_j^{(k)}\mathbf{Q}\mathbf{m}_j^{(k)\top})\right) - \lambda_j(\mathbf{R}_{y,m+1}^{(k)}) \right| \\
&\quad + 2\|\mathbf{S}_{23,r}\| + \sum_{j=2}^3 \|\mathbf{S}_{jj,r} - \mathbf{Q}\mathbf{m}_j^{(k)}\mathbf{Q}\mathbf{m}_j^{(k)\top}\| + \sum_{i \neq j} \|\text{diag}^{-1}(\boldsymbol{\Sigma}_{y,m+1}^{(k)})\check{\mathbf{S}}_{ij}^{(k)}\| \\
&= 2\|\mathbf{S}_{23,r}\| + \sum_{j=2}^3 \|\mathbf{S}_{jj,r} - \mathbf{Q}\mathbf{m}_j^{(k)}\mathbf{Q}\mathbf{m}_j^{(k)\top}\| + \sum_{i \neq j} \|\text{diag}^{-1}(\boldsymbol{\Sigma}_{y,m+1}^{(k)})\check{\mathbf{S}}_{ij}^{(k)}\|,
\end{aligned} \tag{S.63}$$

if we can show that for large enough  $T, d_k$ , in probability, for  $j \in [d_k]/[r_k]$ ,

$$\lambda_j\left(\mathbf{S}_{11,r} + \sum_{j=2}^3 (\mathbf{S}_{1j,r} + \mathbf{S}_{j1,r} + \mathbf{Q}\mathbf{m}_j^{(k)}\mathbf{Q}\mathbf{m}_j^{(k)\top})\right) = \lambda_j(\mathbf{R}_{y,m+1}^{(k)}). \tag{S.64}$$

With (S.63), from the rates obtained in previous arguments, we then have

$$|\lambda_j(\mathbf{S}_{y,m+1}^{(k)}) - \lambda_j(\mathbf{R}_{y,m+1}^{(k)})| = O_P\left(d_k^{\alpha_{k,1}} g_s^{-1} \left\{ \sqrt{\frac{(r_e + d_k)d_k}{T}} + \frac{K\sqrt{r(r_e + d_k)d}}{T} + \frac{K^2 r d}{T^2} \right\}\right),$$

which is the third statement of the lemma. Hence it remains to show (S.64). To this end, define the kernel of  $\mathbf{A}_k^\top \text{diag}^{-1/2}(\boldsymbol{\Sigma}_{y,m+1}^{(k)})$  to be

$$\mathcal{N} := \{\mathbf{w} : \mathbf{A}_k^\top \text{diag}^{-1/2}(\boldsymbol{\Sigma}_{y,m+1}^{(k)})\mathbf{w} = \mathbf{0}\} \equiv \{\mathbf{w} : \mathbf{A}_k^\top \mathbf{w} = \mathbf{0}\}$$

in probability by (S.57) and assumption (RE2), and define  $\mathcal{N}^\perp$  the corresponding row space, which has  $\mathcal{N}^\perp = \text{span}(\mathbf{A}_k)$  in probability by the above argument. Hence by Assumption (L1) that  $\mathbf{A}_k$  is of full rank, the dimension of  $\mathcal{N}^\perp$  is exactly  $r_k$  (in probability), and the rank-nullity theorem implies that, in

probability, the dimension of  $\mathcal{N}$  is exactly  $d_k - r_k$ . Consider, by (S.55) and Assumption (L1), for some constant  $c > 0$ , we have in probability that

$$\lambda_{r_k}(\mathbf{S}_{11,r}) \geq \check{g}_{-k}^{(m)} \min_{\|\mathbf{w}\|=1: \mathbf{w} \in \mathcal{N}^\perp} \left\| \mathbf{w}^\top \text{diag}^{-1/2}(\boldsymbol{\Sigma}_{y,m+1}^{(k)}) \mathbf{A}_k \right\|^2 \lambda_r(T^{-1} \mathbf{Z}_f^{(k)} \mathcal{A}_{f,T} \mathbf{M}_T \mathcal{A}_{f,T}^\top \mathbf{Z}_f^{(k)\top}) \geq c d_k^{\alpha_k, r_k}.$$

At the same time, we have

$$\left\| \mathbf{Qm}_2^{(k)} \mathbf{Qm}_2^{(k)\top} + \mathbf{Qm}_3^{(k)} \mathbf{Qm}_3^{(k)\top} \right\| = O_P((\check{g}_{-k}^{(m)})^{-1}) = O_P(d_k^{\alpha_k, 1} g_s^{-1}),$$

and earlier calculations show that

$$\|\mathbf{S}_{12,r}\| = O_P\left(d_k^{\alpha_k, 1} g_s^{-1/2} \left(\sqrt{\frac{rd_k}{T}} + \frac{Kr\sqrt{d}}{T}\right)\right), \quad \|\mathbf{S}_{13,r}\| = O_P\left(d_k^{\alpha_k, 1} g_s^{-1/2} \sqrt{\frac{rr_e}{T}}\right).$$

Hence in probability, the eigenvalues of  $\mathbf{S}_{11,r}$  are dominating those in

$$\mathbf{G} := \sum_{j=2}^3 (\mathbf{S}_{1j,r} + \mathbf{S}_{j1,r} + \mathbf{Qm}_j^{(k)} \mathbf{Qm}_j^{(k)\top}).$$

Hence the  $r_k$  eigenvectors corresponding to the largest  $r_k$  eigenvalues of  $\mathbf{G} + \mathbf{S}_{11,r}$  coincide with those of  $\mathbf{S}_{11,r}$ 's for large enough  $d_k$ , which are all necessarily in  $\mathcal{N}^\perp$ . This means that the  $(r_k + 1)$ -th largest eigenvalue of  $\mathbf{G} + \mathbf{S}_{11,r}$  and beyond will have eigenvectors in  $\mathcal{N}$  since the dimension of  $\mathcal{N}^\perp$  is  $r_k$ . Take  $\mathbf{w} \in \mathcal{N}$ , then it is easy to see that

$$\mathbf{w}^\top (\mathbf{G} + \mathbf{S}_{11,r}) \mathbf{w} = \mathbf{w}^\top (\mathbf{Qm}_2^{(k)} \mathbf{Qm}_2^{(k)\top} + \mathbf{Qm}_3^{(k)} \mathbf{Qm}_3^{(k)\top}) \mathbf{w} = \mathbf{w}^\top \mathbf{R}_{y,m+1}^{(k)} \mathbf{w},$$

showing that  $\lambda_j(\mathbf{G} + \mathbf{S}_{11,r}) = \lambda_j(\mathbf{R}_{y,m+1}^{(k)})$  for  $j \in [d_k]/[r_k]$ , which is (S.64). This completes the proof of the lemma.  $\square$

*Proof of Theorem 5.2.* For  $j \in [r_k]$  and  $k \in [K]$ , we can decompose

$$\begin{aligned} \lambda_j(\widehat{\mathbf{R}}_{y,m+1}^{(k)}) &\geq \lambda_j(\mathbf{R}_{y,m+1}^{(k)}) \left\{ 1 - \left| \frac{\lambda_j(\widehat{\mathbf{R}}_{y,m+1}^{(k)})}{\lambda_j(\mathbf{S}_{y,m+1}^{(k)})} - 1 \right| - \left| \frac{\lambda_j(\mathbf{S}_{y,m+1}^{(k)})}{\lambda_j(\mathbf{R}_{y,m+1}^{(k)})} - 1 \right| \right. \\ &\quad \left. - \left| \frac{\lambda_j(\widehat{\mathbf{R}}_{y,m+1}^{(k)})}{\lambda_j(\mathbf{S}_{y,m+1}^{(k)})} - 1 \right| \cdot \left| \frac{\lambda_j(\mathbf{S}_{y,m+1}^{(k)})}{\lambda_j(\mathbf{R}_{y,m+1}^{(k)})} - 1 \right| \right\} \\ &= \lambda_j(\mathbf{R}_{y,m+1}^{(k)}) (1 + O_P(r_k d_k^{2\alpha_k, 1 - \alpha_k, j - 1} a_T(0) + a_T(\alpha_{k,1}))) \\ &\succeq_P r_k d_k^{1 - \alpha_k, 1 + \alpha_k, j} (1 + O_P(r_k d_k^{2\alpha_k, 1 - \alpha_k, j - 1} a_T(0) + a_T(\alpha_{k,1}))), \end{aligned}$$

where the second last line used the results in Lemma 8, and the last line used the result from Theorem 5.1, noting that

$$\begin{aligned} a_T(0) &= \sqrt{\frac{r}{T}} \left[ 1 + g_s^{-1/2} \left( r_e^{1/2} + d_k^{1/2} + K \sqrt{\frac{rd}{T}} \right) + g_s^{-1} \frac{K^2 r^{1/2} d}{T^{3/2}} \right] \\ &= O\left( \sqrt{\frac{r}{T}} \left[ 1 + K \sqrt{\frac{rd}{Tg_s}} + \frac{K^2 r^{1/2} d}{T^{3/2} g_s} \right] \right). \end{aligned}$$

Similarly, for  $j \in [d_k]/[r_k]$ ,

$$\begin{aligned} \lambda_j(\widehat{\mathbf{R}}_{y,m+1}^{(k)}) &\leq \lambda_j(\mathbf{R}_{y,m+1}^{(k)}) \left( 1 + \left| \frac{\lambda_j(\widehat{\mathbf{R}}_{y,m+1}^{(k)})}{\lambda_j(\mathbf{S}_{y,m+1}^{(k)})} - 1 \right| \right) + |\lambda_j(\mathbf{S}_{y,m+1}^{(k)}) - \lambda_j(\mathbf{R}_{y,m+1}^{(k)})| \\ &\quad + |\lambda_j(\mathbf{S}_{y,m+1}^{(k)}) - \lambda_j(\mathbf{R}_{y,m+1}^{(k)})| \cdot \left| \frac{\lambda_j(\widehat{\mathbf{R}}_{y,m+1}^{(k)})}{\lambda_j(\mathbf{S}_{y,m+1}^{(k)})} - 1 \right| \\ &= o_P(1) \cdot (1 + o_P(1)) + O_P(b_T) \leq 1 + O_P(b_T), \end{aligned}$$

where the equality used the results of Lemma 8 and (S.62). This completes the proof of the theorem.  $\square$

*Proof of Theorem 5.3.* To prove the theorem, we note that the proofs of Theorem 5.1 and 5.2 will go through for the Bootstrapped fibres and a redefined  $\Sigma_{y,m+1}^{(k)}$  in (5.3) (with  $\check{\mathbf{q}}_k^{(m)}$  replaced by  $\mathbf{W}_b \mathbf{W}_b^T \check{\mathbf{q}}_k^{(m)}$ ) if the following hold:

I.  $\|\mathbf{W}_b\| < \infty$ ;

II.  $\check{\mathbf{q}}_k^{(m)T} \mathbf{W}_b \mathbf{W}_b^T \mathbf{A}_{-k} \mathbf{A}_{-k}^T \mathbf{W}_b \mathbf{W}_b \check{\mathbf{q}}_k^{(m)} \asymp_P g_d / d_k^{\alpha_{k,1}}$ .

I. holds because we restrict the number of times a fibre can be chosen in a Bootstrap sample to be at most 8, so that  $\|\mathbf{W}_b\| \leq 8 < \infty$ .

Using the fact that  $\|\check{\mathbf{q}}_k^{(m)} - \mathbf{U}_{-k,(1)}\| = o_P(1)$ , similar to (S.47), II. can be proved if we can show that

$$\mathbf{U}_{-k,(1)}^T \mathbf{W}_b \mathbf{W}_b^T \mathbf{U}_{-k} \mathbf{G}_{-k} \mathbf{U}_{-k}^T \mathbf{W}_b \mathbf{W}_b^T \mathbf{U}_{-k,(1)} \asymp_P g_s / d_k^{\alpha_{k,1}}.$$

With Assumption (L1'), this boils down to showing that

$$Q := \mathbf{U}_{-k,(1)}^T \mathbf{W}_b \mathbf{W}_b^T \mathbf{U}_{-k,(1)} > c > 0 \quad (\text{S.65})$$

with high probability for some constant  $c > 0$  as  $T, d_k \rightarrow \infty$ . To do this, define  $\mathbf{U}_{-k,(1)} =: (u_j)_{j \in [d_k]}$ , and

$\mathcal{P} :=$  Information on the position of  $\xi_i^{(b)}$ ,  $i \in [d_k]$  on each column of  $\mathbf{W}_b$ ;

$\mathcal{V} :=$  Information on the values of the  $\xi_i^{(b)}$ ,  $i \in [d_k]$ ;

$\mathcal{D} :=$  Information on the values of  $u_j$ .

Let  $X_j$  be the number of  $\xi_i^{(b)}$ 's assigned at the  $j$ th row of  $\mathbf{W}_b$  (say at column number  $i_1, \dots, i_{X_j}$ ). Hence the  $j$ th diagonal value of  $\mathbf{W}_b \mathbf{W}_b^T$  is

$$(\mathbf{W}_b \mathbf{W}_b^T)_{jj} = \sum_{\ell=1}^{X_j} \xi_{i_\ell}^{(b)}, \text{ meaning that } E[(\mathbf{W}_b \mathbf{W}_b^T)_{jj} | \mathcal{P}] = \frac{X_j}{2}, \text{ var}[(\mathbf{W}_b \mathbf{W}_b^T)_{jj} | \mathcal{P}] = \frac{X_j}{4}.$$

But since  $\sum_{j=1}^{d_k} X_j = d_k$ , and each  $X_j$  is identically distributed, we have

$$E(X_j) = 1, \text{ corr}(X_i, X_j) = -\frac{1}{d_k - 1} \text{ for } i \neq j.$$

Hence

$$E(Q | \mathcal{D}) = E[E(Q | \mathcal{P}, \mathcal{D}) | \mathcal{D}] = E\left[\sum_{j=1}^{d_k} u_j^2 E[(\mathbf{W}_b \mathbf{W}_b^T)_{jj} | \mathcal{P}] | \mathcal{D}\right] = \frac{E(X_j)}{2} \sum_{j=1}^{d_k} u_j^2 = \frac{1}{2},$$

$$\text{var}(Q | \mathcal{D}) = E[\text{var}(Q | \mathcal{D}, \mathcal{P}) | \mathcal{D}] + \text{var}(E(Q | \mathcal{D}, \mathcal{P}) | \mathcal{D}),$$

$$\begin{aligned} &= E\left(\sum_{j=1}^{d_k} u_j^4 \cdot \frac{X_j}{4} | \mathcal{D}\right) + \text{var}\left(\sum_{j=1}^{d_k} u_j^2 \cdot \frac{X_j}{2} | \mathcal{D}\right) \\ &= \frac{1}{4} \sum_{j=1}^{d_k} u_j^4 + \sum_{j=1}^{d_k} \frac{u_j^4}{4} \text{var}(X_j) + \frac{1}{2} \sum_{i < j} u_i^2 u_j^2 \left(-\frac{\text{var}(X_j)}{d_k - 1}\right) \\ &= \frac{1}{4} \sum_{j=1}^{d_k} u_j^4 + \frac{\text{var}(X_j)}{4(d_k - 1)} \left(d_k \sum_{j=1}^{d_k} u_j^4 - 1\right). \end{aligned}$$

But  $\sum_{i=1}^{d_k} u_i^2 = 1$  and the same moment structure up to the 4th order imply that

$$Eu_j^2 = \frac{1}{d_k}, \quad Eu_j^4 = \frac{1}{d_k} - (d_k - 1)E(u_i^2 u_j^2) = \frac{1}{d_k^2} + o(d_k^{-1}) = o(d_k^{-1}).$$

Hence  $E(Q) = E(E(Q|\mathcal{D})) = 1/2$ , and

$$\text{var}(Q) = E(\text{var}(Q|\mathcal{D})) = \frac{d_k}{4}Eu_j^4 + \frac{\text{var}(X_j)}{4(d_k - 1)}(d_k^2 Eu_j^4 - 1) = o(1),$$

since  $0 \leq X_j \leq 8$  means that  $\text{var}(X_j) < \infty$ . With  $E(Q) = 1/2$  and  $\text{var}(Q) = o(1)$ , we have established (S.65). This completes the proof of the theorem.  $\square$

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